

Systematic Review and Meta-analysis

Which Graft Is Associated With Better Outcomes in ACL Reconstruction? A Systematic Review and Network Meta-analysis of Randomized Controlled Trials

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Abstract

Background In clinical practice, different surgical methods, including autografts, allografts, and artificial ligaments, have been used in reconstruction after ACL injuries. However, the available evidence is conflicting in terms of which approach has yielded better patient-reported outcomes, joint stability, and risk of graft reinjury. A network meta-analysis allows the comparison of approaches that have not been compared head-to-head in individual RCTs. Although several network

meta-analyses have been performed on this topic, they are hampered by methodological limitations.

Questions/purposes We performed an updated network meta-analysis of RCTs to answer the following questions: Considering different grafts applied in ACL reconstruction, which graft is superior in terms of (1) patient-reported outcomes (International Knee Documentation Committee [IKDC] objective score, Lysholm score, and Tegner score), (2) knee stability results (Lachman test, pivot-shift test, and KT-1000/2000 arthrometer side-to-side difference), and (3) risk of graft reinjury?

Methods We searched the PubMed, Embase, Cochrane Library, and Web of Science databases from January 2000 to November 2024 for completed studies. We reviewed 30,976 papers, and 27 met inclusion criteria, which were as follows: RCTs published in English, RCTs that involved ACL reconstruction with different grafts (autografts, allografts, and artificial ligaments) with fully described surgical techniques, RCTs that reported on at least one of the relevant outcome measures (preoperative and post-operative IKDC objective score, Lysholm score, Tegner score, Lachman test, pivot-shift test, KT-1000/2000 arthrometer side-to-side difference, and risk of graft reinjury), and RCTs that reported a minimum follow-up period of 12 months with at least 80% follow-up completeness. The Cochrane risk-of-bias tool (RoB 2.0) was used to assess the quality of the included studies, with 18 studies judged as being at a low risk of bias and 9 studies assessed as being at an unclear risk of bias. In total, 2572 patients were treated with 16 different types of grafts. The mean length of follow-up was 47 months. The mean age of the included patients was 28 years, and 71% of the patients were male. A Bayesian network meta-analysis was performed via the “gemtc” and “coda” packages of R Studio, version 4.2.1, and the “networkplot” command of Stata 15

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software was used to plot network relationships for direct and indirect comparisons between different interventions. The surface under the cumulative ranking curve (SUCRA) was calculated to rank the intervention effects of different interventions. A SUCRA value approaching 1 (100%) indicates a greater probability that the corresponding treatment is superior in terms of efficacy or safety. The most common treatment comparisons were the single-bundle 4-strand semitendinosus with gracilis tendon autograft versus the bone–patellar tendon–bone autograft and the single-bundle 4-strand semitendinosus with gracilis tendon autograft versus the double-bundle 4-strand semitendinosus with gracilis tendon autograft.

Results Although there were no differences when comparing different grafts for traditional pairwise comparisons of patient-reported outcomes, joint stability, and graft reinjury, the network meta-analysis allows ranking of interventions by efficacy or safety through the SUCRA. In terms of patient-reported outcomes, the double-bundle 4-strand semitendinosus with gracilis tendon autograft had the highest ranking for the IKDC objective score (SUCRA 77%), the single-bundle 3-strand semitendinosus tendon autograft was ranked most favorably for the Lysholm score (SUCRA 78%), and the single-bundle 4-strand semitendinosus with gracilis tendon autograft with synthetic artificial ligament had the highest ranking for the Tegner score (SUCRA 90%). Concerning knee stability, the double-bundle 4-strand semitendinosus with gracilis tendon autograft ranked first (SUCRA 58%) on the Lachman test, the single-bundle 5-strand semitendinosus with gracilis tendon autograft ranked first (SUCRA 64%) in terms of the pivot-shift test, and the bone–patellar tendon–bone autograft was ranked most favorably for the KT-1000/2000 arthrometer side-to-side difference (SUCRA 91%). The most effective treatment in terms of graft reinjury was the double-bundle 4-strand semitendinosus with gracilis tendon autograft (SUCRA 70%). Again, a higher SUCRA value (close to 1) represents higher probability of a treatment being ranked among the most effective interventions.

Conclusion In ACL reconstruction, based on the results of SUCRA rankings, patients should be counseled that the double-bundle 4-strand semitendinosus with gracilis tendon autograft and the bone–patellar tendon–bone autograft represent potentially superior surgical techniques for simultaneously enhancing patient comfort, restoring knee anterior stability, and having low reinjury risk, and the single-bundle 5-strand semitendinosus with gracilis tendon autograft has the potential advantage in restoring rotational knee stability. To achieve best patient-reported outcomes, the single-bundle 3/4-strand semitendinosus tendon autograft and the single-bundle 4-strand semitendinosus with gracilis tendon autograft with synthetic artificial ligament are also recommended. Further comparisons of different grafts should be performed to determine whether

meaningful differences exist between men and women in terms of these surgical outcomes.

Level of Evidence Level I, therapeutic study.

Introduction

An ACL injury is one of the most common sports injuries of the knee, with > 120,000 surgical procedures performed to treat it each year in the United States [23, 27]. An ACL injury may result in recurrent knee instability, decreased activity, meniscal injuries, and increased risk of knee osteoarthritis [18]. ACL reconstruction is widely used for patients who experience severe subjective instability during specific activities and who wish to return to certain sports [12, 18, 33]. In clinical practice, many operative methods, including different autografts, allografts, and artificial ligaments, have been applied for ACL reconstruction, but each of these approaches possesses its own set of advantages and disadvantages [5, 8, 10, 13, 22, 35, 39, 40, 47, 48].

Despite the numerous RCTs and meta-analyses that have been conducted to establish the superiority of one graft over another, there is still no consensus on which approach leads to the best postoperative outcomes [3, 6, 9, 11, 17, 26, 37, 55]. Furthermore, the large number of different grafts makes it impossible to compare all treatments. A network meta-analysis allows for the comparison of multiple treatments in the same study, taking into account all existing direct and indirect evidence and potentially ranking them, even if they have not previously been compared in head-to-head RCTs [7]. Although several previous network meta-analyses have been conducted on this topic, none have adequately differentiated the distinct clinical impact of different bundles and strands of grafts [29, 51].

We therefore performed an updated network meta-analysis of RCTs to answer the following questions: Considering different grafts applied in ACL reconstruction, which graft is superior in terms of (1) patient-reported outcomes (International Knee Documentation Committee [IKDC] objective score, Lysholm score, and Tegner score), (2) knee stability results (Lachman test, pivot-shift test, and KT-1000/2000 arthrometer side-to-side difference), and (3) risk of graft reinjury?

Materials and Methods

This systematic review and network meta-analysis was performed and reported in line with the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) Statement [32]. The protocol was prospectively registered with a prespecified methodology on PROSPERO (CRD42024601097).

Search Strategy and Eligibility Criteria

The inclusion criteria for this study were as follows: (1) RCTs published in English, (2) ACL reconstruction using a graft or artificial ligament (autografts, allografts, and artificial ligaments) with the surgical techniques fully described, (3) at least one of the relevant outcome measures reported (preoperative and postoperative IKDC objective score, Lysholm score, Tegner score, Lachman test, pivot-shift test, KT-1000/2000 arthrometer side-to-side difference, and risk of graft reinjury), and (4) minimum follow-up period of 12 months and at least 80% follow-up completeness at the minimum follow-up time.

The exclusion criteria were as follows: (1) non-RCTs (cohort studies, case-control studies, case reports, or review articles), (2) duplicate publications, (3) full text that could not be obtained, (4) a follow-up period of < 12 months or a follow-up completeness of < 80% at the minimum follow-up time, (5) studies that did not directly compare the outcomes of different grafts, and (6) studies in which patients had fractures on the same side or previous knee surgery (except diagnostic arthroscopy or meniscus surgery).

Information Sources and Search Strategy

Five authors (CFC, SYQ, TL, LY, YX) independently and systematically searched four databases (PubMed, Embase, Cochrane Library, and Web of Science) to identify studies published from January 2000 to November 2024. The base terms used included “ACL,” “transplants,” and “graft” (Appendices 1-4; <http://links.lww.com/CORR/B432>).

Study Selection

We imported all the retrieved studies into Endnote and then excluded duplicates. After screening the title or abstract, we filtered out preliminary studies, downloaded the full text, read the full text, and screened out the original studies that fit this systematic review. According to the inclusion and exclusion criteria, three authors (CFC, SYQ, LW) independently selected all the articles and cross-checked them by reviewing the full texts. Any disagreements at the inclusion stage were resolved by discussion with the corresponding author (JL).

Data Extraction

Data were extracted via a standardized extraction spreadsheet that included title, first author, year of publication, study type, author country, treatment method, and

outcomes. The outcomes evaluated included the preoperative and postoperative IKDC objective score, Lysholm score, Tegner score, Lachman test, pivot-shift test, KT-1000/2000 arthrometer side-to-side difference, and frequency of graft reinjury.

Study Characteristics

A total of 30,976 records were obtained by searching keywords, including PubMed (n = 9521), Embase (n = 11,327), the Cochrane Library (n = 1214), and the Web of Science (n = 8914). A total of 127 studies underwent full-text review, of which 27 studies [1, 2, 4, 14-16, 19-21, 24, 25, 28, 30, 31, 34, 36, 38, 41-43, 45, 46, 49, 50, 52-54] were ultimately included in the final analysis (Fig. 1).

The 27 studies included in this review were published in English and included a total of 2572 patients (Table 1). Most papers were two-arm studies (93% [25 of 27]), with only two studies (7% [2 of 27]) [25, 28] having three arms. The mean length of follow-up was 47 months. The mean age of the combined patients was 28 years, and 71% of the patients were male. Given that most studies reported that there were no differences in gender composition between the experimental group and the control group, and only four studies [15, 20, 38, 43] analyzed postoperative indicators with gender differences as a subgroup, it was not feasible to include gender differences analysis in this network meta-analysis. The IKDC objective score was evaluated in 14 studies with 1342 patients (Fig. 2A), the Lysholm score was evaluated in 9 studies with a total of 885 patients (Fig. 2B), and the Tegner score was evaluated in 7 studies with a total of 964 patients (Fig. 2C). The Lachman test was evaluated in 16 studies with 1446 patients (Fig. 3A), the pivot-shift test was evaluated in 18 studies with 1710 patients (Fig. 3B), and the KT-1000/2000 side-to-side difference was evaluated in 7 studies with 526 patients (Fig. 3C). Risk of graft reinjury was evaluated in 13 studies with 1210 patients (Fig. 4). The most common treatment comparisons (which contributed to each network analysis) were single-bundle 4-strand semitendinosus with gracilis tendon autograft versus bone-patellar tendon-bone autograft and single-bundle 4-strand semitendinosus with gracilis tendon autograft versus the double-bundle 4-strand semitendinosus with gracilis tendon autograft. A total of 16 different graft types were used for ACL reconstruction in these patients (Supplemental Table 1; <http://links.lww.com/CORR/B432>).

Risk of Bias and Publication Bias Assessment

Two researchers (CFC, TL) assessed the quality of the included studies with the Cochrane risk-of-bias tool (RoB 2.0). The studies were categorized as “low risk,” “unclear

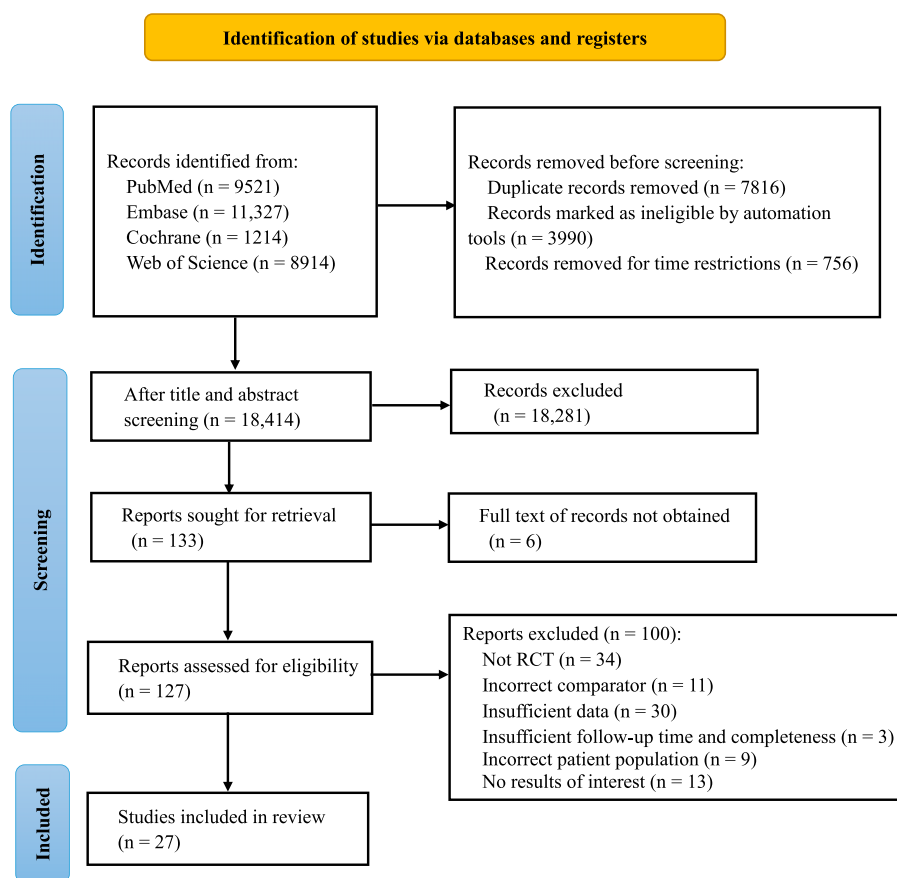


Fig. 1 Flow chart of article selection process.

risk,” or “high risk,” and any discrepancies in the assessment were judged after discussion with a third researcher (JL). The reviewers assessed articles for publication bias in five areas: (1) bias arising from the randomization process, (2) bias from deviations from the intended intervention, (3) bias from missing outcome data, (4) bias in outcome measures, and (5) bias in the choice of outcome report. The Network package of Stata 15 was used to draw funnel plots for rough qualitative judgment of publication bias.

In the assessment of the individual trials, 18 studies were judged to be at a low risk of bias, which had low risk of bias in at least four major categories of bias, and 9 studies were judged to be at an unclear risk of bias (Fig. 5A). A cumulative evaluation of the bias revealed that the quality of the studies was overall deemed to be good (Fig. 5B). In the evaluation of the risk of bias across trials, the domains of randomization process, missing outcome data, and selection of the reported results were deemed to be at a low risk of bias. On the contrary, the domains relating to blinding (patients, surgeons, and evaluators) were considered to be at an unclear risk of bias, which was mainly attributed to the nature of surgical interventions. The results of the funnel plots demonstrate that almost all the

points are situated within the funnel and exhibit greater symmetry between the left and right sides, suggesting a low likelihood of publication bias (Supplemental Fig. 1; <http://links.lww.com/CORR/B432>).

Effect Measures

We used relative risk for dichotomous data and we used mean difference for continuous variable data. Both types of indicators are expressed as 95% confidence intervals. In addition, a p value < 0.05 denoted statistical significance. A Bayesian network meta-analysis was performed via the “gemtc” and “coda” packages of R Studio, version 4.2.1, and the “networkplot” command of Stata 15 software was used to plot network relationships for direct and indirect comparisons between different interventions. The consistency model was established with an annealing number of 20,000 and an iteration number of 50,000, and the deviation information criterion (DIC) results of the iteration results were further compared with the DIC results of the inconsistency model. If the difference value was < 5 , the data conformed to consistency, and the results of indirect

Table 1. Summarized characteristics of included studies

Author [ref.]	Year	Country	Type of graft	Sample size	Age in years	Included outcome	Mean follow-up time in months
Jansson et al. [20]	2003	Finland					
Treatment Group 1			Bone–patellar tendon–bone autograft	51		1, 2, 3, 4, 5	24
Treatment Group 2			Single-bundle 4-strand semitendinosus with gracilis tendon autograft	48		1, 2, 3, 4, 5	24
Gobbi et al. [15]	2005	Italy					
Treatment Group 1			Single-bundle 4-strand semitendinosus tendon autograft	50	31	3	36
Treatment Group 2			Single-bundle 4-strand semitendinosus with gracilis tendon autograft	47	29	3	36
Ibrahim et al. [19]	2005	Kuwait					
Treatment Group 1			Single-bundle 4-strand semitendinosus with gracilis tendon autograft	40	22	1, 2	81
Treatment Group 2			Double-bundle 4-strand semitendinosus with gracilis tendon autograft	45	22	1, 2	81
Laxdal et al. [25]	2005	Sweden					
Treatment Group 1			Bone–patellar tendon–bone autograft	40	28	4, 5, 7	26
Treatment Group 2			Single-bundle 3-strand semitendinosus tendon autograft	39	24	4, 5, 7	26
Treatment Group 3			Single-bundle 4-strand semitendinosus with gracilis tendon autograft	39	26	4, 5, 7	26
Järvelä [21]	2007	Finland					
Treatment Group 1			Double-bundle 4-strand semitendinosus with gracilis tendon autograft	35	33 ± 9.0	1, 3, 4, 7	14
Treatment Group 2			Single-bundle 4-strand semitendinosus with gracilis tendon autograft	30	33 ± 9.0	1, 3, 4, 7	14
Siebold et al. [41]	2008	Germany					
Treatment Group 1			Single-bundle 4-strand semitendinosus with gracilis tendon autograft	35	28	6, 7	19
Treatment Group 2			Double-bundle 4-strand semitendinosus with gracilis tendon autograft	35	29	6, 7	19
Aglietti et al. [1]	2010	Italy					
Treatment Group 1			Single-bundle 4-strand semitendinosus with gracilis tendon autograft	35	28	6, 7	24
Treatment Group 2			Double-bundle 4-strand semitendinosus with gracilis tendon autograft	35	28	6, 7	24

Table 1. continued

Author [ref.]	Year	Country	Type of graft	Sample size	Age in years	Included outcome	Mean follow-up time in months
Barenius et al. [2]	2010	Sweden					
Treatment Group 1			Bone–patellar tendon–bone autograft	78	33 ± 6.3	3, 4, 5, 7	100.8
Treatment Group 2			Single-bundle 4-strand semitendinosus tendon autograft	75	35 ± 7.5	3, 4, 5, 7	100.8
Ghalayini et al. [14]	2010	UK					
Treatment Group 1			Synthetic artificial ligament	24	31.7 ± 6.7	2, 3, 7	60
Treatment Group 2			Bone–patellar tendon–bone autograft	26	30.9 ± 6.7	2, 3, 7	60
Volpi et al. [49]	2010	Italy					
Treatment Group 1			Bone–patellar tendon–bone autograft	20	27 ± 5.7	3	31
Treatment Group 2			Double-bundle 4-strand semitendinosus with gracilis tendon autograft	20	30 ± 4.9	3	28
Sun et al. [45]	2011	China					
Treatment Group 1			Single-bundle 4-strand semitendinosus with gracilis tendon autograft	91	29.6 ± 6.9	1, 2, 3, 4, 5	91.2
Treatment Group 2			Single-bundle 4-strand semitendinosus with gracilis tendon allograft	95	31.2 ± 8.3	1, 2, 3, 4, 5	94.8
Noh et al. [31]	2011	Korea					
Treatment Group 1			Single-bundle 4-strand semitendinosus with gracilis tendon autograft	33	23	1, 2, 3, 4, 5	28.1
Treatment Group 2			Single-bundle 2-strand Achilles tendon allograft	32	22	1, 2, 3, 4, 5	31.6
Sajovic et al. [36]	2011	Slovenia					
Treatment Group 1			Single-bundle 4-strand semitendinosus with gracilis tendon autograft	32	36	1, 2, 7	132
Treatment Group 2			Bone–patellar tendon–bone autograft	32	38	1, 2, 7	132
Lui et al. [28]	2012	China					
Treatment Group 1			Bone–patellar tendon–bone autograft	14	26.5 ± 5.0	2	12
Treatment Group 2			Single-bundle 4-strand semitendinosus with gracilis tendon autograft	21	23.2 ± 3.9	2	12
Treatment Group 3			Double-bundle 4-strand semitendinosus with gracilis tendon autograft	17	26.4 ± 4.9	2	12
Noh et al. [30]	2013	Korea					
Treatment Group 1			Single-bundle 2-strand Achilles tendon with bone block allograft	33	23	1, 2, 3, 4, 6	30.9

Table 1. continued

Author [ref.]	Year	Country	Type of graft	Sample size	Age in years	Included outcome	Mean follow-up time in months
Treatment Group 2	2013	Korea	Single-bundle 2-strand Achilles tendon allograft	34	22	1, 2, 3, 4, 6	30.9
Song et al. [43]			Double-bundle 4-strand tibialis tendon allograft	65	30.3	1, 2, 7	65.6
Treatment Group 1			Single-bundle 2-strand tibialis tendon allograft	65	35.5	1, 2, 7	68.4
Treatment Group 2	2014	Iran	Single-bundle 2-strand tibialis tendon allograft	65	35.5	1, 2, 7	68.4
Razi et al. [34]			Bone–patellar tendon–bone autograft	46	30.8 ± 4.5	1, 2, 3	36
Treatment Group 1			Double-bundle 4-strand semitendinosus with gracilis tendon autograft	41	28.2 ± 3.7	1, 2, 3	36
Treatment Group 2	2016	Japan	Double-bundle 4-strand semitendinosus with gracilis tendon autograft	76	28.2 ± 12.6	1, 5, 6, 7	24
Sasaki et al. [38]			Bone–patellar tendon–bone autograft	74	27.0 ± 11.9	1, 5, 6, 7	24
Treatment Group 1			Double-bundle 4-strand semitendinosus with gracilis tendon autograft	78	30.5 ± 4.9	1, 2, 3, 4, 5	55.2
Tian et al. [46]	2016	China	Double-bundle 4-strand semitendinosus with gracilis tendon autograft	79	29.9 ± 6.1	1, 2, 3, 4, 5	55.2
Treatment Group 1			Double-bundle 4-strand semitendinosus with gracilis tendon allograft	79	29.9 ± 6.1	1, 2, 3, 4, 5	55.2
Treatment Group 2			Double-bundle 4-strand semitendinosus with gracilis tendon allograft	79	29.9 ± 6.1	1, 2, 3, 4, 5	55.2
Yang et al. [50]	2017	China	Single-bundle 4-strand semitendinosus with gracilis tendon autograft	30	34.9 ± 4.3	1	34.8
Treatment Group 1			Double-bundle 4-strand gracilis tendon autograft	30	34.8 ± 4.2	1	36.2
Treatment Group 2			Double-bundle 4-strand gracilis tendon autograft	30	34.8 ± 4.2	1	36.2
Yoo et al. [52]	2017	Korea	Single-bundle 4-strand semitendinosus with gracilis tendon autograft	71	30	1, 2, 3, 7	32.8
Treatment Group 1			Single-bundle 2-strand tibialis tendon allograft	70	24	1, 2, 3, 7	34.5
Treatment Group 2			Single-bundle 2-strand tibialis tendon allograft	70	24	1, 2, 3, 7	34.5
Bi et al. [4]	2018	China	Single-bundle 4-strand peroneus longus tendon autograft	62	29.1 ± 6.5	1, 7	29.7
Treatment Group 1			Single-bundle 4-strand peroneus longus tendon autograft	62	29.1 ± 6.5	1, 7	29.7
Treatment Group 2			Single-bundle 4-strand semitendinosus tendon autograft	62	27.9 ± 6.7	1, 7	30.3
Zhang et al. [54]	2019	China	Single-bundle 4-strand semitendinosus with gracilis tendon autograft	78	27.6 ± 7.3	1, 2, 3, 4, 5, 6	28.2

Table 1. continued

Author [ref.]	Year	Country	Type of graft	Sample size	Age in years	Included outcome	Mean follow-up time in months
Treatment Group 2			Double-bundle 4-strand semitendinosus with gracilis tendon autograft	78	25.9 ± 5.2	1, 2, 3, 4, 5, 6	28.2
Smith et al. [42]	2020	USA					
Treatment Group 1			Single-bundle 4-strand semitendinosus tendon autograft	32	17.6 ± 2.5	6, 7	24
Treatment Group 2			Bone–patellar tendon–bone autograft	32	17.8 ± 2.3	6, 7	24
Guglielmetti et al. [16]	2021	Brazil					
Treatment Group 1			Single-bundle 4-strand semitendinosus with gracilis tendon autograft	31	24.6 ± 5.3	1, 2, 3, 4, 6	24
Treatment Group 2			Bone–patellar tendon–bone autograft	31	25.2 ± 5.5	1, 2, 3, 4, 6	24
Krishna et al. [24]	2021	Singapore					
Treatment Group 1			Single-bundle 4-strand semitendinosus with gracilis tendon autograft	28	26.3 ± 7.3	1, 2, 3, 4, 6	24
Treatment Group 2			Single-bundle 5-strand semitendinosus with gracilis tendon autograft	28	27.6 ± 7.3	1, 2, 3, 4, 6	24
Zaid et al. [53]	2023	China					
Treatment Group 1			Single-bundle 4-strand semitendinosus with gracilis tendon autograft	47	26.4 ± 4.0	1, 2, 4, 5, 7	18
Treatment Group 2			Single-bundle 4-strand semitendinosus with gracilis tendon autograft with synthetic artificial ligament	47	26.1 ± 3.5	1, 2, 4, 5, 7	18

Data presented as the number or mean ± SD. 1 = pivot-shift test; 2 = Lachman test; 3 = IKDC objective score; 4 = Lysholm score; 5 = Tegner score; 6 = KT-1000/2000 arthrometer side-to-side difference; 7 = graft reinjury.

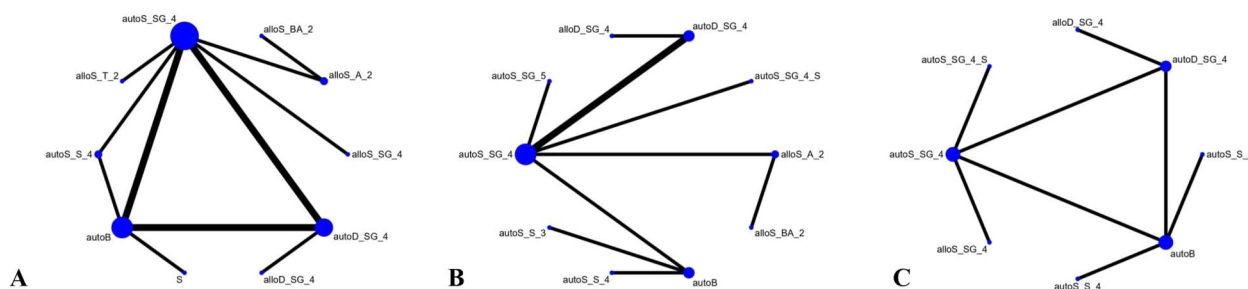


Fig. 2 Network plots for clinical outcomes, including (A) the IKDC objective score, (B) the Lysholm score, and (C) the Tegner score. Abbreviations and full names of the grafts are listed in Supplemental Table 1 (<http://links.lww.com/CORR/B432>).

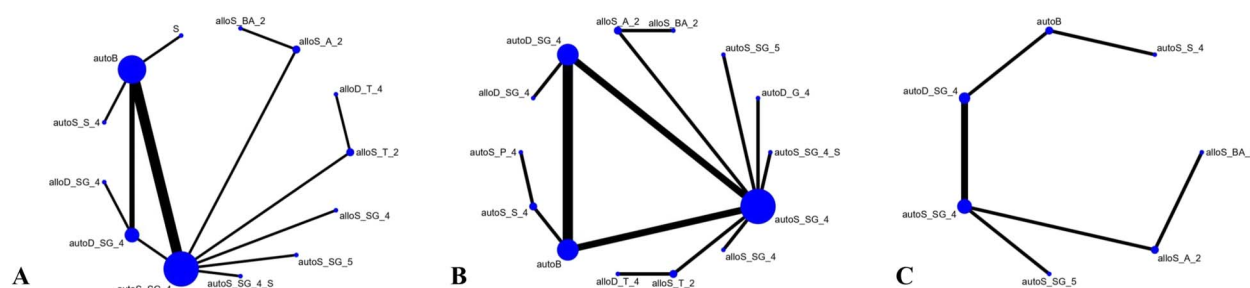


Fig. 3 Network plots for knee stability results, including (A) the Lachman test, (B) the pivot-shift test, and (C) the KT-1000/2000 side-to-side difference. Abbreviations and full names of the grafts are listed in Supplemental Table 1 (<http://links.lww.com/CORR/B432>).

comparisons of the network meta-analysis were calculated to be not significantly different from results of the direct comparisons. The surface under the cumulative ranking curve (SUCRA) was calculated to rank the intervention effects of different interventions. A SUCRA value approaching 1 (100%) indicates a greater probability that the corresponding treatment is superior in efficacy or safety rankings.

Heterogeneity Assessment

We used the I^2 test to evaluate heterogeneity between studies in both consistency and inconsistency models. I^2 values $> 50\%$ were deemed as substantial heterogeneity. Conversely, I^2 values $< 50\%$ were classified as low

heterogeneity. The heterogeneity assessment results demonstrated that the values were $< 50\%$ in both consistency and inconsistency models, indicating low heterogeneity in comparisons among studies (Supplemental Table 2; <http://links.lww.com/CORR/B432>). Although heterogeneity among the studies in this network meta-analysis was low, we employed a random-effects model for the analysis because it explicitly accounted for potential minimal heterogeneity.

Inconsistency Assessment

In this study, a local inconsistency test was performed using the node-splitting method for studies with closed loops. The results of local inconsistency analyses revealed no statistically significant deviations in closed-loop comparisons ($p > 0.05$), thus suggesting the validity of the consistency model (Supplemental Figs. 2-6; <http://links.lww.com/CORR/B432>).

Sensitivity Assessment

The sensitivity assessment of seven outcomes (IKDC, Lysholm score, Tegner score, Lachman test, pivot-shift test, KT-1000/2000 side-to-side difference, risk of graft reinjury) was conducted via consistency and inconsistency models. The results of posterior mean deviance (Dbar), effective number of parameters (pD), and DIC indicated that the values were similar between the two models (Supplemental Table 2; <http://links.lww.com/CORR/B432>), suggesting no substantial differences in model fit. The results of both inconsistency assessment and sensitivity assessment indicated that the studies included were comparable and that the data were stable, demonstrating consistent effect estimates across alternative model specifications (consistency and inconsistency models), which reflected the good ability

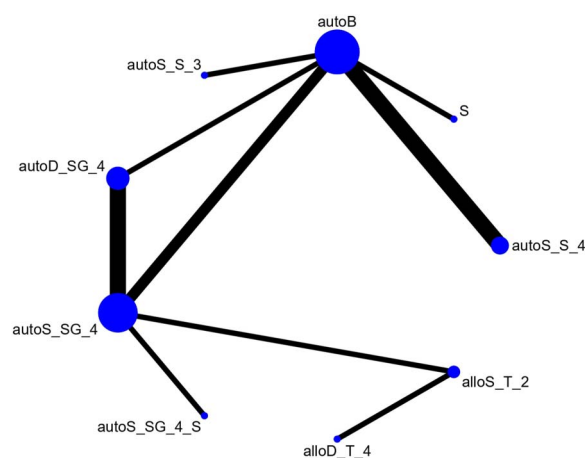


Fig. 4 Network plots for risk graft reinjury. The straight line denotes directly comparable studies, the thickness of the line segment signifies the quantity of directly comparable studies, the dots indicate the intervention, and the size of the dots represents the sample size for that intervention. Abbreviations and full names of the grafts are listed in Supplemental Table 1 (<http://links.lww.com/CORR/B432>).

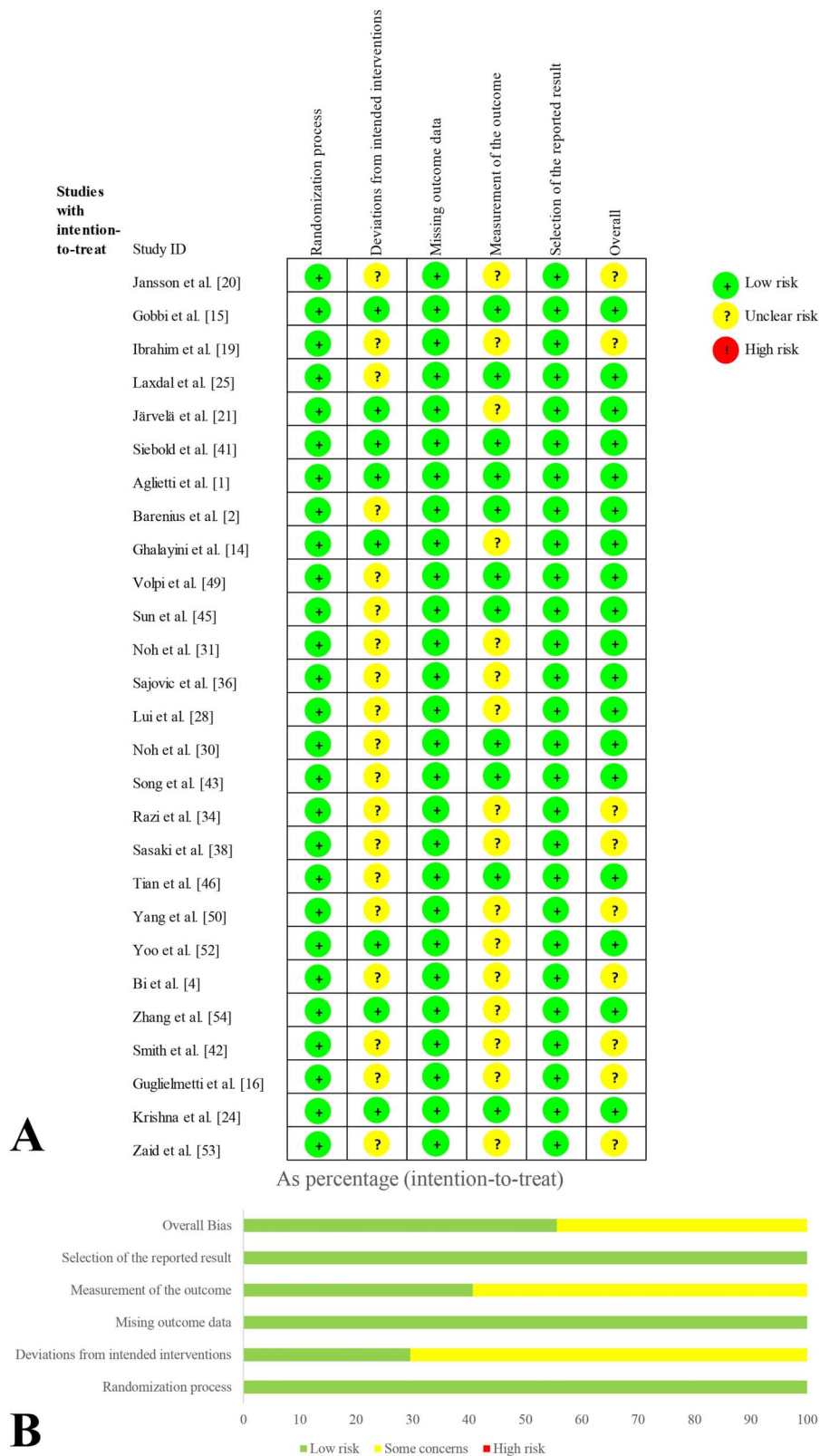


Fig. 5 (A) Risk of bias graph for each study and (B) summary of the risk of bias for the 27 RCTs. A color image accompanies the online version of this article.

of the findings to resist model assumptions and potential uncertainties in the data.

Results

Although there were no statistical differences when comparing different grafts for traditional pairwise comparisons of patient-reported outcomes, knee joint stability, and risk of graft reinjury (Supplemental Tables 3-9; <http://links.lww.com/CORR/B432>), the network meta-analysis-based SUCRA rankings provides a quantitative basis for ranking the efficacy or safety of different grafts.

Patient-reported Outcomes

The double-bundle 4-strand semitendinosus with gracilis tendon autograft and the bone–patellar tendon–bone autograft were ranked most favorably for the IKDC objective score (SUCRA 77% and 72%, respectively), the single-bundle 3-strand semitendinosus tendon autograft and the single-bundle 4-strand semitendinosus with gracilis tendon autograft were ranked most favorably for the Lysholm score (SUCRA 78% and 68%, respectively), and the single-bundle 4-strand semitendinosus with gracilis tendon autograft with synthetic artificial ligament and the single-bundle 4-strand semitendinosus with gracilis tendon autograft were ranked most favorably for the Tegner score (SUCRA 90% and 62%, respectively) (Table 2). Again, a higher SUCRA value (close to 1) represents higher probability of a treatment being ranked among the most effective interventions.

Knee Stability Results

With the Lachman test, the double-bundle 4-strand semitendinosus with gracilis tendon autograft ranked first (SUCRA 58%). The single-bundle 5-strand semitendinosus with gracilis tendon autograft and the double-bundle 4-strand gracilis tendon autograft were ranked most favorably (SUCRA 64% and 61%, respectively) in terms of the pivot-shift test. The bone–patellar tendon–bone autograft and the double-bundle 4-strand semitendinosus with gracilis tendon autograft were ranked most favorably for the KT-1000/2000 arthrometer side-to-side difference (SUCRA 91% and 71%, respectively) (Table 3).

Risk of Graft Reinjury

The three most effective treatments in terms of low risk of graft reinjury were the double-bundle 4-strand

semitendinosus with gracilis tendon autograft (SUCRA 70%), the synthetic artificial ligament (64%), and the bone–patellar tendon–bone autograft (61%) (Table 4).

Discussion

In clinical practice, different methods (autografts, allografts, and artificial ligaments) have been used for ACL reconstruction. Despite an abundance of RCTs and resultant meta-analyses, there is still no consensus on which approach leads to the best postoperative outcomes. Moreover, the substantial number of various grafts makes it unfeasible to compare all treatments in two-arm RCTs or traditional meta-analyses. We therefore conducted a network meta-analysis to compare different grafts for ACL reconstruction, as many had not previously been compared in head-to-head RCTs. This was achieved by pooling all study data and summarizing results of both direct and indirect comparisons, along with the rank of the included treatments. While traditional pairwise comparisons (similar to traditional meta-analyses) showed no differences in postoperative outcomes among grafts, the network meta-analysis enables ranking of therapeutic interventions by efficacy or safety using the SUCRA, which is a key advantage of this methodology. We found that the double-bundle 4-strand semitendinosus with gracilis tendon autograft had the highest probability of being superior in terms of the IKDC objective score, Lachman test, and risk of graft reinjury; the single-bundle 3-strand semitendinosus tendon autograft and the single-bundle 4-strand semitendinosus with gracilis tendon autograft with synthetic artificial ligament were ranked most favorably for the Lysholm and Tegner scores, respectively; the single-bundle 5-strand semitendinosus with gracilis tendon autograft and the bone–patellar tendon–bone autograft ranked first in terms of the pivot-shift test and the KT-1000/2000 arthrometer side-to-side difference, respectively. Notably, SUCRA rankings should be interpreted as probable trends rather than definitive conclusions. Clinicians should balance SUCRA rankings with direct pairwise comparisons where those are available. For example, if two grafts have similar SUCRA values and no differences in direct comparisons, the choice may reasonably depend on surgeon experience or the patient's subjective preferences rather than model-derived probabilities alone.

Limitations

This network meta-analysis had several limitations. First, given that most studies merely reported that there was no statistically significant difference in gender composition between the experimental group and the control group, and

Table 2. Summarized results of SUCRA values for patient-reported outcomes in the included studies

Rank	Patient-reported outcome		
	IKDC objective score	Lysholm score	Tegner score
1	Double-bundle 4-strand semitendinosus with gracilis tendon autograft (77%)	Single-bundle 3-strand semitendinosus tendon autograft (78%)	Single-bundle 4-strand semitendinosus with gracilis tendon autograft with synthetic artificial ligament (90%)
2	Bone–patellar tendon–bone autograft (72%)	Single-bundle 4-strand semitendinosus with gracilis tendon autograft (68%)	Single-bundle 4-strand semitendinosus with gracilis tendon autograft (62%)
3	Single-bundle 4-strand semitendinosus tendon autograft (67%)	Single-bundle 5-strand semitendinosus with gracilis tendon autograft (66%)	Single-bundle 3-strand semitendinosus tendon autograft (57%)
4	Single-bundle 2-strand Achilles tendon allograft (65%)	Single-bundle 2-strand Achilles tendon allograft (55%)	Bone–patellar tendon–bone autograft (53%)
5	Single-bundle 4-strand semitendinosus with gracilis tendon autograft (55%)	Single-bundle 4-strand semitendinosus with gracilis tendon autograft with synthetic artificial ligament (55%)	Single-bundle 4-strand semitendinosus with gracilis tendon allograft (49%)
6	Single-bundle 2-strand tibialis tendon allograft (49%)	Single-bundle 2-strand Achilles tendon with bone block allograft (45%)	Double-bundle 4-strand semitendinosus with gracilis tendon autograft (44%)
7	Single-bundle 4-strand semitendinosus with gracilis tendon allograft (46%)	Single-bundle 4-strand semitendinosus tendon autograft (41%)	Double-bundle 4-strand semitendinosus with gracilis tendon allograft (35%)
8	Double-bundle 4-strand semitendinosus with gracilis tendon allograft (31%)	Bone–patellar tendon–bone autograft (39%)	Single-bundle 4-strand semitendinosus tendon autograft (12%)
9	Single-bundle 2-strand Achilles tendon with bone block allograft (26%)	Double-bundle 4-strand semitendinosus with gracilis tendon allograft (28%)	NA
10	Synthetic artificial ligament (12%)	Double-bundle 4-strand semitendinosus with gracilis tendon autograft (25%)	NA

A higher SUCRA value (approaching 100%) indicates a greater probability that the corresponding treatment is superior in terms of efficacy or safety. In other words, a higher SUCRA value indicates a better ranking. For example, double-bundle 4-strand semitendinosus with gracilis tendon autograft has a highest SUCRA of 77% in IKDC objective score, which means that it has a 77% probability of being the best treatment for the IKDC objective score (Rank 1); bone–patellar tendon–bone autograft has a SUCRA of 72%, positioning it as the second best treatment in terms of the IKDC objective score (Rank 2). NA = not available.

only four studies [15, 20, 38, 43] analyzed postoperative indicators with gender differences as a subgroup, it was not feasible to include gender differences analysis in this network meta-analysis. Thus, the reader should be cautious when applying the results of this study to both genders. Second, we acknowledge that a proportion of the included studies were at varying degrees of unclear risk of bias. The domain related to blinding was identified as having an unclear risk of bias, primarily because of the difficulty of blinding surgeons, which is a methodological constraint common to surgical interventions. It is important to note that most of the assessments of postoperative outcome assessments were conducted by nonsurgical clinicians or

rehabilitation specialists, thereby mitigating potential bias introduced by surgeon unblinding. However, because of awareness of treatment allocation, some assessors' evaluation of subjective endpoints, such as Lachman tests and pivot-shift tests, may have potential risk. This highlights the need for caution in interpreting findings in clinical applications. Third, inconsistencies in postoperative rehabilitation protocols and follow-up time may have introduced confounding factors into this study. Nevertheless, the minimum follow-up time of all studies was > 12 months, the average follow-up time was 47 months, and all patients underwent a series of standardized rehabilitation exercises according to different rehabilitation

Table 3. Summarized results of SUCRA values for knee stability results in the included studies

Rank	Knee stability results		
	Lachman test	Pivot-shift test	SSD
1	Double-bundle 4-strand semitendinosus with gracilis tendon autograft (58%)	Single-bundle 5-strand semitendinosus with gracilis tendon autograft (64%)	Bone–patellar tendon–bone autograft (91%)
2	Double-bundle 4-strand semitendinosus with gracilis tendon allograft (56%)	Double-bundle 4-strand with gracilis tendon autograft (61%)	Double-bundle 4-strand semitendinosus with gracilis tendon autograft (71%)
3	Single-bundle 4-strand semitendinosus with gracilis tendon autograft (56%)	Single-bundle 4-strand semitendinosus with gracilis tendon autograft with synthetic artificial ligament (58%)	Single-bundle 4-strand semitendinosus tendon autograft (59%)
4	Single-bundle 4-strand semitendinosus with gracilis tendon autograft with synthetic artificial ligament (54%)	Single-bundle 4-strand semitendinosus with gracilis tendon autograft (58%)	Single-bundle 5-strand semitendinosus with gracilis tendon autograft (46%)
5	Double-bundle 4-strand tibialis tendon allograft (54%)	Single-bundle 4-strand semitendinosus with gracilis tendon allograft (55%)	Single-bundle 4-strand semitendinosus with gracilis tendon autograft (33%)
6	Bone–patellar tendon–bone autograft (55%)	Single-bundle 2-strand Achilles tendon allograft (55%)	Single-bundle 2-strand Achilles tendon allograft (27%)
7	Single-bundle 4-strand semitendinosus tendon autograft (53%)	Single-bundle 2-strand Achilles tendon with bone block allograft (54%)	Single-bundle 2-strand Achilles tendon with bone block allograft (23%)
8	Single-bundle 4-strand semitendinosus with gracilis tendon allograft (53%)	Bone–patellar tendon–bone autograft (51%)	NA
9	Single-bundle 2-strand tibialis tendon allograft (51%)	Double-bundle 4-strand semitendinosus with gracilis tendon allograft (44%)	NA
10	Synthetic artificial ligament (51%)	Double-bundle 4-strand tibialis tendon allograft (42%)	NA
11	Single-bundle 5-strand semitendinosus with gracilis tendon autograft (50%)	Double-bundle 4-strand semitendinosus with gracilis tendon autograft (42%)	NA
12	Single-bundle 2-strand Achilles tendon allograft (34%)	Single-bundle 4-strand peroneus longus tendon autograft (41%)	NA
13	Single-bundle 2-strand Achilles tendon with bone block allograft (30%)	Single-bundle 4-strand semitendinosus tendon autograft (38%)	NA
14	NA	Single-bundle 2-strand tibialis tendon allograft (37%)	NA

A higher SUCRA value (approaching 100%) indicates a greater probability that the corresponding treatment is superior in terms of efficacy or safety. In other words, a higher SUCRA value indicates a better ranking. For example, double-bundle 4-strand semitendinosus with gracilis tendon autograft has a highest SUCRA of 58% in the Lachman test, which means that it has a 58% probability of being the best treatment for the Lachman test (Rank 1); double-bundle 4-strand semitendinosus with gracilis tendon allograft has a SUCRA of 56%, positioning it as the second best treatment in terms of the Lachman test (Rank 2). SSD = KT-1000/2000 arthrometer side-to-side difference.

programs. Therefore, this limitation does not seem disqualifying to us. To improve the quality of meta-analyses, we recommend that future studies include high-quality

RCTs with similar follow-up periods and homogeneous rehabilitation programs. Finally, there were no statistical differences in traditional pairwise comparisons for all

Table 4. Summarized results of SUCRA values for risk of graft reinjury in the included studies

Rank	Risk of graft reinjury
1	Double-bundle 4-strand semitendinosus with gracilis tendon autograft (70%)
2	Synthetic artificial ligament (64%)
3	Bone–patellar tendon–bone autograft (61%)
4	Single-bundle 2-strand tibialis tendon allograft (50%)
5	Single-bundle 4-strand semitendinosus with gracilis tendon autograft (48%)
6	Single-bundle 4-strand semitendinosus tendon autograft (42%)
7	Single-bundle 3-strand semitendinosus tendon autograft (40%)
8	Single-bundle 4-strand semitendinosus with gracilis tendon autograft with synthetic artificial ligament (39%)
9	Double-bundle 4-strand tibialis tendon allograft (36%)

A higher SUCRA value (approaching 100%) indicates a greater probability that the corresponding treatment is superior in terms of efficacy or safety. In other words, a higher SUCRA value indicates a better ranking. For example, double-bundle 4-strand semitendinosus with gracilis tendon autograft has a highest SUCRA of 70% in risk of graft reinjury, which means that it has a 70% probability of being the best treatment for the risk of graft reinjury (Rank 1); synthetic artificial ligament has a SUCRA of 64%, positioning it as the second best treatment in terms of the risk of graft reinjury (Rank 2).

postoperative outcomes among various grafts, and the ranking results of SUCRA were based on point estimate probabilities. As a consequence, it is important to consider higher ranking results as potential superiority rather than definitive evidence.

Patient-reported Outcomes

We found no difference in patient-reported outcomes among different grafts. However, the SUCRA ranking was performed by aggregation of nondirect evidence in our study, and the results showed that the double-bundle 4-strand semitendinosus with gracilis tendon autograft, the single-bundle 3/4-strand semitendinosus tendon autograft, and the single-bundle 4-strand semitendinosus with gracilis tendon autograft with synthetic artificial ligament represent potentially superior

surgical techniques in terms of the IKDC objective score, the Lysholm score, and the Tegner score, respectively. Therefore, we recommend that the grafts mentioned above be potentially considered for all patients concerned about postoperative comfort. Because the current postoperative assessment scale lacks an assessment of the patient's postoperative proprioceptive recovery, we suggest that attention should be paid to the inclusion of this indicator in the future, as it is one of the most important factors in the maintenance of the dynamic stability of the knee joint.

Knee Stability Results

This network meta-analysis revealed no difference in knee stability results among various grafts. In contrast to our findings, Sun et al. [44] found that ACL reconstruction with synthetic ligaments achieved better instrumented laxity than autografts. The difference may be a function of the use of two different autografts (bone–patellar tendon–bone graft and 4-strand hamstring tendon graft) in their study to compare with synthetic ligaments, which introduced a confounding factor into the analysis. Based on SUCRA ranking, the double-bundle 4-strand semitendinosus with gracilis tendon autograft and the bone–patellar tendon–bone autograft may reasonably be recommended for individuals who regularly participate in sharp cutting exercises, as evidenced by the SUCRA ranking, which indicates that these grafts are the most likely to be advantageous in restoring anterior stability to patients' knees.

Risk of Graft Reinjury

Overall, risk of graft reinjury did not differ across all grafts. Based on the SUCRA ranking results, three graft types (the double-bundle 4-strand semitendinosus with gracilis tendon autograft, the synthetic artificial ligament, and the bone–patellar tendon–bone autograft) are potentially recommended for primary ACL reconstruction in terms of low graft reinjury risk. The difference in reinjury risk should be considered as a component in a broader conversation with each patient about graft selection, which should also include potential differences in rehabilitation time, donor site morbidity, recovery of proprioception, and other complications. Moreover, further studies should focus on whether meaningful differences mentioned above are present in both men and women.

Conclusion

In ACL reconstruction, based on the results of SUCRA rankings, patients should be counseled that the double-bundle 4-strand semitendinosus with gracilis tendon

autograft and the bone–patellar tendon–bone autograft represent potentially superior surgical techniques for simultaneously enhancing patient comfort, restoring knee anterior stability, and minimizing reinjury risk, and the single-bundle 5-strand semitendinosus with gracilis tendon autograft has the potential advantage in restoring rotational knee stability. To achieve best patient-reported outcomes, the single-bundle 3/4-strand semitendinosus tendon autograft and the single-bundle 4-strand semitendinosus with gracilis tendon autograft with synthetic artificial ligament are also recommended for clinicians. Considering these findings, comparisons of different grafts should be further conducted to determine whether meaningful differences exist between men and women in terms of these surgical outcomes.

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Hamstring Tendon Versus Bone-Patellar Tendon-Bone Autograft for ACL Reconstruction with Concurrent Lateral Extra-Articular Procedure

A Systematic Review and Network Meta-Analysis of Randomized Controlled Trials

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Background: No studies have yet evaluated the clinical outcomes of different anterior cruciate ligament (ACL) autografts in combination with a lateral extra-articular procedure (LEP). Thus, we systematically reviewed randomized controlled trials (RCTs) and conducted a network meta-analysis (NMA) to compare graft options for ACL reconstruction (ACLR) with concurrent LEP.

Methods: A systematic search in the PubMed, Embase, Cochrane Library, and Google Scholar databases identified RCTs on primary ACLR plus LEP. Data on ACLR failure, residual knee instability, patient-reported outcome measures, and complications were analyzed using NMA.

Results: On the basis of 13 studies (1,690 patients), ACLR with a hamstring tendon (HT) autograft plus LEP was associated with significantly lower odds of graft rupture (odds ratio [OR]: 0.28; 95% confidence interval [CI]: 0.16 to 0.50), graft failure (OR: 0.27; 95% CI: 0.15 to 0.47), clinical failure (OR: 0.48; 95% CI: 0.36 to 0.65), and residual pivot shift (OR: 0.43; 95% CI: 0.22 to 0.84) compared with isolated ACLR. Bone-patellar tendon-bone (BPTB) autograft plus LEP was associated with a significantly lower clinical failure rate (OR: 0.30; 95% CI: 0.12 to 0.80) compared with isolated ACLR. Both HT (mean difference [MD]: 2.40; 95% CI: 1.25 to 3.55) and BPTB (MD: 3.70; 95% CI: 0.85 to 6.55) autograft plus LEP were associated with higher Lysholm scores compared with isolated ACLR. The outcomes did not differ between the graft types when combined with LEP.

Conclusions: The odds of graft rupture, graft failure, and clinical failure after ACLR with HT autograft plus LEP were lower by 72%, 73%, and 52%, respectively, than the odds after isolated ACLR. ACLR with BPTB autograft plus LEP significantly lowered only the odds of clinical failure, by approximately 70%, potentially due to the smaller sample size. Both grafts remain viable options for ACLR plus LEP, with the benefit of LEP requiring further validation.

Level of Evidence: Therapeutic Level I. See Instructions for Authors for a complete description of levels of evidence.

Graft selection in anterior cruciate ligament (ACL) reconstruction (ACLR) is critical, as it affects clinical outcomes^{1,2}. The graft failure rate after primary ACLR is approximately 7 times higher for allografts than for bone-patellar tendon-bone (BPTB) and hamstring tendon (HT) autografts³⁻⁶. However, the risk of revision is twice as high for HT autografts than for BPTB or quadriceps tendon autografts⁷⁻⁹.

Graft failure and residual rotational instability remain important concerns, with rates of up to 14% and 30%, respectively¹⁰⁻¹². Lateral extra-articular procedures (LEPs), such as anterolateral ligament reconstruction (ALLR) and lateral extra-articular tenodesis (LET), address these issues by providing secondary stabilization, mitigating the pivot shift phenomenon¹³⁻¹⁵. Recent meta-analyses showed that combining ACLR

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Disclosure: The **Disclosure of Potential Conflicts of Interest** forms are provided with the online version of the article (<http://links.lww.com/JBJS/I827>).

with LEP reduces graft failure and rotational instability more effectively than ACLR alone¹⁶⁻²¹.

Despite the importance of graft choice and the success of ACLR plus LEP, the optimal graft for combined ACLR and LEP remains unknown. No previous studies have compared different types of autografts used for ACLR with concurrent LEP. Therefore, we conducted a systematic review and network meta-analysis (NMA) to evaluate failure rates and outcomes according to graft type in ACLR plus LEP, hypothesizing superiority of BPTB over HT.

Materials and Methods

Search Strategy

The review was registered in PROSPERO (CRD42023485705) and conducted according to the PRISMA (Preferred Reporting Items for Systematic reviews and Meta-Analyses) guidelines. A systematic search was performed in the PubMed, Embase, Cochrane Library, and Google Scholar databases for articles published through November 21, 2024, using the following terms: (“Anterior Cruciate Ligament” [MeSH] OR “Anterior cruciate ligament” OR

“ACL”) AND (“Tenodesis” OR “Lateral plasty” OR “Extra articular” OR “Anterolateral ligament” OR “ALL”).

Eligibility

The study inclusion criteria were (1) skeletally mature individuals with an ACL rupture, (2) primary single-bundle ACLR plus LET or ALLR, (3) a Level-I randomized controlled trial (RCT) in English, (4) ≥12-month follow-up, (5) specified graft types, and (6) reported outcomes, such as graft failure rate, revision rate, and/or patient-reported outcome measures (PROMs). The exclusion criteria were studies (1) with incomplete data, (2) of revision ACLR, and (3) with Level-II to V evidence. Two independent reviewers screened the results; disagreements were resolved through discussion.

Data Extraction

Two reviewers collected the data, including the study details (author, year, country, design, sample size, follow-up), patient characteristics (sex, age, body mass index), and outcomes (failure rate, residual pivot shift, anteroposterior laxity, International

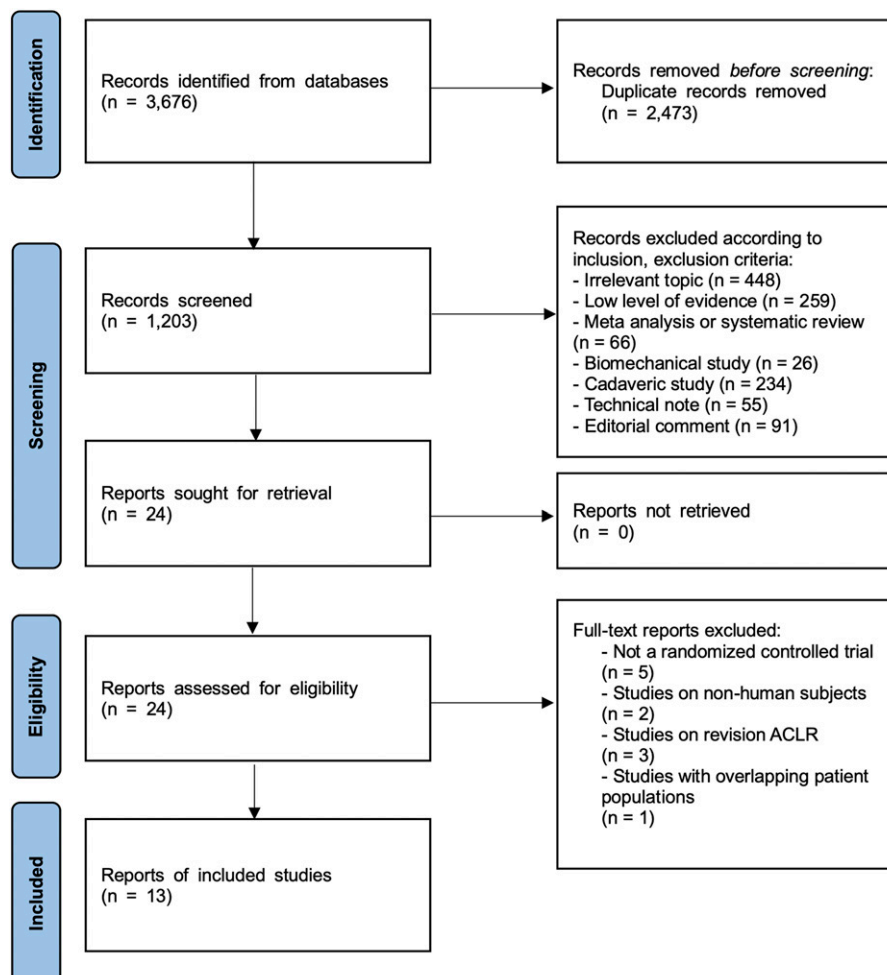


Fig. 1

PRISMA flow diagram for the systematic review. ACLR = anterior cruciate ligament reconstruction.

Knee Documentation Committee [IKDC] score, Lysholm score, Tegner score, complications). Missing or unclear data were clarified by contacting the authors. The risk of bias was assessed using the Cochrane Collaboration's risk-of-bias tool (RoB 2), version 2²². Disagreements were adjudicated by a senior reviewer.

Statistical Analysis

All statistical analyses and visualizations were performed using R software (version 4.2.1; R Foundation for Statistical Computing), with significance set at $p < 0.05$. Means and standard deviations of numerical variables were recorded, with missing standard deviations estimated using the method proposed by Furukawa et al.²³. For graphical data without numerical values, extraction was performed using the approach described by Gheibi et al.²⁴. To avoid patient overlap among studies by the same group, only the largest study from the group was included in the meta-analysis; however, the others were retained in the review.

Continuous outcomes were analyzed using mean differences (MDs) with 95% confidence intervals (CIs), and dichotomous outcomes were analyzed using odds ratios (ORs) with 95% CIs. Heterogeneity was assessed using the I^2 statistic, with substantial heterogeneity defined as $I^2 > 50\%$; a pooled synthesis was not conducted in such cases. Additionally, subgroup analyses were performed on the basis of the graft type used in isolated ACLR and the LEP type to explore potential sources of heterogeneity.

An NMA compared the treatment effects of HT and BPTB autografts for ACLR plus LEP versus isolated ACLR (control). Outcomes included ACLR failure, residual pivot shift, and the Lysholm score. ACLR failure was categorized as graft rupture, graft failure (rupture or revision), or clinical failure (rupture, revision, pivot shift grade of 2 or 3, or residual

anteroposterior laxity of >5 mm). When clinical failure components overlapped within a study, the highest reported number was used for analysis. Separate NMAs were performed for each failure definition.

NMA was conducted using the netmeta package (version 2.9-2) in R. Transitivity—a key NMA assumption requiring similarity in effect modifiers for valid comparisons²⁵—was evaluated qualitatively and via meta-regression on patient age, sex, and inclusion of high-risk individuals. A global consistency test and a node-splitting analysis were not feasible due to the absence of closed loops.

The surface under the cumulative ranking curve (SU-CRA) was calculated to rank treatments from most to least effective. SUCRA values (0 to 1) represent the cumulative probability distribution of treatment rankings, with higher values indicating greater efficacy.

Results

Study Characteristics

Thirteen studies comprising 1,690 patients were included in the final analysis (Fig. 1). Tables I and II present the detailed study characteristics, and Appendix Supplementary Table 1 summarizes concomitant injuries and surgeries.

Methodological Quality Assessment

The methodological quality of the included Level-I RCTs was assessed and presented as a risk-of-bias summary (Fig. 2-A) and a traffic light plot (Fig. 2-B). Qualitatively, the studies included in the NMA were comparable across key effect modifiers. Additionally, meta-regression analyses showed no significant association of these modifiers with any outcomes (see Appendix Supplementary Fig. 1), supporting the transitivity assumption. Subgroup analyses for each outcome showed no

TABLE I Overview of the Included Randomized Controlled Trials*

Study	Year	Country	No. of Patients	Graft Used for ACLR	Type of Lateral Procedure
Mogoş et al. ²⁶	2023	Italy	57	Autologous HT	ALLR (autologous HT)
El-Azab et al. ³⁴	2023	Egypt	100	Autologous HT	LET
Grassi et al. ²⁷	2022	Italy	59	Autologous HT	LET
Hamido et al. ³⁶	2021	Kuwait	107	Autologous HT	ALLR (autologous HT)
Castoldi et al. ³¹	2020	France	120	Autologous BPTB	LET
Getgood et al. ³⁰	2020	Canada	618	Autologous HT	LET
Porter and Shadbolt ²⁹	2020	Australia	55	Autologous HT	LET
Sonnery-Cottet et al. ²⁸	2020	France	224	Autologous HT	ALLR (autologous HT)
Ibrahim et al. ³⁸	2017	Kuwait	110	Autologous HT	ALLR (autologous HT)
Trichine et al. ³⁵	2014	Algeria	120	Autologous BPTB	LET
Vadalà et al. ³²	2013	Italy	60	Autologous HT	LET
Zaffagnini et al. ³⁷	2006	Italy	75	Autologous HT	LET
Anderson et al. ³³	2001	United States	105	Autologous HT	LET

*ACLR = anterior cruciate ligament reconstruction, HT = hamstring tendon, ALLR = anterolateral ligament reconstruction, LET = lateral extra-articular tenodesis, BPTB = bone-patellar tendon-bone.

TABLE II Patient Demographics and Inclusion and Exclusion Criteria Used in the Included Studies*

Graft Used for ACLR	Study	Follow-up Duration†	Sex, M:F‡	Age†† (yr)	BMI†† (kg/m ²)	Inclusion Criteria	Exclusion Criteria
Autologous HT	Mogoş et al., 2023 ²⁶	12 mo	17:8/27:5	31 ± 7.1	NR	(1) Unilateral primary ACL tear (2) Participation in high-grade pivoting sports (3) Preoperative pivot-shift test result showing grade-2 or 3 instability	(1) Recurrent ACL tear (2) Previous knee surgery (3) Presence of multiligament knee injuries (4) Skeletal immaturity (5) Requirement for additional extra-articular procedures (except ALL reconstruction) or PCL reconstruction (6) Diagnosis of rheumatoid arthritis
	El-Azab et al., 2023 ³⁴	24 mo	41:7/39:8	27 ± 5.85/ 28 ± 6.0	NR	(1) ACL-deficient knee confirmed by physical examination and MRI (2) Patient age of 18-45 years (3) Skeletal maturity (4) Pivot-shift test result showing grade-2 or 3 instability	(1) History of revision surgery or ACL tear in the contralateral knee (2) Presence of multiligament knee injuries or polytrauma (3) Generalized ligamentous laxity (4) Symptomatic articular cartilage defects requiring treatment (Outerbridge grade of >II). (5) Varus or valgus malalignment exceeding 3°
	Grassi et al., 2022 ²⁷	25.0 ± 5.6 mo	13:2/20:4	23.4 ± 4.7/ 24.1 ± 6.8	NR	(1) ACL deficiency (2) Patient age of 18-65 years	(1) Previous knee surgery (2) Presence of concomitant ligamentous injuries
	Hamido et al., 2021 ³⁶	Median, 60 mo (range, 55-65 mo)	52:0/50:0	26 (18-40)/ 24 (18-33)	NR	(1) Combined ACL and ALL tears and grade-3 pivot shift in a male athlete (2) Presence of a Segond fracture (3) Participation in high-level sports or sports involving frequent pivoting movements	(1) Presentation for revision ACLR surgery (2) Previous knee surgery or knee dislocation (3) Presence of multiligament knee injuries (4) Presence of osteoarthritis
	Getgood et al., 2020 ³⁵	2 yr	151:161/ 151:155	18.8 ± 3.2/ 19.1 ± 3.3	23.8 ± 3.7/ 24 ± 3.8	(1) ACL deficiency (2) Patient age of 14-25 years (3) Considered at higher risk of reinjury based on the presence of at least 2 of the following factors: (a) participation in competitive pivoting sports, (b) preoperative pivot-shift test result showing grade-2 or 3 instability, and/or (c) generalized ligamentous laxity (Beighton score of ≥4) or genu recurvatum of >10°	(1) Previous ACLR of either knee (2) Presence of multiligament knee injuries (3) Skeletal immaturity (4) Symptomatic articular cartilage defects requiring treatment (5) Varus malalignment exceeding 3° (6) Patient inability or unwillingness to comply with a 2-year postoperative follow-up
	Porter and Shadbolt, 2020 ³⁸	2 yr	15:12/17: 11	22.3 ± 3.7/ 21.8 ± 4.1	21.1 ± 1.8/ 21 ± 1.5	(1) ACL rupture confirmed by MRI and arthroscopic surgery (2) Skeletal maturity (3) Non-contact ACL injury sustained during twisting or pivoting sports (4) ACLR performed within 6 weeks after injury (5) Pivot-shift test result at least 1 instability grade higher compared with the contralateral knee after ACLR	(1) Previous ACL injury in either knee (2) Presence of other ligament injuries with grade of >1 or reparable meniscal tears (3) Diagnosis of rheumatoid arthritis, connective tissue disorder, or autoimmune disease (4) Unwillingness to be randomized into a treatment group (5) ACL injury sustained >6 weeks prior to surgery (6) Patient not fit for general anesthesia
	Sonnery-Cottet et al., 2020 ³⁹	12.3 ± 1.9 mo	96:16/78: 34	25.9 ± 4.7/ 24.7 ± 4.4	NR	(1) ACL deficiency requiring reconstruction (2) Patient age of 18-35 years	(1) Previous ligament surgery in either knee (2) Presence of multiligament knee injuries (3) BMI of <18.5 or >30 kg/m ² (4) Professional athlete (5) Articular cartilage defects requiring treatment (6) Connective tissue disorder or congenital disease (7) Patient refusal to participate in the study
	Ibrahim et al., 2017 ⁴⁰	27 mo	50:0/53:0	26 (20-30)/ 26 (21-32)	NR	(1) Unilateral primary ACL tear in a male athlete	(1) Presentation for revision ACLR (2) Multiligament knee injuries
	Vadalà et al., 2013 ³⁷	44.6 mo	0:28/0:27	27 (15-40)	NR	(1) Surgery for chronic ACL instability in a female patient (2) Patient age of <40 years (3) Preoperative pivot-shift test result showing grade-2 or 3 instability (4) Minimum of 2 months between trauma and surgery	(1) Previous ligament surgery in either knee (2) Presence of osteoarthritis (3) Imaging evidence of grade-III or IV chondral damage on both patellar surfaces or medial and lateral femoral condyles, according to the International Cartilage Repair Society (4) Presence of concomitant systemic disease
	Zaffagnini et al., 2006 ³⁷	5 yr	15:10/18: 7	31.3 (26-49)/26.7 (15-44)	NR	(1) ACL rupture (2) Patient age of <50 years (3) Intact PCL and maximum grade-1 MCL injury (4) Patient involved in stopping sports at the competitive or amateur level	(1) Previous knee surgery (2) Presence of meniscal injury (3) Abnormal contralateral knee (4) Presence of articular cartilage injury (5) Presence of osteoarthritis (6) History of patellofemoral pain
	Anderson et al., 2001 ³⁸	35.4 ± 11.6 mo	46:24/22: 13	21.9 (14-44)/22 (14-40)	NR	(1) Unilateral ACL tear (2) Patient age of <50 years (3) Patient willingness to be subjected to randomization (4) Normal contralateral knee (5) Absence of substantial injury to other ligamentous structures (less than grade-1 injury to the MCL)	(1) Previous knee surgery other than meniscectomy (2) Presence of osteoarthritis in either knee (3) History of patellofemoral pain or considerable patellofemoral crepitation

continued

TABLE II (continued)

Graft Used for ACLR	Study	Follow-up Duration†	Sex, M:F‡	Age†§ (yr)	BMI†§ (kg/m ²)	Inclusion Criteria	Exclusion Criteria
Autologous BPTB	Castoldi et al., 2020 ³¹	19.4 yr	43:18/47:13	25.9 (16-40)/26.2 (15-40)	NR	(1) Complete isolated primary ACL rupture with a plan for arthroscopic ACLR	(1) History of ACL repair or reconstruction (2) Presence of multiligament knee injuries
	Trichine et al., 2014 ³²	24.5 mo	60:0/60:0	28.6 ± 4.7/27.7 ± 4.75	NR	(1) ACL rupture with advanced chronic anterior laxity (defined as a positive Lachman test result [graded 3+], grade 3+ pivot shift, and a positive anterior drawer test result at 90° of flexion [graded 2+ or 3+]) (2) Patient willingness to be subjected to randomization	(1) Previous knee surgery (2) Presence of multiligament knee injuries (3) Injury to the contralateral knee (4) Skeletal immaturity

*ACL = anterior cruciate ligament, ACLR = ACL reconstruction, BMI = body mass index, HT = hamstring tendon, NR = not reported, ALL = anterolateral ligament, MRI = magnetic resonance imaging, MCL = medial collateral ligament, PCL = posterior cruciate ligament, BPTB = bone-patellar tendon-bone. †The values are given as the mean, the mean ± standard deviation, or the median (range).
‡For isolated ACLR/for ACLR + lateral extra-articular procedure, or overall. Numbers of patients may be less than in Table I due to loss to follow-up.

significant differences according to graft type in isolated ACLR or according to LEP type (see Appendix Supplementary Figs. 2 and 3).

Graft Rupture

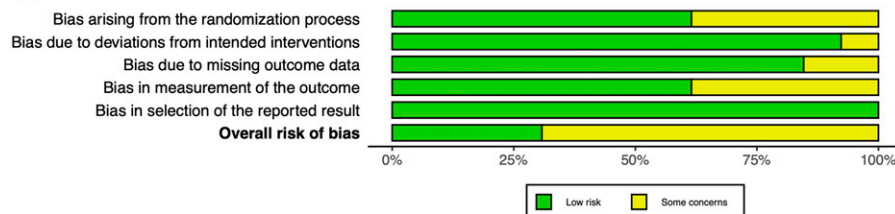
The NMA on graft rupture included 9 studies (Fig. 3-A)²⁶⁻³⁴. Low heterogeneity was observed ($I^2 = 0\%$, $\tau^2 = 0.0$, $p = 0.98$). Adding LEP to ACLR with HT autograft significantly reduced the odds of graft rupture (OR: 0.28; 95% CI: 0.16 to 0.50) compared with isolated ACLR, while the effect of adding LEP to ACLR with BPTB

failed to reach significance (OR: 0.26; 95% CI: 0.03 to 2.41). No difference was observed between graft types in ACLR plus LEP. The SUCRA rankings placed ACLR plus LEP higher (HT: 0.73; BPTB: 0.72) than isolated ACLR (0.05).

Graft Failure

Eleven studies reported graft failure, including rupture or revision surgery (Fig. 3-B)²⁶⁻³⁶. Low heterogeneity was observed ($I^2 = 0\%$, $\tau^2 = 0.0$, $p = 0.98$). Adding LEP to ACLR with HT autograft significantly reduced the odds of graft failure (OR: 0.27; 95% CI:

(a) Summary of RoB 2.0 assessments



(b) Traffic light plot of RoB 2.0 assessments

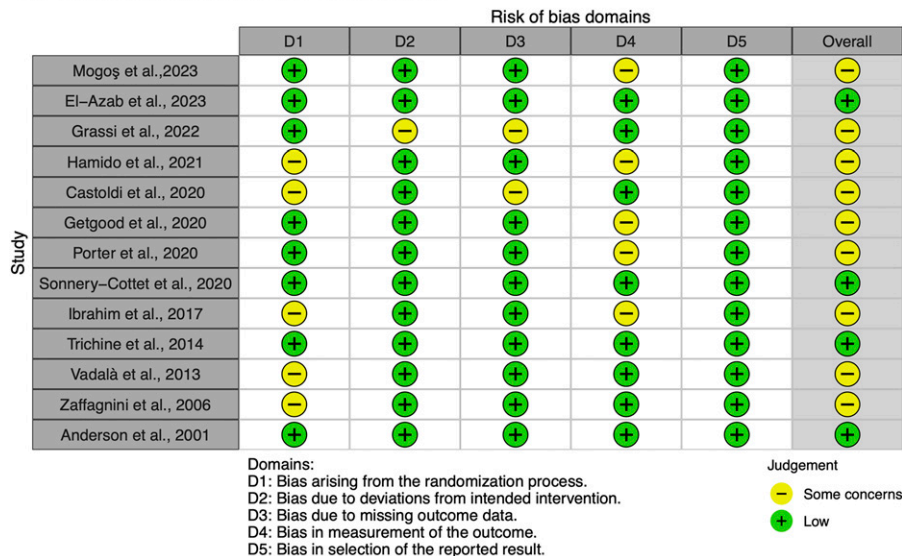


Fig. 2
Risk-of-bias (RoB) assessments for the included studies. **Fig. 2-A** Summary of the RoB 2.0 results. **Fig. 2-B** Traffic light plot illustrating the RoB 2.0 results.

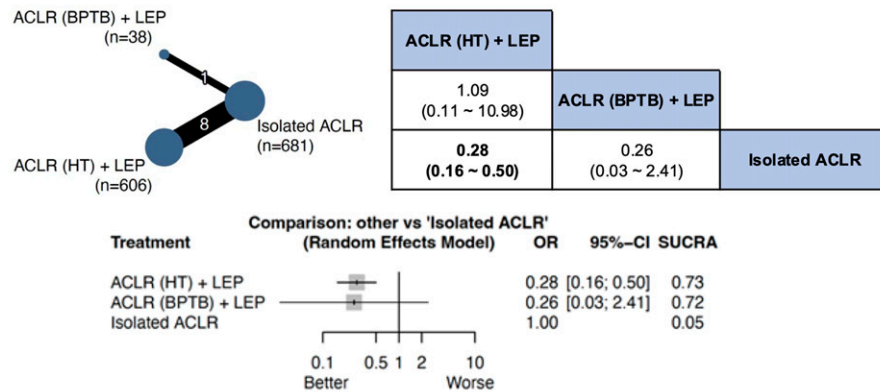
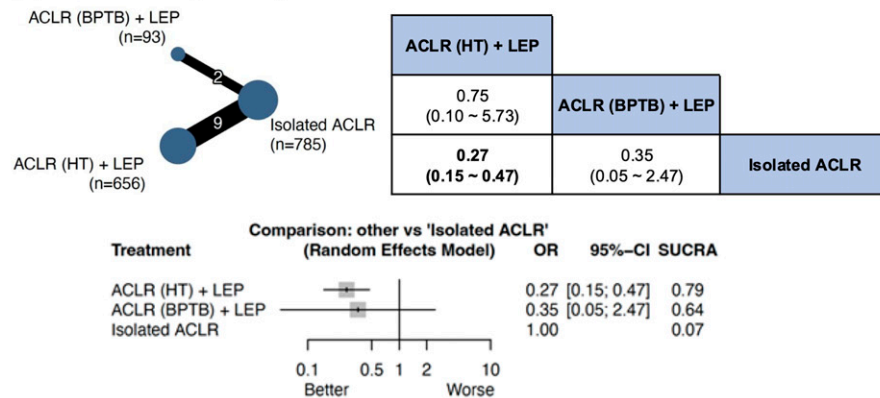
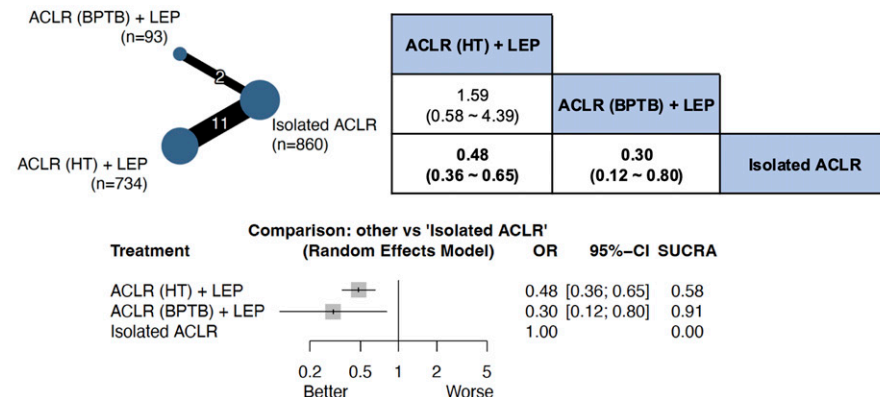
(a) Graft rupture ($I^2 = 0\%$)**(b) Graft failure ($I^2 = 0\%$)****(c) Clinical failure ($I^2 = 0\%$)**

Fig. 3

Edge plots, league tables, and forest plots for graft rupture (**Fig. 3-A**), graft failure (**Fig. 3-B**), and clinical failure (**Fig. 3-C**). The I^2 statistic represents heterogeneity. In the edge plot, the width of each line is proportional to the number of trials comparing the pair of treatments, the number on the line represents the number of these trials, and the size of each circle is proportional to the number of cases. The league plot shows the standardized mean difference and 95% confidence interval for each treatment pair; significant findings are in bold. ACLR = anterior cruciate ligament reconstruction, HT = hamstring tendon, BPTB = bone-patellar tendon-bone, LEP = lateral extra-articular procedure, OR = odds ratio, CI = confidence interval, SUCRA = surface under the cumulative ranking curve.

0.15 to 0.47) compared with isolated ACLR, while the effect of adding LEP to ACLR with BPTB failed to reach significance (OR: 0.35; 95% CI: 0.05 to 2.47). No difference was observed between

graft types in ACLR plus LEP. The SUCRA rankings placed ACLR with HT autograft plus LEP first (0.79), followed by BPTB autograft plus LEP (0.64) and isolated ACLR (0.07).

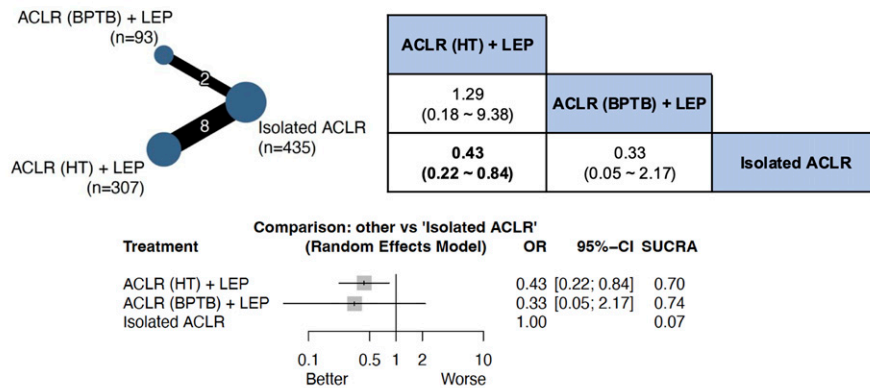
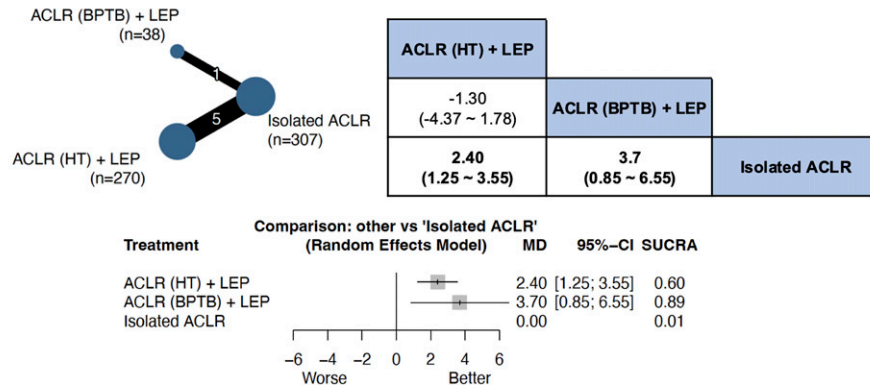
(a) Residual pivot shift ($I^2 = 0\%$)**(b) Lysholm score ($I^2 = 0\%$)**

Fig. 4
Edge plots, league tables, and forest plots for residual pivot shift (Fig. 4-A) and Lysholm score (Fig. 4-B). The I^2 statistic represents heterogeneity. In the edge plot, the width of each line is proportional to the number of trials comparing the pair of treatments, the number on the line represents the number of these trials, and the size of each circle is proportional to the number of cases. The league plot shows the standardized mean difference and 95% confidence interval for each treatment pair; significant findings are in bold. ACLR = anterior cruciate ligament reconstruction, HT = hamstring tendon, BPTB = bone-patellar tendon-bone, LEP = lateral extra-articular procedure, OR = odds ratio, MD = mean difference, CI = confidence interval, SUCRA = surface under the cumulative ranking.

Clinical Failure

The NMA on clinical failure—defined as graft rupture, revision surgery, grade-2 or 3 pivot shift, or residual anteroposterior laxity of >5 mm—included 13 studies (Fig. 3-C)²⁶⁻³⁸. Clinical failure was defined by pivot shift in 7 of these studies^{29,30,32-36}, by residual anteroposterior laxity in 3^{31,37,38}, and by graft failure in 3²⁶⁻²⁸. Low heterogeneity was observed ($I^2 = 0\%$, $\tau^2 = 0.0$, $p = 0.79$). Adding LEP to ACLR with BPTB (OR: 0.30; 95% CI: 0.12 to 0.80) or with HT (OR: 0.48; 95% CI: 0.36 to 0.65) significantly reduced the odds of clinical failure compared with isolated ACLR. No differences were found between graft types in ACLR plus LEP. The SUCRA rankings placed BPTB autograft plus LEP first (0.91), followed by HT autograft plus LEP (0.58) and isolated ACLR (0.00).

Residual Pivot Shift

Ten studies reported residual pivot shift (grade 2 or 3) (Fig. 4-A)^{26,29,31-38}. Low heterogeneity was observed ($I^2 = 0\%$, $\tau^2 = 0.0$, $p = 0.58$). ACLR with HT autograft plus LEP significantly reduced the odds of a residual pivot shift (OR: 0.43; 95% CI: 0.22 to 0.84) compared with isolated ACLR, while the effect of adding LEP to

BPTB failed to reach significance (OR: 0.33; 95% CI: 0.05 to 2.17). No differences were found between graft types in ACLR plus LEP. The SUCRA rankings placed BPTB autograft plus LEP first (0.74), followed by HT autograft plus LEP (0.70) and isolated ACLR (0.07).

Lysholm Score

Six studies reported the Lysholm score (Fig. 4-B)^{28,29,31,32,36,38}. Low heterogeneity was observed ($I^2 = 0\%$, $\tau^2 = 0.0$, $p = 0.40$). Adding LEP was associated with significantly higher Lysholm scores compared with isolated ACLR for both BPTB (MD: 3.70; 95% CI: 0.85 to 6.55) and HT (MD: 2.40; 95% CI: 1.25 to 3.55). No differences were found between graft types in ACLR plus LEP. The SUCRA rankings placed BPTB autograft plus LEP first (0.89), followed by HT autograft plus LEP (0.60) and isolated ACLR (0.01).

Other Clinical Outcomes

Residual anteroposterior laxity and clinical scores (IKDC and Tegner activity) were extracted (Tables III and IV), but significant heterogeneity precluded pooled analysis.

TABLE III ACLR Failure, Residual Pivot Shift, and Residual Anteroposterior Laxity Reported in the Included Studies*

Graft Used for ACLR	Study	Graft Rupture (no. [%])		Graft Failure (no. [%])		Clinical Failure (no. [%])		Residual Pivot Shift (no. [%])		Residual AP Laxity (mm)		
		Isolated ACLR	ACLR + LEP	Isolated ACLR	ACLR + LEP	Isolated ACLR	ACLR + LEP	Isolated ACLR	ACLR + LEP	Isolated ACLR†	ACLR + LEP†	Method†
Autologous HT	Mogoş et al., 2023 ³⁶	1 (4.0%)	0 (0%)	1 (4.0%)	0 (0%)	1 (4.0%)	0 (0%)	0 (0%)	0 (0%)	0.24 ± 0.43	0.13 ± 0.34	Rolimeter
	El-Azab et al., 2023 ³⁴	3 (6.3%)	1 (2.1%)	3 (6.3%)	1 (2.1%)	4 (8.4%)	1 (2.1%)	4 (8.3%)	1 (2.1%)	NR	NR	
	Grassi et al., 2022 ²⁷	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	NR	NR	1.5 ± 1.6	1.7 ± 2.2	KT-1000
	Hamido et al., 2021 ³⁶	NR	NR	0 (0%)	0 (0%)	5 (9.6%)	0 (0%)	5 (9.6%)	0 (0%)	2.5 ± 0.7	1.2 ± 0.7	KT-1000
	Getgood et al., 2020 ³⁰	34 (11.4%)	11 (3.8%)	34 (11.4%)	11 (3.8%)	120 (40.3%)	72 (24.7%)	NR	NR	NR	NR	
	Porter and Shadbolt, 2020 ²⁹	4 (17.4%)	0 (0%)	4 (17.4%)	0 (0%)	6 (26.1%)	4 (14.3%)	6 (26.1%)	4 (14.3%)	NR	NR	
	Sonnery-Cottet et al., 2020 ²⁸	6 (5.4%)	1 (0.9%)	6 (5.4%)	1 (0.9%)	6 (5.4%)	1 (0.9%)	NR	NR	1 ± 1	1 ± 1	Rolimeter
	Ibrahim et al., 2017 ³⁸	NR	NR	NR	NR	3 (6%)	1 (1.9%)	2 (4%)	1 (1.9%)	1.8 ± 0.8	1.3 ± 0.2	KT-1000
	Vadalà et al., 2013 ³²	2 (7.1%)	0 (0%)	2 (7.1%)	0 (0%)	4 (14.3%)	0 (0%)	4 (14.3%)	0 (0%)	2.8 ± 0.8	2.7 ± 0.9	KT-1000
	Zaffagnini et al., 2006 ³⁷	NR	NR	NR	NR	5 (20.0%)	3 (12.0%)	4 (16.0%)	0 (0%)	NR	NR	
Autologous BPTB	Anderson et al., 2001 ³³	2 (2.9%)	0 (0%)	2 (2.9%)	0 (0%)	8 (11.4%)	7 (20.0%)	8 (11.4%)	7 (20.0%)	3.1 ± 2.3	2.6 ± 2.2	KT-1000
	Castoldi et al., 2020 ³¹	4 (9.5%)	1 (2.6%)	4 (9.5%)	1 (2.6%)	16 (38.1%)	6 (15.8%)	1 (2.4%)	0 (0%)	NR	NR	
	Trichine et al., 2014 ³⁵	NR	NR	0 (0%)	0 (0%)	3 (5.8%)	1 (1.8%)	3 (5.8%)	1 (1.8%)	4.4 ± 4.8	4.1 ± 4.7	Telos

*AP = anteroposterior, ACLR = anterior cruciate ligament reconstruction, LEP = lateral extra-articular procedure, HT = hamstring tendon, NR = not reported, BPTB = bone-patellar tendon-bone. †The values are given as the mean ± standard deviation. ‡Rolimeter (Aircast), KT-100 (MEDmetric), or Telos (Telos).

Complications

Various complications were reported in studies comparing isolated ACLR to ACLR with HT autograft plus LEP. Hamido et al.³⁶ noted 2 cases of arthrofibrosis in their ALLR group, which resolved within 4 weeks with nonoperative management. Getgood et al.³⁰ reported more hardware irritation cases in their LET group (14) than in their isolated ACLR group (4). Sonnery-Cottet et al.²⁸ found a higher rate of reoperations for cyclops syndrome in their isolated ACLR group (10 of 112 versus 0 of 112, $p = 0.001$). Anderson et al.³³ reported additional procedures, such as meniscectomy or staple removal, after 6 procedures with LET and 5 with isolated ACLR. Porter and Shadbolt²⁹ noted similar meniscal tear rates in groups with and without LET, and no other complications. Ibrahim et al.³⁸ reported 1 wound infection in their ACLR plus ALLR group, whereas Mogoş et al.²⁶ found no complications in the ACLR plus ALLR group. Four studies^{27,32,34,37} reported no clinically important complications.

In a study comparing isolated ACLR to ACLR with BPTB autograft plus LEP, Trichine et al.³⁵ reported no complications or additional surgeries. However, Castoldi et al.³¹ observed higher lateral tibiofemoral osteoarthritis in their

ACLR with BPTB autograft plus LET group (59% versus 22%, $p = 0.02$) but no significant differences in medial or patellofemoral osteoarthritis.

Discussion

A key finding from our study is that adding LEP to ACLR with either HT or BPTB autograft was associated with significantly lower odds of ACLR failure compared with isolated ACLR. HT autografts plus LEP had 72%, 73%, and 52% lower odds of graft rupture, graft failure, and clinical failure, respectively. HT autografts also demonstrated superior rotational stability when used with LEP, as indicated by the pivot-shift test. The addition of LEP to ACLR with BPTB autograft yielded 70% lower odds of clinical failure; however, the impacts on graft rupture, graft failure, and the incidence of a pivot shift did not reach significance. These results highlight the need for additional studies to clarify the role of LEP when used with BPTB autografts. Notably, no significant differences were observed between HT and BPTB autografts when combined with LEP.

Graft rupture or revision surgery following isolated ACLR has been extensively studied, with the focus on choosing between

TABLE IV Clinical Outcome Scores Reported in the Included Studies*

Graft Used for ACLR	Study	Lysholm Score		IKDC Score		Tegner Score	
		Isolated ACLR	ACLR + LEP	Isolated ACLR	ACLR + LEP	Isolated ACLR	ACLR + LEP
Autologous HT	Mogoş et al., 2023 ²⁶	NR	NR	87 ± 9	91.5 ± 11	NR	NR
	El-Azab et al., 2023 ³⁴	NR	NR	87.9 ± 7.19	91.9 ± 8.9	NR	NR
	Grassi et al., 2022 ²⁷	NR	NR	NR	NR	NR	NR
	Hamido et al., 2021 ³⁶	94 ± 4.5	96 ± 5	NR	NR	7.8 ± 1.4	7.9 ± 0.8
	Getgood et al., 2020 ³⁰	NR	NR	86.6 ± 0.8	87.3 ± 0.8	NR	NR
	Porter and Shadbolt, 2020 ²⁹	92.5 ± 4.8	96.8 ± 8	90.9 ± 10.7	94.2 ± 11.2	7 ± 2.3	8 ± 1.9
	Sonnery-Cottet et al., 2020 ²⁸	88 ± 14	92 ± 10	81.2 ± 15.3	86.8 ± 12.3	6 ± 2	6 ± 2
	Ibrahim et al., 2017 ³⁸	96 ± 5.3	98 ± 7.6	NR	NR	8 ± 1.5	8 ± 1.5
	Vadalà et al., 2013 ³²	94.5 ± 6.7	95.8 ± 4	87 ± 1.8	89 ± 1.5	6.7 ± 1.4	7.5 ± 1.2
	Zaffagnini et al., 2006 ³⁷	NR	NR	76 ± 8	89 ± 13	7.1 ± 1.3	8.5 ± 1.9
	Anderson et al., 2001 ³³	NR	NR	NR	NR	NR	NR
Autologous BPTB	Castoldi et al., 2020 ³¹	86.6 ± 7.2	90.3 ± 5.8	81.1 ± 7.9	82.4 ± 6.7	NR	NR
	Trichine et al., 2014 ³⁵	NR	NR	92.1 ± 9.5	88.2 ± 8	NR	NR

*The values are given as the mean ± standard deviation. IKDC = International Knee Documentation Committee, ACLR = anterior cruciate ligament reconstruction, LEP = lateral extra-articular procedure, HT = hamstring tendon, NR = not reported, BPTB = bone-patellar tendon.

HT and BPTB grafts. Registry studies have reported a 1.6 to 6.1 times higher risk of graft rupture or revision with HT autografts compared with BPTB autografts^{9,39-42}, and a recent meta-analysis also found a 1.6 times higher risk⁴³. However, some other meta-analyses found no difference in the failure rates between the 2 grafts⁴⁴⁻⁴⁶. These discrepancies may result from study design heterogeneity and statistical fragility in ACLR research^{43,47}. Nevertheless, the significantly higher failure rates associated with HT autografts, or trends toward them, suggest that HT is not superior to BPTB in isolated ACLR.

Adding LEP to isolated ACLR seemingly mitigated or eliminated outcome differences between HT and BPTB autografts. The fact that, of the 3 types of failure assessed in this study, only clinical failure was found to have a significantly lower rate when LEP was added to the BPTB autografts may be attributable to their superior control of residual anteroposterior laxity, which is 1 of the criteria used to define clinical failure. BPTB autografts are reported to provide superior anteroposterior stability compared with HT autografts^{48,49}, and this inherent graft characteristic likely influenced our findings. Although the ACLR failure rate did not differ between the grafts when combined with LEP, these findings underscore the potential of LEP to enhance outcomes regardless of the graft type. However, these findings should be interpreted with caution, as factors such as age, sex, and sports participation may influence the outcomes. Maletis et al.⁹ reported that patients aged <21 years who underwent isolated ACLR with HT autografts had a 1.6 times higher revision risk than those with BPTB autografts. Salem et al.⁵⁰ also found higher graft rupture rates after isolated ACLR with HT autografts than with BPTB autografts in females aged 15 to 20 years (17.5% versus 6.4%). However, there is limited evidence regarding the comparative

outcomes of HT and BPTB autografts in combined ACLR and LEP, and we found none with risk stratification by age and sex. One study focused exclusively on high-risk patients aged <25 years³⁰, 1 included only females³², and 3 included only males^{35,36,38}. Thus, our findings reflect outcomes in a general ACL injury population rather than in specific age or risk groups. Further research examining the impacts of age and sex on graft outcomes in combined ACLR and LEP is needed.

Improved rotational stability, demonstrated by a decreased pivot shift phenomenon, is a key advantage of adding LEP to ACLR compared with ACLR alone. Biomechanical studies suggest that LEPs, including ALLR and LET, enhance rotational stability, leading to better clinical outcomes, such as fewer positive pivot shifts and lower graft failure rates^{51,52}. One meta-analysis showed that, overall, ALLR or LET reduced the graft failure risk by approximately 64% and improved rotational stability compared with ACLR alone⁵³. Similarly, Park et al., using NMA, reported that ACLR plus LEP outperformed isolated ACLR in terms of graft failure and positive pivot shift rates¹⁷. This is consistent with our study, which showed that combining LEP with ACLR reduced the graft failure risk and positive pivot shift rates by approximately 70% and 60%, respectively. While prior studies compared different types of LEPs, our analysis focused on graft type, finding that HT and BPTB grafts yielded similar graft failure and pivot shift outcomes when combined with LEP.

The indications for LEP remain debated and inconclusive^{21,54}. LEP is commonly used in revision ACLR, particularly in cases with a high-grade pivot shift, where it effectively reduces failure rates⁵⁵. Its use in primary ACLR is expanding, especially in high-risk patients, such as those who have a high-grade pivot shift, are <25 years old, participate in pivoting

sports, have an elevated tibial slope, or have generalized ligamentous laxity^{56,57}. Sonnery-Cottet et al.⁵⁸ recommended adding LEP to primary ACLR for patients with a chronic ACL injury, Second fracture, or lateral femoral notch sign. Most studies in our analysis (9 of 13) focused on high-risk patients, primarily those with pivot shifts or sports participation^{26,29,30,32,34-38}. However, our meta-regression analysis showed no significant impact of including high-risk patients on outcomes, thereby supporting LEP as a standard adjunct for improving outcomes of primary ACLR regardless of risk status, particularly when using HT and BPTB autografts. This suggests the potential for expanding its indications.

PROMs demonstrated an advantage of augmentation with LEP. In our study, the Lysholm score was significantly higher for HT and BPTB autografts plus LEP than for isolated ACLR (MD: 3 points). However, the observed MD was below the minimal clinically important difference (MCID) of 8.9 for the Lysholm score⁵⁹, and therefore did not reach meaningful clinical impact. Similar patterns have been observed in other meta-analyses, where combined surgery improved subjective IKDC, Lysholm, and Tegner scores but none exceeded the MCID thresholds^{17,53}.

Complications related to graft donor-site morbidity are key concerns in autograft selection for ACLR. BPTB autografts are associated with anterior knee pain, kneeling pain, patellar fracture, patellar tendon rupture, and quadriceps weakness, while HT autografts carry risks of infection and reduced hamstring strength^{46,60}. In our analysis, donor-site morbidity was rarely reported but should still be considered a concern given its importance. Hardware irritation was observed after use of HT autografts, and lateral compartment osteoarthritis was observed with BPTB autografts; however, these complications appeared more related to the LET than to the graft harvesting.

Our study had some limitations. First, as direct comparisons of BPTB and HT autografts were absent, we relied on NMA for indirect comparisons; hence, inconsistency between direct and indirect evidence could not be assessed. Although we performed a transitivity assessment and an additional meta-regression analysis to partially address this limitation, these methods do not fully substitute for formal consistency testing. Second, the limited number of RCTs, particularly those directly comparing BPTB autografts plus LEP to isolated ACLR, resulted in a smaller pool of studies. Third, data on outcomes such as anteroposterior knee stability, various PROMs, strength recovery, and return to sports are crucial in graft selection but could not be analyzed due to study heterogeneity. Fourth, in

each included study, some patients had concomitant injuries or surgeries involving meniscal or cartilage defects in addition to ACLR. These cases have been summarized in Appendix Supplementary Table 1 to aid result interpretation but may have introduced variability into the outcomes. Fifth, variability in inclusion criteria, especially in definitions of high-risk populations and in age criteria across studies, could have introduced bias. Lastly, future research on optimal graft choice should explore alternative grafts, such as the quadriceps tendon and iliotibial band, as viable options.

In conclusion, concurrent ACLR with HT autograft plus LEP was associated with 72%, 73%, and 52% lower odds of graft rupture, graft failure, and clinical failure, respectively, compared with isolated ACLR. Adding LEP to ACLR with BPTB autograft was significantly associated only with a reduction in clinical failure, by approximately 70%, potentially due to the smaller sample size. Both grafts remain viable options for ACLR plus LEP, with the benefit of LEP requiring further validation.

Appendix

 Supporting material provided by the authors is posted with the online version of this article as a data supplement at <http://links.lww.com/JBJS/1828>. ■

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BMJ Open Comparative efficacy and safety of different surgical approaches for the treatment of anterior cruciate ligament injury: a Bayesian network meta-analysis protocol

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ABSTRACT

Introduction Anterior cruciate ligament (ACL) rupture can be treated surgically or non-surgically, with several surgical interventions available at present. However, the comparatively effective surgical intervention with relatively fewer side effects remains unknown. This study aims to fill in this gap by conducting a Bayesian network meta-analysis (NMA) and provide a theoretical basis for the clinical application.

Method and analysis We will perform a Bayesian NMA and will include randomised controlled trials (RCTs) published in English or Chinese that compare surgical intervention (ie, standard ACL reconstruction, ACL remnant-preserving reconstruction and ACL repair with suture augmentation to conservative therapy or studies that compare one surgical intervention to another for the symptom relief and function recovery of patients with ACL rupture. Primary outcome will be the proportion of patients with symptomatic and functional improvement measured by the Knee Injury and Osteoarthritis Outcome Score before and 6 months after treatment, with scores ranging from 0 (worst) to 100 (best). Secondary outcomes will be knee-specific quality of life (ACL QoL), return to activity and level of sport participation (Tegner or modified Tegner score), health-related QoL (EuroQol Group 5-Dimension 5-Level, EQ-5D-5L), resource use, intervention-related complications and patient satisfaction. We have developed search strategies for PubMed, Embase, the Cochrane Library and Web of Science, retrieving RCTs that meet the inclusion criteria from database inception to 1 December 2023. The methodological quality of the included RCTs will be assessed based on the Cochrane risk of bias table. The relative ranking probability of the best intervention will be estimated using the surface under the cumulative ranking curve. The Bayesian NMA will be conducted by using WinBUGS V.1.4.3. The Grading of Recommendations Assessment, Development and Evaluation approach will be applied to determine our confidence in an overall treatment ranking from the NMA.

Ethics and dissemination Ethical approval for this study is not required because no private or confidential patient data will be used in this study. Findings of this study would be disseminated through the publication in a peer-reviewed medical journal.

PROSPERO registration number CRD42023437115.

STRENGTHS AND LIMITATIONS OF THIS STUDY

- ⇒ Rigorous methods will be followed, and the quality of all included articles will be assessed using validated tools.
- ⇒ Literature published in both English and Chinese will be included which broaden the searching scope and knowledge span.
- ⇒ Only randomised controlled trials will be included for analysis, which could limit the evidence by excluding observational studies.
- ⇒ Due to the long-time span of the included literature, multiple factors may contribute to the heterogeneity of target indicators and affect the results.

INTRODUCTION

Anterior cruciate ligament (ACL) rupture predominantly affects young, athletically active individuals, with over 200 000 cases annually in the USA, incurring costs exceeding US\$7 billion.¹ Risk factors of ACL rupture include female sex (with risk two times higher than male),² young age (with a peak at 16 to 18 years), and earlier, more intense, and more frequent participation in sports.³ ACL injuries are often complicated by concomitant injury of the medial collateral ligament (19%–38%) and lateral (20%–45%) or medial (0%–28%) meniscal tears,⁴ leading to symptoms like pain, activity limitation and knee joint swelling.⁵ Diagnosis is effectively done through MRI, which boasts high accuracy.⁶ The injury significantly impacts knee stability and function, decreasing activity levels and life quality, and increasing the risk of long-term osteoarthritis if not properly treated.^{5–7} Despite its prevalence, treatment varies greatly due to the scarcity of high-quality evidence guiding management strategies.

While surgical intervention is frequently favoured for treating ACL ruptures, owing to

its generally better functional outcomes, the choice of the specific surgical technique remains a subject of debate due to the distinct advantages and drawbacks associated with each method.⁸ Common surgical techniques include standard ACL reconstruction, ACL remnant-preserving reconstruction and ACL repair with suture augmentation are common choices.⁹ Each surgical technique for ACL rupture has its merits in restoring joint stability and function. For example, standard ACL reconstruction proven track record of restoring knee stability and function. It has high success rates in terms of patient satisfaction and return to sports. However, it also faces the challenges of graft failure and longer rehabilitation period. But it is a technically more challenging procedure with the risk of preserving damaged tissue that could impair healing.¹⁰ ACL remnant-preserving reconstruction is believed to enhance proprioceptive recovery, improve vascularisation and healing of the graft and potentially less postoperative pain and faster recovery.¹¹ ACL repair with suture augmentation is minimally invasive compared with full reconstruction. Preservation of original ACL tissue can maintain native anatomy and potentially promote quicker recovery and cause less postoperative pain. But it is suitable mainly for proximal ACL tears with good tissue quality and presents with higher rates of failure reported in some studies.^{12 13} When all three surgical methods can be used to treat the same patient, it often leads to difficulty in decision-making for the doctor. Currently, there is a lack of comparative studies to establish which method is superior in terms of effectiveness and safety, an essential consideration for refining ACL treatment strategies.

To fill the gap in this field and provide optimised interventions to patients with ACL rupture, we intend to conduct a Bayesian network meta-analysis (NMA) to evaluate the comparative efficacy of different surgical approaches for the treatment of ACL rupture and provide important guidance to clinical practice.

METHODS AND ANALYSIS

Literature search

This protocol was registered on the PROSPERO International Prospective Register of Systematic Reviews (registration number: CRD42023437115) and developed in accordance with the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols guidelines.¹⁴ Any modifications to the protocol will be updated and subsequently published alongside the results of the NMA. PubMed database, Web of Science database, the Cochrane Central Register of Controlled Trials (CENTRAL) and EMBASE database will be comprehensively searched from the inception of each database up to 1 December 2023 for relevant randomised controlled trials (RCTs). Additionally, we will manually search the references of retrieved papers and reviews to include further relevant material. Grey literature sources such as GreyNet, SIGLE, BLDSC and registry sites like the ICRT Registry Network will also be explored for pertinent

RCTs. Studies selected for further investigation include those comparing surgical interventions for the treatment of ACL rupture, including standard ACL reconstruction, ACL remnant-preserving reconstruction and ACL repair with suture augmentation. We will include RCTs to assess the efficacy and safety of these surgical approaches. The retrieval of literature and data analysis have been completed. We started this study on 20 May 2023 and anticipate completing the submission process by 31 March 2024. During this time, we will continue to screen and incorporate newly published literature into our analysis. Detailed search strategies are listed in online supplemental appendix 1.

Inclusion and exclusion criteria

RCTs that meet the following criteria will be included in this study: (1) the surgical methods for treating ACL rupture are standard ACL reconstruction, ACL remnant-preserving reconstruction or ACL repair with suture augmentation. (2) Compare surgical interventions to conservative therapies or studies that compare one surgical intervention versus another. (3) Studies with follow-up more than 6 months. (4) Studies published in English or Chinese. Exclusion criteria include (1) Non-RCTs, including case reports, reviews, animal trials, letters to the editor and retrospective studies. (2) RCTs from which data cannot be extracted. (3) RCTs published neither in English nor Chinese. Two independent reviewers will perform the study selection. Qualified RCTs will be identified for inclusion by two independent reviewers based on predefined eligibility criteria. Any disagreements at the selection process will be resolved through discussion, or with a third researcher.

Data extraction

Two independent reviewers will extract pertinent data from the included studies. In cases of incomplete data or when the full text is unavailable, we will try to contact the corresponding author of the potentially relevant study. Data presented solely in visual format within figures will be extracted using GetData V.2.20. Data extracted for analysis will include the name of the first author, year of publication, specific details of the populations, type of surgical interventions, outcomes, follow-up time points and side effects of each study. Any disagreements will be solved through discussion or with a third reviewer.

The risk of bias of included RCTs will be assessed according to Cochrane Handbook for Systematic Reviews of Interventions and bias adjustment methods described by Dias *et al.*¹⁵ Bias caused by random sequence generation, allocation concealment, blinding of participants, blinding of outcome assessment, incomplete outcome data, selective reporting and others (mainly including conflict of interests) will be carefully evaluated. Each risk of bias category will be classified using a three-level rating system: low risk, unclear risk and high risk. Studies identified with three or more high-risk factors will be considered to have poor methodological quality.¹⁶ Following the

analysis, trials with three or more high risk of bias will be excluded to determine if the results would change substantially.

Primary outcome will be the proportion of patients with symptomatic and functional improvement measured by the Knee Injury and Osteoarthritis Outcome Score (KOOS) before and 6 months after treatment, with scores ranging from 0 (worst) to 100 (best). KOOS is a wide-accepted patient-reported outcome measure designed to assess the patient's opinion about their knee and associated problems. It consists of five subscales: Pain, Other Symptoms, Function in Daily Living, Function in Sport and Recreation, and Knee-Related Quality of Life (QoL). KOOS has been proven to be reliable and sensitive in assessing outcomes related to knee injuries and osteoarthritis. We will compare the KOOS scores before and 6 months after treatment to assess the improvement or decline in knee health and associated factors. Secondary outcomes will be knee-specific QoL (ACL QoL), return to activity and level of sport participation (Tegner or modified Tegner score), health-related QoL (EuroQol Group 5-Dimension 5-Level, EQ-5D-5L), resource use, intervention-related complications and patient satisfaction. ACL QoL is a questionnaire specifically designed to assess the QoL of patients with ACL injuries. The Tegner Activity Scale is a tool for grading work and sporting activities on a scale from 0 to 10, with higher scores indicating higher activity levels. EQ-5D-5L is a standardised instrument for measuring generic health status. It provides a simple descriptive profile and a single index value for health status. This provides a broader measure of the treatment's impact on overall health and well-being, beyond the specific focus on the knee. Resource use entails the quantification of all healthcare resources used during the treatment and recovery process, including hospital stays, medication and physiotherapy sessions. Understanding the resource implications is vital for assessing the cost-effectiveness and practicality of the treatment. Considering that safety and risk assessment are crucial for a comprehensive evaluation of the treatment's viability and patient advisability, we analysis the intervention-related complications which include any adverse events or complications directly related to the treatment. Patient satisfaction is a subjective measure of the patients' satisfaction with the treatment outcomes and the process. It is an important indicator of treatment success and can influence future treatment choices and adherence.

Statistical analyses

The statistical analyse approaches have been described in detail in our previous paper.¹⁷ Given that Bayesian NMA can narrow the width of the credible interval (CrI) of the estimate by incorporating more studies into each group and shows strength in comparing the efficacy and safety of different surgical approaches,^{18–20} we decided to conduct a Bayesian NMA to identify the most effective and safest intervention.

The posterior density of unknown variables will be estimated using the Markov Chain Monte Carlo method and the variance between RCTs will be accommodated by employing a random effects model.^{20–22} Two Markov chains will be designed to run simultaneously. Each of them starts with a different set of initial values that are selected arbitrarily for convergence. For each set of initial values, a total of 50 000 simulations will be conducted with the first 10 000 simulations being abandoned as the burn-in period. The WinBUGS code can be found at <http://www.bristol.ac.uk/social-community-medicine/projects/mpes/>.

The pooled effect sizes, STD mean difference (SMD) or risk ratio (RRs) will be calculated from the median of the posterior distribution. The 97.5th and 2.5th percentiles of the posterior distribution will represent the upper and lower limit of the 95% CrI, respectively. A 95% CrI including 0 for SMD or 1 for RR will be interpreted as indicating a non-statistically significant difference. In our models, each trial arm will contribute one independent data point, which will help in assessing the model's fit to the data. Posterior mean residual deviance was calculated to measure the fit of the model to the data.²³ If the mean of the residual deviance will be equal or similar to the number of data points of the model, the model fitted the data well.²³ The minimum clinically important difference (MCID) is defined as the smallest change in an outcome that constitutes a clinically meaningful treatment effect,²⁴ which is fundamental to interpreting clinical trial outcomes, making clinical decisions and designing studies with sufficient statistical power to detect any such effect.²⁵ MCID will be defined as 0.5 SD greater change of the related 95% CI.

Risk of bias will be assessed using the Collaboration's 'risk of bias' tool as described above and bias adjustment methods described by Dias *et al* will be applied to evaluate internal validity.¹⁵ We will make use of the model fit and heterogeneity parameters to choose suitable model.²⁶ We will assess the potential publication bias by considering the completeness of the literature search and use funnel plot and regression tests to specify publication bias assessments. Egger's weighted regression and Begg's rank correlation tests will be used to evaluate publication bias. To obtain a meaningful comparison-adjusted funnel plot, we will order the intervention groups by assuming that small studies are likely to favour advanced support surfaces. The certainty of the evidence from the NMA for each network contrast and the ranking of intervention groups was assessed following the GRADE (Grading of Recommendations Assessment, Development and Evaluation) approach.^{27 28} The certainty of evidence from direct comparisons in NMA will be rated as high, moderate, low or very low based on risk of bias, imprecision, inconsistency and publication bias. For indirect comparisons, the quality of evidence will be assessed using GRADE categories and transitivity considerations, with the lower of the two direct comparison ratings applied to the indirect evidence. Confidence in each NMA effect estimate

will be based on the higher rating of direct and indirect evidence but may be downgraded if inconsistencies are found between the two.

Surgical operations will be ranked based on the posterior probability of their effect sizes, with the ranking outcomes presented by the surface under the cumulative ranking curve (SUCRA). A SUCRA value of 100% signifies that the intervention is the most effective treatment, whereas a value of 0% denotes the least effective treatment.^{29 30} We will perform the NMA using WinBUGS (V.1.4.3, MRC Biostatistics Unit, Cambridge, UK). SUCRA figures will be drawn in Stata software (V.15.1, StataCorp).

Patient and public involvement

None.

DISCUSSION

Previous research has indicated that standard ACL reconstruction,³¹ ACL remnant-preserving reconstruction^{32 33} and ACL repair with suture augmentation³⁴ are effective in restoring joint stability of patients with ACL rupture. However, earlier systematic reviews and meta-analyses typically focused on pooling data from trials comparing just two surgical interventions, leaving a gap in comprehensive comparative analysis and an unclear understanding of the relative merits of each method.^{35–37} No previous study has systematically evaluated the symptom alleviation and side effects across all these surgical options. To address this gap, we are conducting this NMA, which allows for an extended comparison across multiple surgical interventions. Bayesian NMA is able to compare the efficacy among different surgical interventions and narrow the width of CrI of the estimate via increasing the number of studies of each group. Our study is pioneering in its approach as the first NMA to compare these surgical interventions and would provide a theoretical basis for clinical application.

ETHICS AND DISSEMINATION

Ethical approval for this study is not required, for there are no private or confidential patient data included in this study. The results would be published in a peer-reviewed medical journal.

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Contributors WZ conceptualised the study, acquired funding and administered the project. YF and YH wrote the original draft. YF and YH reviewed and edited the manuscript.

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Competing interests None declared.

Patient and public involvement Patients and/or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

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The Cochrane Library (Wiley Online Library)

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#6 (anterior cruciate ligament reconstruction (ACLR) with remnant preservation or anterior cruciate ligament reconstruction with remnant preservation):ti,ab,kw

#7 (anterior cruciate ligament (ACL) remnant-preserving reconstruction or ACLR-R):ti,ab,kw

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#6 TS=(anterior cruciate ligament (ACL) reconstruction with remnant-preserving)

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#8 TS=(anterior cruciate ligament (ACL) repair with suture augmentation)

#9 #2 OR #3 OR #7 OR #8

#10 TS=(randomized) OR TS=(placebo) OR TS=(controlled) OR TS=(random*) OR TS=(trial*) OR TS=(groups)

#11 TS=(singl*) OR TS=(doubl*) OR TS=(tripl*)

#12 TS=(mask*) OR TS=(blind*)

#13 #11 AND #12

#14 #10 OR #13

#15 #1 AND #9 AND #14

Grey literature: Websites searched

Key words as “anterior cruciate ligament reconstruction” OR “anterior cruciate ligament repair” OR “ACL reconstruction” OR “ACL repair” will be searched in the following grey literature websites.

OpenGrey (www.opengrey.eu/)

GreyNet International (<http://greynet.org/>)

OpenDOAR(<http://www.opendoar.org/>)

WHO (www.who.int)

Bielefeld Academic Search Engine (<http://www.base-search.net/about/en/BASE>)

The National Technical Information Service (<http://www.ntis.gov/>)

NHS UK (www.nhs.uk)

Duke University Medical Center Library (<http://guides.mclibrary.duke.edu/greyliterature>)

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item
ADMINISTRATIVE INFORMATION		
Title:		
Identification	1a	Identify the report as a protocol of a systematic review
Update	1b	If the protocol is for an update of a previous systematic review, identify as such
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number
Authors:		
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments
Support:		
Sources	5a	Indicate sources of financial or other support for the review
Sponsor	5b	Provide name for the review funder and/or sponsor
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol
INTRODUCTION		
Rationale	6	Describe the rationale for the review in the context of what is already known
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)
METHODS		
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated
Study records:		
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review

Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2 , Kendall's τ)
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

SYSTEMATIC REVIEW

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The Influence of Different Autologous Tendons on the Recovery of Knee Muscle Strength Following Anterior Cruciate Ligament Reconstruction: A Systematic Review and Network Meta-analysis

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Abstract

Background Outcomes of anterior cruciate ligament reconstruction (ACLR) vary significantly depending on autograft selection, particularly in restoring knee flexor and extensor strength, which are essential for athletic performance. The aim of this study was to investigate the impact of ACLR with different autografts on the recovery of knee flexor and extensor muscle strength. Subgroup analyses by follow-up duration were performed to characterize recovery across postoperative phases.

Methods A frequentist network meta-analysis of randomized controlled trials (RCT) and observational studies was conducted. Databases (PubMed, Embase, Web of Science, Cochrane Library) were searched from inception to January 2025. Eligible studies reported isokinetic strength outcomes after ACLR. The primary outcome was the limb symmetry index (LSI) for peak torque, which represented the ratio of injured-limb strength to uninjured-limb strength, reflecting inter-limb symmetry. Secondary outcomes were peak torque, the hamstring-to-quadriceps (H/Q) ratio and side-to-side deficit. Risk of bias (RoB) was assessed using the Cochrane RoB and the Risk of Bias Assessment Tool for Non-randomized Studies (RoBANS). Certainty of evidence was graded using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) framework. Graft performance was ranked using surface under the cumulative ranking curve (SUCRA) probabilities, and mean differences (MDs) with 95% confidence intervals (CIs) were calculated.

Results Thirty-one studies involving 2384 patients were included in the quantitative analysis. For knee extensor strength, assessed using LSI at 60°/s, 4-strand semitendinosus tendon (4S-ST) and 4-strand hamstring tendon (4S-HT) autografts outperformed bone-patella tendon-bone (BPTB) and quadriceps tendon (QT) autografts. At 180°/s, the 4S-HT autograft showed higher knee extensor strength LSI than the QT autograft, significantly outperformed the BPTB autograft at long-term follow-up (≥ 18 months), and consistently ranked highest according to SUCRA. The

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BPTB autograft also showed higher knee extensor strength LSI than the QT autograft. For knee flexor strength (LSI, at 60°/s and 180°/s), the QT autograft outperformed the 4S-HT autograft, with SUCRA consistently identifying the QT autograft as the top-ranked option and the 4S-HT autograft as the lowest-ranked option. No significant intergroup LSI differences were observed at short-term follow-up (≤ 6 months), whereas QT autograft's superiority in knee flexor strength became evident in the mid-term follow-up (6–12 months). The 4S-ST autograft may be superior to 4S-HT autograft in knee flexor strength LSI. Network inconsistency tests showed no significant inconsistency. However, GRADE assessments rated the certainty of evidence for comparisons of knee extensor and flexor strength between autografts as very low to low.

Conclusions The 4S-HT autograft was associated with greater knee extensor strength LSI, whereas the QT autograft had the most consistent advantage in knee flexor strength LSI. The 4S-ST autograft ranked between these grafts, but may provide greater knee flexor strength LSI than the 4S-HT autograft. The BPTB autograft was associated with greater knee extensor strength LSI than the QT autograft. Given the very low to low certainty of evidence, these findings should be interpreted with caution.

PROSPERO registration number CRD42024530838.

Key points

- The 4S-HT autograft was associated with greater knee extensor strength LSI.
- The QT autograft had the most consistent advantage in knee flexor strength LSI.
- The 4S-ST autograft may provide greater knee flexor strength LSI than the 4S-HT autograft.
- The BPTB autograft was associated with greater knee extensor strength LSI than the QT autograft.
- The certainty of evidence for these comparisons was very low to low.

Keywords Anterior cruciate ligament reconstruction, Autograft, Muscle strength, Network meta-analysis

Introduction

Anterior cruciate ligament (ACL) injuries, which account for more than 50% of knee injuries [1], are common among people aged 16–39, with an incidence of approximately 85 cases per 100,000 individuals [2]. Athletes, particularly those involved in high-risk sports such as soccer, football, basketball, and lacrosse, have an increased risk of ACL injury, with more than three-quarters of such injuries requiring surgical intervention [3, 4]. ACL reconstruction (ACLR) is the standard surgical procedure for these injuries [5].

Varying degrees of muscle strength decline occur in the muscles surrounding the knee joint following ACLR [6]. Knee extensor strength remains significantly reduced at 1 year, with limited improvement up to and beyond 5 years after surgery [6]. Different autografts may have distinct effects on recovery of knee flexor and extensor muscle strength after ACLR [7–10]. Bone-patellar tendon-bone (BPTB) and hamstring tendon (HT) autografts are the most commonly used grafts [11, 12]. BPTB autografts have historically been the gold standard for patients with significant functional requirements due to their benefits in knee joint stability and improved return-to-sports (RTS) [13, 14]. They may also lead to better knee flexor strength recovery than HT autografts [15],

but may cause anterior knee pain and reduce knee extensor strength [15, 16]. HT autografts allow for a small incision and avoid disruption of the extensor mechanism, but may increase the risk of bone tunnel widening and infection [13]. Additionally, quadriceps tendon (QT) autografts are gaining popularity [17]. QT autografts provide a large graft diameter and can also achieve a small incision, although long-term outcome data remain limited [13]. Due to the advantages and disadvantages of each tendon autograft, controversy persists regarding the optimal choice for ACLR. Therefore, to ensure the best treatment strategy for RTS, adjustments should be made according to the specific needs of the injured athlete [13].

Safe and timely RTS is a primary goal for all athletes [18] and is one of the most frequently used endpoints for evaluating surgical techniques [19]. A systematic review by Ardern et al. [20] found that 82% of patients had returned to some form of sports participation, 63% returned to their preinjury level, and 44% had resumed competitive sports at the final follow-up. Additionally, beyond simply returning to sport, many athletes are concerned about whether they can regain their pre-injury level of sport participation and performance [21, 22]. Muscle strength is fundamental to enhancing force-time characteristics (e.g. rate of force development and

external mechanical power) [23–25], optimizing sport skills such as jumping, sprinting and changing direction [26–28], and sport-specific performance [29, 30]. As such, it is an important indicator of athletic performance [31]. In a survey by the Pediatric Research in Sports Medicine (PRiSM) Society, the most commonly used testing methods were manual muscle testing (60%), followed by isokinetic (28%) and isometric (12%) tests [32].

The isokinetic muscle strength test quantifies muscle strength at specific angular velocities and is considered the "gold standard" for evaluating muscle strength [32]. It also exhibits high reproducibility at angular velocities of 60°/s and 180°/s, making it an effective tool for assessing both knee flexor and extensor strength [33].

One recent systematic review compared isokinetic muscle strength between QT, BPTB and HT autografts at different angular velocities and follow-up times [8]. QT autografts showed superior knee flexion strength compared to HT autografts and similar results to BPTB autografts at 3, 6 and 12 months [8]. Conversely, HT autografts demonstrated superior knee extensor strength at 6 and 12 months compared with QT autografts [8]. However, that review was limited to pairwise comparisons between two graft types at a time and did not include a network analysis across multiple autografts [8]. Therefore, a network meta-analysis is required to comprehensively compare the postoperative muscle strength recovery outcomes of multiple autograft types, thereby enabling indirect comparisons among various autografts and providing treatment ranking evidence to support clinical surgical decision-making.

This present study aimed to compare the effects of different autografts on postoperative recovery level of knee flexor and extensor muscle strength after ACLR using network meta-analysis. Furthermore, by integrating findings from this network meta-analysis with existing evidence, clinicians may select the most appropriate autograft according to the specific muscle strength recovery needs of individual patients.

Methods

This frequentist network meta-analysis was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta Analyses-Network Meta Analyses (PRISMA-NMA) [34]. The protocol was registered in the International Prospective Registry of Systematic Reviews (PROSPERO, CRD42024530838).

Search Strategy

A systematic literature search was performed in PubMed, Embase, Web of Science and the Cochrane Library from database inception to January 2025. No language restrictions were applied, and grey literature was excluded.

Search terms were derived from the Medical Subject Headings (MeSH) database and related keywords for ACLR and rehabilitation. Boolean operators and database-specific filters were applied. The full search strategy is provided in Supplementary Table S1.

Inclusion Criteria

Studies were included according to the PICOS framework [35]: (1) patients of any age who had undergone primary ACLR surgery with no ligament injuries in contralateral knee; (2) interventions involving different tendon autografts for ACLR; (3) comparisons between different tendon autografts; (4) muscle strength assessments using isokinetic testing at angular velocities of 60°/s or 180°/s; and (5) RCTs or observational studies.

Exclusion Criteria

Studies were excluded if they met any of the following criteria: (1) duplicates; (2) meta-analyses or reviews; (3) studies not retrieved (those that may meet the inclusion criteria but remain undetected due to limitations in search strategies, insufficient database coverage, language barriers, and other factors); (4) incomplete outcome data (e.g. mean values without standard deviations [SDs]) or unavailable follow-up data; or (5) studies whose interventions, populations, outcomes, or measurement methods were found to not meet the inclusion criteria after reviewing the full texts.

Literature Screening

The literature screening was conducted independently by two authors, first based on the title and abstracts, followed by full-text review. Inter-rater agreement was assessed using the Cohen's kappa coefficient. Discrepancies were resolved by a third reviewer. The PRISMA 2020 flow diagram for systematic reviews [36] was used to document the study selection process.

Risk of Bias and Quality of Evidence

Risk of bias (RoB) was assessed independently by two researchers, with discrepancies resolved through discussion with a third researcher. RCTs were evaluated using the Cochrane RoB tool [37], which examines seven potential sources of bias: randomization sequence generation, allocation concealment, blinding of both patients and personnel, blinding of outcome assessment, incomplete outcome data, selective reporting and other biases. Each domain was classified as low, high or unclear risk. Review Manager (RevMan) [Computer program]. Version 5.4. The Cochrane Collaboration, 2020 was used to facilitate a visual summary of RoB across studies. Observational studies were assessed using the Risk of Bias Assessment Tool for Non-randomized Studies (RoBANS)

[38], which evaluates six potential sources of bias: patients selection, confounding variables, measurement of exposure, blinding of outcome assessment, incomplete outcome data and selective outcome reporting. Each domain was classified as low, high or unclear risk.

The certainty of evidence for network meta-analysis estimates was evaluated using the method proposed by Izcovich et al. [39], which integrates the confidence of direct, indirect, and network estimates.

Additionally, the certainty of evidence for all outcome measures was assessed using the Grading Recommendations Assessment, Development and Evaluation (GRADE) system [40, 41]. Evidence was categorized as high, moderate, low or very low based on risk of bias, inconsistency, indirectness, imprecision and publication bias.

Data Extraction

Two researchers independently extracted relevant study characteristics, with discrepancies resolved by a third researcher. Data were collected using standardized extraction forms to ensure the validity and reliability of the results. Extracted information included first author, year of publication, study design, country, type of tendon autograft, patient demographics, number of drop-outs, duration of follow-up, and outcome measures. For pooled analyses of baseline mean age and sex distribution, the overall mean age was calculated using a sample size-weighted method, and the pooled proportion of male participants was calculated by summing the number of male participants and dividing by the total sample size. Studies with unavailable or incomplete baseline age or sex data were excluded from these demographic syntheses.

Because general patient characteristics may differ (e.g., athletes typically demonstrate greater muscle strength than the general population), peak torque values alone may be influenced by individual heterogeneity. Therefore, the limb symmetry index (LSI) for peak torque was selected as the primary outcome for assessing muscle strength recovery after ACLR with different tendon autografts. LSI was calculated as the ratio of injured-limb strength to uninjured-limb strength and reflects inter-limb strength symmetry rather than absolute strength; therefore, a higher LSI does not necessarily indicate greater absolute strength in the affected limb. Secondary outcomes included peak torque, the hamstring-to-quadriceps (H/Q) ratio and side-to-side deficit. Peak torque refers to the maximum torque generated by a muscle group during joint movement, reflecting maximal muscle strength and serving as a gold standard indicators in the isokinetic test [42]. The H/Q ratio was defined as the ratio of knee flexion strength to extension strength,

while side-to-side deficit was calculated as the difference between non-injured and injured limb torque, divided by the non-injured limb torque.

Network Meta-analysis Methodology

Network meta-analysis is an extension of traditional pairwise meta-analysis that allows indirect comparisons between interventions not directly compared in the included studies. Data were extracted from study tables or text, including sample size, means, and SD. The “network” suite of commands in Stata (USA, Texas, 18.0) was utilized to generate network plots for each outcome, illustrating the autograft types compared across studies. Inconsistency was assessed using the inconsistency model to evaluate global inconsistency. If global inconsistency was detected, a node-splitting approach was applied to assess local inconsistency in closed-loop structures. Forest plots of mean differences (MDs), reported together with their corresponding 95% confidence intervals (CIs), were constructed to visualize individual study results and pooled estimates. Autograft types were ranked using the surface under the cumulative ranking (SUCRA), and ranking tables were generated to compare the relative effect differences among the autograft types. Funnel plots were employed to assess potential publication bias.

Postoperative recovery of knee extensor and flexor strength after ACLR is time dependent, with rapid improvement up to 12–18 months and minimal gains thereafter [6]. Accordingly, subgroup analyses were performed based on follow-up duration: short-term (≤ 6 months), mid-term (6–12 months), and long-term (≥ 18 months). No studies reported follow-up between 12 and 18 months, so this interval was not analyzed. A two-sided P -value < 0.05 was considered statistically significant.

Results

Literature Selection

A comprehensive search of databases identified 1434 studies. Following strict inclusion and exclusion criteria, a total of 31 studies were finally included. The inter-rater agreement between the two reviewers during literature selection was excellent, with a Cohen's kappa coefficient of 0.966. The screening process is shown in Fig. 1, and the specific reasons for exclusion are presented in Supplementary Table S3.

General Information and Characteristics of Included Studies

The general characteristics of the 31 included studies are summarized in Supplementary Table S4. Of these, eleven were RCTs [43–53] and 20 were observational

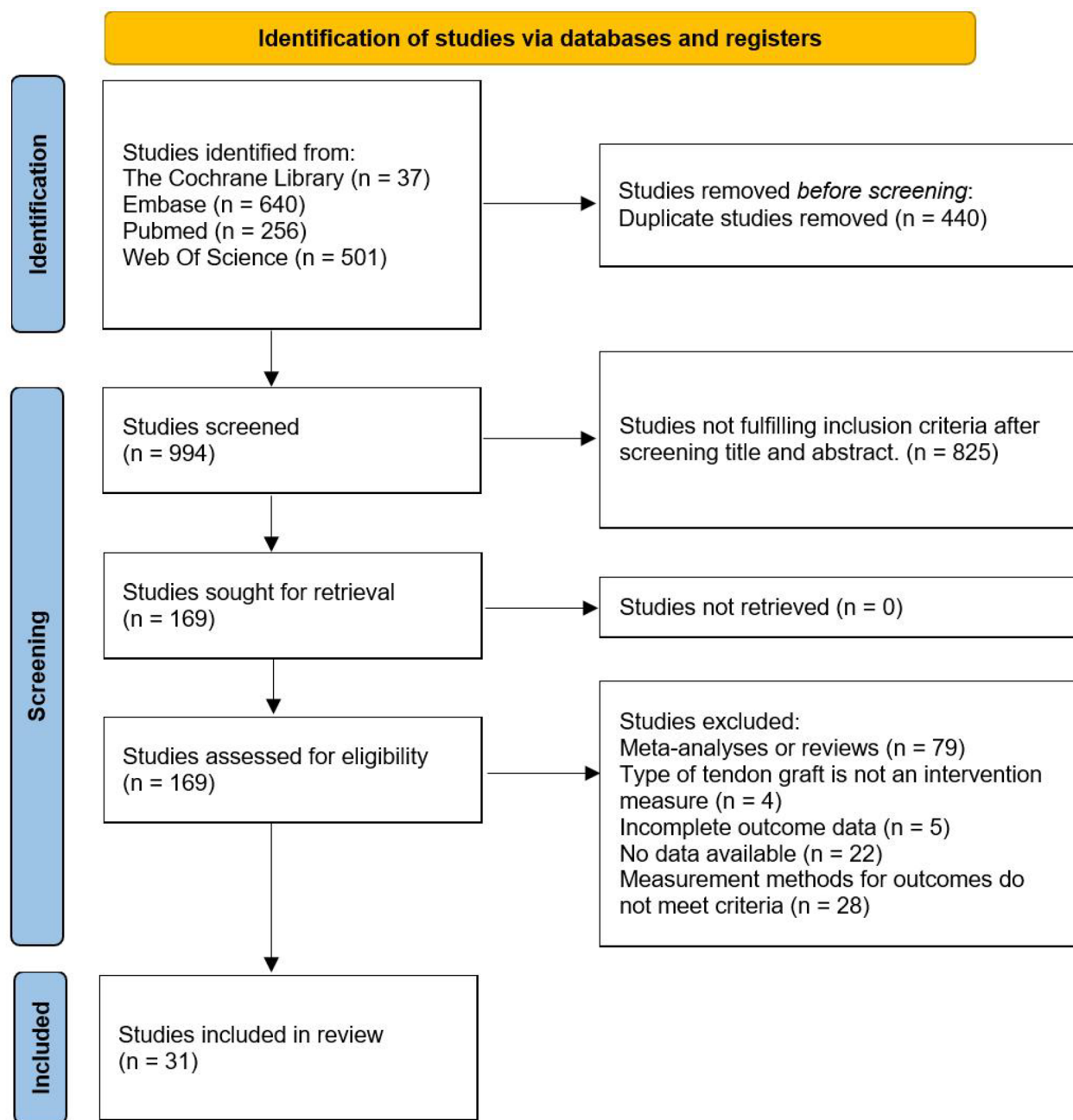


Fig. 1 PRISMA flow diagram for comparison of muscle strength recovery after ACLR using different autografts

studies [54–73], with a total of 2384 patients from all over the world. Interventions are referred to the different autograft types being compared. The distribution of autograft types was as follows: bone-patellar tendon-bone autograft (BPTB autograft, $n = 546$), patellar tendon autograft without a bone block (PT autograft, $n = 51$), four-strand semitendinosus tendon autograft (4S-ST autograft, $n = 299$), five-strand semitendinosus tendon autograft (5S-ST autograft, $n = 18$),

four-strand hamstring tendon autograft using semitendinosus and gracilis (4S-HT autograft, $n = 1120$) and quadriceps tendon autograft (QT autograft, $n = 350$). Male patients comprised 64.2% of the total (excluding 4 studies with missing sex data), and the mean age was 27.0 years (excluding 3 studies with missing age data). The mean follow-up duration was 26.1 months (median, 12.0 months; range, 6–128.4 months). Ten studies reported loss to follow-up, with a total of 292

patients affected. Data extraction results are presented in Supplement Table S2.

Risk of Bias Assessment

The risk of bias (RoB) of the included RCTs was evaluated using the Cochrane RoB tool (Supplementary Figure S1). The independent reviewers' judgments concerning each bias risk item are presented as percentages across all studies, as illustrated in Supplementary Figure S1a. Supplementary Figure S1b presents an overview of the assessments conducted by independent reviewers for each risk of bias item across all studies included in the analysis. Most studies (91%) reported random participant selection, though specific randomization methods were often unclear. Blinding of participants, personnel and outcome assessors was generally not feasible due to differences in surgical procedures, resulting in high risk of performance and detection bias. While 64% of studies adopted adequate allocation concealment, 45% reported incomplete outcome data, and selective reporting risk was unclear. For observational studies, risk of bias was evaluated using RoBANS (Supplementary Table S5). Most studies were retrospective, with high risk of selection bias (85%) and high risk of bias in outcome assessment due to lack of blinding. Confounding variables were adequately controlled in 60% of studies, and standardized

isokinetic measurements minimized exposure bias. Loss to follow-up affected 35% of studies, and selective reporting risk was unclear. Overall, the main biases were related to selection and outcome assessment, whereas measurement reliability and confounding control were generally satisfactory.

Knee Extensor LSI at 60°/s

All Included Studies

Network meta-analysis for knee extensor LSI at 60°/s included 14 studies with 1608 patients and 4 autograft types: BPTB, 4S-ST, 4S-HT and QT [44, 48, 52, 53, 55, 58, 61, 63–65, 67, 68, 70, 71] (Fig. 2a). The largest group was 4S-HT autograft (n=813). Forest plot analysis presented MDs with 95% CIs among the above-mentioned autograft types, and all detailed comparison results were displayed in Fig. 2b; among these comparisons, except for the one between 4S-HT and 4S-ST, all others showed significant differences: BPTB versus 4S-ST (MD=−6.76, 95% CI−12.10 to −1.42; certainty of evidence, very low in both observational studies and RCTs), QT versus 4S-ST (MD=−10.14, 95% CI−15.44 to −4.85; certainty, very low in both observational studies and RCTs), BPTB versus 4S-HT (MD=−5.32, 95% CI−7.55 to −3.08; certainty, very low in observational studies and low in RCTs), QT versus 4S-HT (MD=−8.70, 95% CI−10.82 to −6.59; certainty, very low in observational studies and low in RCTs), QT versus BPTB (MD=−3.39, 95% CI−6.41 to −0.37)

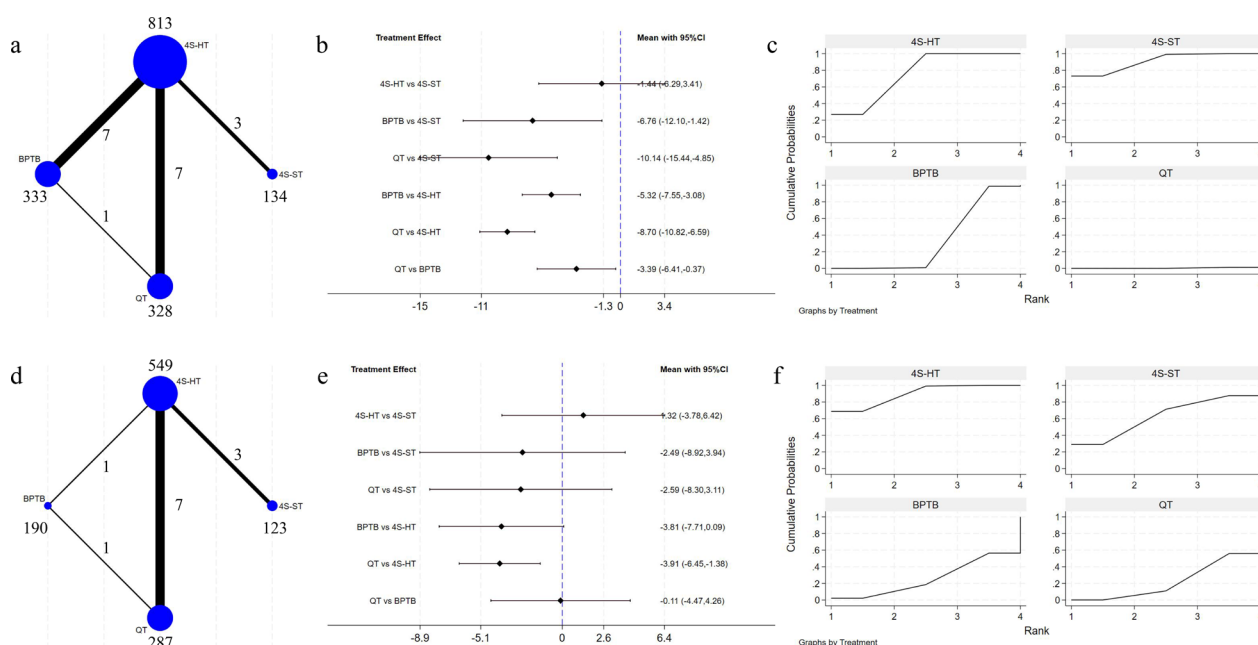


Fig. 2 Network analysis of knee extensor LSI. **a** Network map of knee extensor LSI at 60°/s. Blue dots represent sample size, line thickness indicates the number of comparisons, and numbers indicate the exact sample. **b** Forest plot (mean difference) of knee extensor LSI at 60°/s. **c** Probability rankings of knee extensor LSI at 60°/s. **d** Network map of knee extensor LSI at 180°/s. **e** Forest plot (mean difference) of knee extensor LSI at 180°/s. **f** Probability rankings of knee extensor LSI at 180°/s. Abbreviations: 4S-HT autograft, four-strand hamstring tendon autograft; 4S-ST autograft, four-strand semitendinosus tendon autograft; BPTB autograft, bone–patella tendon–bone autograft; QT autograft, quadriceps tendon autograft

to -6.59 ; certainty, very low in observational studies and low in RCTs), and QT versus BPTB ($MD = -3.39$, 95% CI -6.41 to -0.37 ; certainty, very low in both observational studies and RCTs). SUCRA rankings (Fig. 2c) were 4S-ST 90.8%, 4S-HT 75.6%, BPTB 33.2%, and QT 0.4%. The network showed no significant inconsistency ($P = 0.185$). The results of the GRADE evidence grading were presented in Supplementary Table S6.

Subgroup 1: Follow-Up ≤ 6 months

Subgroup 1 included 6 studies involving 521 patients and 4 autograft types: BPTB, 4S-ST, 4S-HT and QT [58, 61, 64, 67, 68, 71] (Supplementary Figure S2a). The largest group was the 4S-HT autograft ($n = 307$). Forest plot analysis (Supplementary Figure S2b) showed that significant differences were observed between BPTB and 4S-HT ($MD = -7.12$, 95% CI $[-14.21, -0.02]$), between QT and 4S-HT ($MD = -10.56$, 95% CI $[-13.50, -7.61]$), and between QT and 4S-ST ($MD = -13.52$, 95% CI $[-23.59, -3.45]$). The SUCRA rankings (Supplementary Figure S2c) were 4S-ST 89.0%, 4S-HT 74.9%, BPTB 30.0%, and QT 6.1%. Inconsistency assessment was not performed due to the lack of direct comparisons.

Subgroup 2: Follow-Up 6–12 months

Subgroup 2 included 7 studies with a total of 532 patients and 4 autograft types: BPTB, 4S-ST, 4S-HT and QT [44, 48, 52, 53, 58, 67, 68] (Supplementary Figure S2d). The largest group was the 4S-HT autograft ($n = 291$). Forest plot analysis (Supplementary Figure S2e) demonstrated significant differences between 4S-HT and both BPTB ($MD = 6.00$, 95% CI $[1.49, 10.50]$) and QT ($MD = 7.96$, 95% CI $[4.57, 11.35]$). The SUCRA rankings (Supplementary Figure S2f) were 4S-HT 92.3%, 4S-ST 58.4%, BPTB 35.4%, and QT 13.9%. Inconsistency assessment was not performed due to the lack of direct comparisons.

Subgroup 3: Follow-Up ≥ 18 months

Subgroup 3 included 5 studies with a total of 555 patients and 4 autograft types: BPTB, 4S-ST, 4S-HT and QT [48, 55, 63, 65, 70] (Supplementary Figure S2g). The largest group was the 4S-HT autograft ($n = 215$). Forest plot analysis (Supplementary Figure S2h) showed that significant differences existed between BPTB and both 4S-HT ($MD = -4.35$, 95% CI $[-7.14, -1.57]$) and 4S-ST ($MD = -8.03$, 95% CI $[-15.74, -0.32]$). The SUCRA rankings (Supplementary Figure S2i) were 4S-ST 88.5%, 4S-HT 60.0%, QT 44.0%, and BPTB 7.4%. The inconsistency test indicated no significant inconsistency ($P = 0.8840$), confirming a reliable network.

Knee Extensor LSI at 180°/s

All Included Studies

The network meta-analysis of knee extensor LSI at 180°/s (Fig. 2d) included 10 studies with 1149 patients and 4 autograft types: BPTB, 4S-ST, 4S-HT and QT [44, 47, 55, 59, 61, 63, 65, 67, 70, 73]. The 4S-HT autograft had the largest sample ($n = 549$). Forest plot analysis (Fig. 2e) showed that a significant difference was observed between QT and 4S-HT ($MD = -3.91$, 95% CI $[-6.45, -1.38]$). The certainty of evidence for this comparison was very low in observational studies and low in RCTs. SUCRA rankings (Fig. 2f) were 4S-HT 89.3%, 4S-ST 62.6%, BPTB 25.7%, and QT 22.3%. Inconsistency and heterogeneity tests confirmed a reliable network ($P = 0.3015$). The results of the GRADE evidence grading were presented in Supplementary Table S6.

Subgroup 1: Follow-Up ≤ 6 months

Subgroup 1 included 4 studies with 323 patients and 3 autograft types: 4S-ST, 4S-HT and QT [47, 61, 67, 73] (Supplementary Figure S3a), with 4S-HT autograft having the largest sample ($n = 181$). Forest plot analysis (Supplementary Figure S3b) showed a significant difference between QT and 4S-HT ($MD = -8.01$, 95% CI $[-13.00, -3.03]$). SUCRA rankings were 4S-HT 84.0%, 4S-ST 62.5%, and QT 3.5% (Supplementary Figure S3c). No inconsistency assessment was performed due to limited direct comparisons.

Subgroup 2: Follow-Up 6–12 months

Subgroup 2 included 3 studies with 279 patients and 2 autograft types: 4S-HT and QT [44, 67, 70] (Supplementary Figure S3d), with 4S-HT having the largest sample ($n = 155$). Forest plot analysis (Supplementary Figure S3e) indicated that a significant difference existed between QT and 4S-HT ($MD = -5.12$, 95% CI $[-8.38, -1.86]$), and SUCRA rankings were 4S-HT 100% versus QT 0% (Supplementary Figure S3f). No inconsistency assessment was conducted.

Subgroup 3: Follow-Up ≥ 18 months

Subgroup 3 included 5 studies with 547 patients and 4 autograft types: BPTB, 4S-ST, 4S-HT and QT [55, 59, 63, 65, 70] (Supplementary Figure S3g), with 4S-HT autograft again having the largest sample ($n = 213$). Forest plot analysis (Supplementary Figure S3h) demonstrated the presence of a significant difference between BPTB and 4S-HT ($MD = -3.33$, 95% CI $[-5.74, -0.92]$). SUCRA rankings were 4S-HT 73.9%, QT 63.8%, 4S-ST 50.9%, and BPTB 11.4%

(Supplementary Figure S3i). Inconsistency testing confirmed a reliable network ($P=0.5928$).

Knee Flexor LSI at 60°/s

All Included Studies

The network analysis comparing knee flexor LSI at 60°/s was presented in Fig. 3a. 13 studies with 1477 patients and 4 autograft types (BPTB, 4S-ST, 4S-HT and QT) [44, 48, 52–54, 58, 61, 63–65, 67, 68, 70] were included, with 4S-HT autograft having the largest sample size ($n=749$). Forest plot analysis (Fig. 3b) showed a significant difference between QT and 4S-HT (MD=9.20, 95% CI [2.80, 15.59]). The certainty of evidence for this comparison was very low in observational studies and low in RCTs. SUCRA rankings (Fig. 3c) were QT 84.9%, 4S-ST 65.2%, BPTB 44.8%, and 4S-HT 5.1%. No inconsistency assessment was performed, and the GRADE evidence grading results were presented in Supplementary Table S6.

Subgroup 1: Follow-Up ≤ 6 months

Subgroup 1 included 5 studies with 410 patients and 4 autograft types: BPTB, 4S-ST, 4S-HT and QT [58, 61, 64, 67, 68] (Supplementary Figure S4a), with 4S-HT as the largest group ($n=235$). No statistically significant difference was observed (Supplementary Figure S4b), and SUCRA rankings were 4S-ST 67.2%, QT 57.0%, BPTB

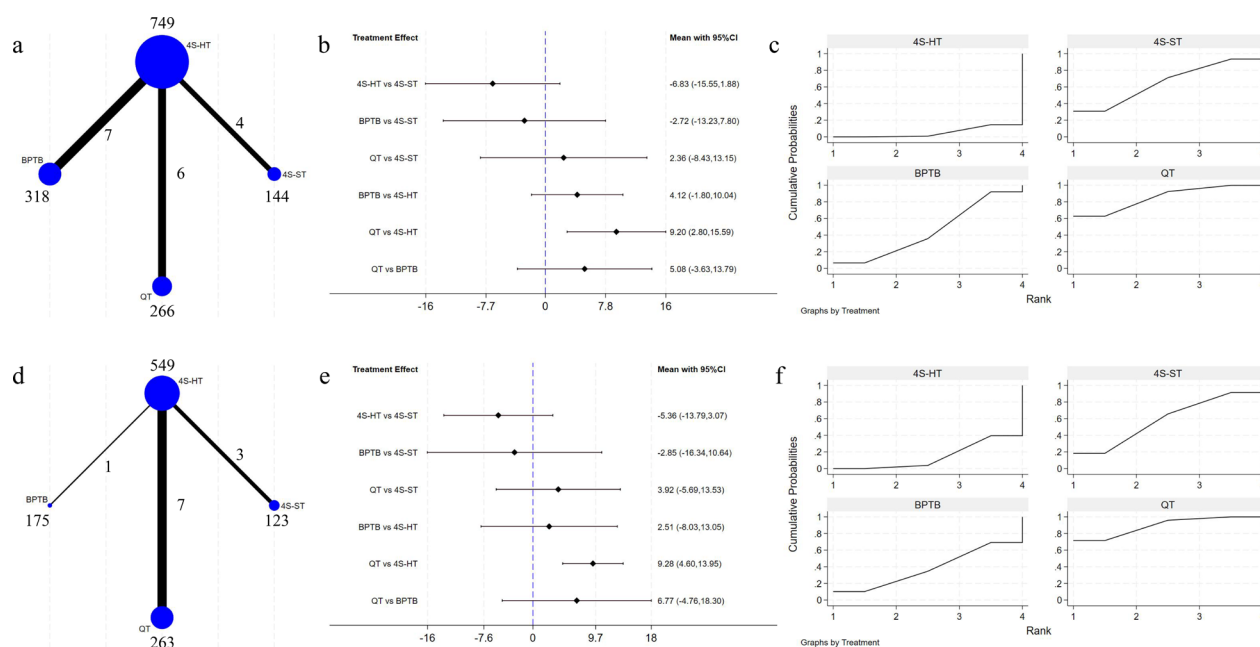
51.3%, and 4S-HT 24.6% (Supplementary Figure S4c). No inconsistency assessment was conducted.

Subgroup 2: Follow-Up 6–12 months

Subgroup 2 included 7 studies with 532 patients and 4 autograft types: BPTB, 4S-ST, 4S-HT and QT [44, 52, 53, 58, 67, 68, 70] (Supplementary Figure S4d), with 4S-HT as the largest group ($n=291$). Forest plot analysis (Supplementary Figure S4e) showed significant differences between 4S-HT and both BPTB (MD=−6.20, 95% CI [−11.10, −1.30]) and QT (MD=−10.84, 95% CI [−15.68, −6.01]). SUCRA rankings were QT 93.5%, BPTB 55.7%, 4S-ST 48.4%, and 4S-HT 2.3% (Supplementary Figure S4f). Inconsistency assessment was not performed due to the lack of direct comparisons.

Subgroup 3: Follow-Up ≥ 18 months

Subgroup 3 included 5 studies with 535 patients and 4 autograft types: BPTB, 4S-ST, 4S-HT and QT [48, 54, 63, 65, 70] (Supplementary Figure S4g), with 4S-HT as the largest group ($n=225$). No statistically significant difference was observed (Supplementary Figure S4h), and SUCRA rankings were QT 76.8%, 4S-ST 58.4%, BPTB 35.7%, and 4S-HT 29.1% (Supplementary Figure S4i). No inconsistency assessment was performed due to limited direct comparisons.



Knee Flexor LSI at 180°/s

All Included Studies

The network meta-analysis of knee flexor LSI at 180°/s is summarized as follows. 9 studies with 1109 patients and 4 autograft types (BPTB, 4S-ST, 4S-HT and QT) [44, 47, 59, 61, 63, 65, 67, 70, 73] (Fig. 3d) were included, with 4S-HT autograft having the largest sample size ($n=549$). Forest plot analysis (Fig. 3e) showed a significant difference between QT and 4S-HT (MD=9.28, 95% CI [4.60, 13.95]). The certainty of evidence for this comparison was very low in observational studies and low in RCTs. SUCRA rankings (Fig. 3f) were QT 89.2%, 4S-ST 58.4%, BPTB 38.0%, and 4S-HT 14.4%. No inconsistency assessment was performed, and Supplementary Table S6 contains the results of the GRADE evidence grading.

Subgroup 1: Follow-Up ≤ 6 months

Subgroup 1 included 4 studies with 323 patients and 3 autograft types: 4S-ST, 4S-HT and QT [47, 61, 67, 70] (Supplementary Figure S5a), with 4S-HT as the largest group ($n=181$). Forest plot analysis (Supplementary Figure S5b) showed that no statistically significant difference was observed. SUCRA rankings were 4S-ST 74.4%, QT 59.4%, and 4S-HT 16.2% (Supplementary Figure S5c). No inconsistency assessment was conducted.

Subgroup 2: Follow-Up 6–12 months

Subgroup 2 included 3 studies with 279 patients and 2 autograft types: 4S-HT and QT [44, 67, 70] (Supplementary Figure S5d), with 4S-HT as the largest group ($n=155$). Forest plot analysis (Supplementary Figure S5e) showed that a significant difference existed between QT and 4S-HT (MD=13.93, 95% CI [9.79, 18.06]). SUCRA rankings were QT 100% and 4S-HT 0% (Supplementary Figure S5f). Inconsistency assessment was not performed due to the lack of direct comparisons.

Subgroup 3: Follow-Up ≥ 18 months

Subgroup 3 included 4 studies with 507 patients and 4 autograft types: BPTB, 4S-ST, 4S-HT and QT [59, 63, 65, 70] (Supplementary Figure S5g), with 4S-HT as the largest group ($n=213$). No statistically significant difference was observed (Supplementary Figure S5h). SUCRA rankings were QT 75.5%, 4S-ST 51.0%, BPTB 47.6%, and 4S-HT 25.8% (Supplementary Figure S5i). No inconsistency assessment was performed due to limited direct comparisons.

Comparisons of Secondary Outcomes

The comparison of peak torque across different angular velocities involved 6 autograft types (BPTB, PT, 4S-ST, 5S-ST, 4S-HT and QT). Forest plots (Supplementary Figures S6–S9) showed no statistically significant

differences in knee extensor or flexor peak torque among the autografts at 60°/s or 180°/s. Notably, some autograft types had small sample sizes (5S-ST, 18 patients; PT, 30 patients).

Postoperative muscle strength recovery assessed by H/Q ratio was analyzed for 4 autograft types at 60°/s and 180°/s. Forest plot analysis (Supplementary Figures S10–S11) indicated that significant differences in H/Q ratio were observed between QT and 4S-HT at both 60°/s (MD=12.79, 95% CI [3.29, 22.29]) and 180°/s (MD=12.99, 95% CI [0.26, 25.73]), and between 4S-ST and 4S-HT at 180°/s (MD=7.99, 95% CI [2.58, 13.41]). Good network consistency was observed at 60°/s ($P=0.2064$).

Side-to-side deficits in extensor and flexor peak torque were compared across 5 autograft types (BPTB, PT, 4S-ST, 4S-HT and QT). Among the findings of this comparison, a significant difference in extensor peak torque side-to-side deficits was observed between BPTB and 4S-HT at 180°/s (MD=3.09, 95% CI [0.71, 5.47]). For flexor peak torque side-to-side deficits at 180°/s, significant differences were noted between QT and each of 4S-HT (MD=−11.90, 95% CI [−20.90, −2.89]), 4S-ST (MD=−9.00, 95% CI [−14.34, −3.66]), and BPTB (MD=−12.90, 95% CI [−24.27, −1.52]). No significant differences were observed at 60°/s. Additionally, the small sample of PT autograft ($n=21$) should be noted (Supplementary Figures S12–S15).

Assessment of Publication Bias

Funnel plots were generated in Stata to assess potential publication bias (Supplementary Figure S16). Plots for knee flexion and extension at different angular velocities showed no significant asymmetry, indicating minimal publication bias in this study.

Discussion

This study provides a comprehensive evaluation of comparisons in knee extensor and flexor muscle strength after ACLR using different autografts. Compared with previous meta-analyses [7, 8], the present meta-analysis included a larger number of studies and autografts with indirect comparisons. By synthesizing data from 6 autograft types, 31 studies and 2384 patients, we assessed postoperative isokinetic muscle strength at 60°/s and 180°/s, thereby offering a holistic perspective on graft-specific recovery of muscle strength. Subgroup analyses by follow-up duration (short-term, ≤ 6 months; mid-term, 6–12 months; long-term, ≥ 18 months) were conducted to explore temporal differences in recovery. In summary, across both angular velocities (60°/s and 180°/s), the 4S-HT autograft showed superior knee extensor strength recovery, whereas the QT autograft

consistently showed the greatest potential in flexor strength recovery. The 4S-ST autograft occupied an intermediate position, exhibiting advantages in certain long-term follow-ups and showing greater knee flexor strength LSI at 60°/s compared with the 4S-HT autograft. Additionally, the BPTB autograft primarily demonstrated greater knee extensor strength recovery than the QT autograft at 60°/s. However, given the very low to low certainty of evidence, these findings should be interpreted with caution.

Knee Extensor LSI at 60°/s and 180°/s

For knee extensor LSI at 60°/s, forest plots and SUCRA rankings suggested that 4S-ST and 4S-HT autografts may confer relative advantages over BPTB and QT autografts across the overall follow-up period. In contrast, Johnston et al. [7] and Herbawi et al. [8] reported that the 4S-HT autograft primarily demonstrated superiority over BPTB and QT autografts in knee extensor strength recovery when follow-up was ≤ 6 months, but no difference was observed between the 4S-HT autograft and BPTB and QT autografts when follow-up was ≥ 12 months [8]. This discrepancy may reflect the combined direct and indirect evidence used in the present network meta-analysis, whereas previous studies relied mainly on direct pairwise comparisons. Specifically, our analysis included both direct 4S-HT versus QT evidence and indirect evidence through BPTB, broadening the evidence base for the 4S-HT versus QT estimate. The discrepancy may also be explained by smaller sample sizes in previous reviews, which included only two studies for each relevant subgroup (170 and 142 patients), limiting statistical power. Although no significant difference was observed between 4S-ST and 4S-HT autografts, the certainty of evidence for this comparison was very low, reducing confidence in the finding. In the overall comparison, the BPTB autograft demonstrated greater knee extensor strength recovery than the QT autograft, while in subgroup analyses, smaller sample sizes may have widened confidence intervals and reduced statistical significance.

In the overall comparison of knee extensor LSI at 180°/s, forest plots and SUCRA rankings indicated that the 4S-HT autograft consistently outperformed other grafts across all follow-up periods. It ranked first in the overall, short-term, mid-term, and long-term analyses, showing stable recovery of knee extensor strength and demonstrating significant superiority over the QT autograft in the overall, short-term and mid-term follow-ups, as well as over the BPTB autograft in the long-term follow-up. These findings are consistent with previous meta-analysis [9].

There were differences in knee extensor strength recovery following QT autograft among different follow-up

subgroups. Although QT autografts performed less favorably in short- and mid-term analyses, their ranking improved substantially at long-term follow-up, rising to second at 180°/s and third at 60°/s and surpassing BPTB and 4S-ST autografts in this subgroup. This pattern is consistent with studies showing greater knee extensor-strength recovery with HT than QT autografts at 6 months but no significant difference among QT, BPTB and HT autografts at 24 months [8]. These patterns may reflect time-dependent recovery, but they may also be affected by the small number of long-term studies, limited sample sizes, non-standardized rehabilitation protocols and incomplete intervention networks (the short-term phase lacked BPTB autografts, and the mid-term phase lacked both 4S-ST and BPTB autografts). Moreover, the certainty of evidence for QT versus BPTB at 60°/s and QT versus 4S-ST at 180°/s was very low, limiting confidence in these subgroup findings. Future studies should follow a cohort prospectively to analyze the temporal trajectory of knee extensor strength recovery in patients with QT autograft for ACLR.

Knee Flexor LSI at 60°/s and 180°/s

In the overall comparison of the LSI for knee flexor strength at 60°/s after ACLR, the QT autograft had the highest SUCRA values in the overall analysis and most subgroups, consistent with previous research [7, 8]. However, no statistically significant intergroup differences were observed in the short-term (follow-up ≤ 6 months) and long-term (follow-up ≥ 18 months) follow-ups; only in the mid-term (follow-up 6–12 months) did QT and BPTB autografts show a trend of superiority over the 4S-HT autograft. Herbawi et al. [8] reported significantly better short-term outcomes with the QT autograft compared with 4S-HT, whereas our results did not confirm this. This discrepancy may be partly explained by the small sample size of the QT group (approximately half the size of the HT group) in Johnston et al. [67], thereby introducing a bias toward the larger comparator group. At long-term follow-up, knee flexor strength recovery with QT was comparable to that with 4S-HT and BPTB, which aligns with Herbawi et al. [8]. The 4S-HT autograft consistently demonstrated relative disadvantages in knee flexor strength recovery, likely due to the reduced hamstring cross-sectional area following tendon harvest [74]. Supporting this, Girdwood et al. [6] reported a 10–15% greater knee flexor strength deficit in hamstring tendon graft recipients compared with other graft types. In the present study, no significant difference was observed between the 4S-ST and 4S-HT autograft in knee flexor strength recovery across the overall sample and all subgroups, while the SUCRA plot indicated a high probability that the 4S-ST autograft was superior to the 4S-HT

autograft in this regard, consistent with prior research that has indicated superior knee flexor strength recovery with 4S-ST over 4S-HT [10, 75, 76]. The absence of statistical significance may reflect methodological differences, as previous studies used conventional pairwise meta-analysis whereas the present study incorporated evidence from multiple comparative pairs. Similarly, no significant difference was observed between BPTB and QT autografts, although SUCRA rankings suggested a high probability that QT was superior for knee flexor-strength recovery.

In the overall comparison of LSI for knee flexor strength at 180°/s after ACLR, results were largely consistent with those observed at 60°/s, except for the absence of data on certain interventions—specifically, the comparison in short-term (Follow-up ≤ 6 Months) lacked data on BPTB autograft, while the comparison in mid-term (Follow-up 6–12 Months) lacked data on both BPTB and 4S-ST autograft.

Comparisons of Secondary Outcomes

In the analysis of H/Q ratio, a higher H/Q ratio for QT autografts indicated better knee flexor-strength recovery than 4S-HT autografts at both 60°/s and 180°/s, consistent with the findings of Salas et al. [77]. Additionally, side-to-side deficit analyses at 180°/s suggested that the QT autograft may outperform the 4S-HT, 4S-ST, and BPTB autograft in knee flexor strength recovery. These results align with the primary outcome measure (LSI). By contrast, peak torque comparisons revealed no significant differences among autografts, likely due to the lack of adjustment for preoperative torque differences, which were not reported in the included studies. Additionally, residual confounding may also arise from unadjusted variables including activity level, body mass index, and sex. Moreover, the extremely small sample sizes for the PT and 5S-ST autograft rendered their comparative outcomes highly unreliable.

Clinical implications

The present findings may provide several cautious clinical implications for graft selection and rehabilitation planning after ACLR. Although graft choice should primarily be individualized according to patient age, activity level, sport-specific demands, anatomical considerations, surgeon experience, and patient preference [78–80], the observed graft-specific patterns of muscle strength recovery may serve as supplementary information during shared decision-making. The relatively favorable knee extensor strength recovery observed with the 4S-HT and 4S-ST autografts suggests that

these grafts may be considered when early or sustained quadriceps strength recovery is a particular clinical priority. Conversely, the QT autograft showed a tendency toward better knee flexor strength recovery, which may be relevant for patients in whom preservation of hamstring function is important, such as those participating in sports requiring repeated sprinting, cutting, or dynamic knee stabilization. However, these findings should not be interpreted as evidence that one graft is universally superior to another. From a rehabilitation perspective, the results highlight the need for graft-specific monitoring and targeted strength training after ACLR. Patients receiving 4S-HT or 4S-ST autografts may require particular attention to knee flexor strength restoration. In contrast, patients receiving QT or BPTB autografts may require careful monitoring of quadriceps strength recovery, especially during the early and mid-term postoperative periods.

Limitations

Several limitations should be acknowledged. First, variability in rehabilitation protocols may have contributed to differences among follow-up subgroups. Rehabilitation is a critical determinant of muscle strength recovery; however, insufficient reporting prevented assessment of rehabilitation-related factors. Previous research indicates that approximately 40% of patients receive supervised rehabilitation for less than 3 months after ACLR [81], and only 40% continue rehabilitation beyond 9 months [82], suggesting that many patients may receive inadequate rehabilitation. Additional RCTs with standardized rehabilitation protocols are therefore needed to strengthen the credibility of future evidence. At present, however, no consensus exists regarding the optimal dose, duration or components of post-ACLR rehabilitation [83]. Second, because few RCTs assessed postoperative knee muscle strength, observational studies were included to support the network meta-analysis. Although methods such as sample matching were used to reduce confounding, the GRADE certainty of evidence remained very low, highlighting the need for more high-quality RCTs. Third, most trials compared BPTB and 4S-HT autografts, leaving other intervention groups with small samples and many absent direct comparisons. Consequently, closed-loop networks were incomplete, limiting statistical assessment of consistency among indirect comparisons [84]. Several effect estimates also had wide confidence intervals, indicating low precision and possible variability in true effects. These limitations mean that the results should be interpreted cautiously, and further well-designed

head-to-head RCTs with standardized rehabilitation protocols are warranted.

Conclusions

Overall, integrating findings from the pooled and subgroup analyses, this network meta-analysis suggests that among commonly used autografts for ACLR, the 4S-HT autograft was associated with greater knee extensor strength LSI, whereas the QT autograft had the most consistent advantage in knee flexor strength LSI. The 4S-ST autograft ranked between these grafts, but may provide greater knee flexor strength LSI than the 4S-HT autograft. Additionally, the BPTB autograft was associated with greater knee extensor strength LSI than the QT autograft. However, given the very low to low certainty of the available evidence, these findings should be interpreted with caution, and further high-quality, large-scale randomized controlled trials are therefore warranted to validate these results and guide optimal graft selection in clinical practice.

Abbreviations

ACLR	Anterior cruciate ligament reconstruction
LSI	Limb symmetry index
RTS	Return to sport
RCT	Randomized controlled trial
BPTB	Bone-patella tendon-bone
PT	Patella tendon
4S-ST	Four-strand semitendinosus tendon
5S-ST	Five-strand semitendinosus tendon
4S-HT	Four-strand hamstring tendon
QT	Quadriceps tendon
H/Q	Hamstring-to-quadriceps
RoB	Risk of bias
GRADE	Grading of Recommendations Assessment, Development and Evaluation
SUCRA	Surface under the cumulative ranking curve
MD	Mean difference
CI	Confidence interval
PRISMA-NMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses-Network Meta-Analyses
AMSTAR	Assessing the Methodological Quality of Systematic Reviews
RoBANS	Risk of Bias Assessment Tool for Non-randomized Studies
IKDC	International Knee Documentation Committee
PRISM	Pediatric Research in Sports Medicine

Supplementary Information

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Additional file 1.

Additional file 2.

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Author Contributions

C.L. and W.L. Wrote original draft. W.X. and Y.L. conceptualized this article and revised the draft. Z.H. and Q.B. suggested revisions. C.L., W.L. and X.L. undertook the literature review and extracted the data. C.L., W.L. and D.Y. coded the statistical analysis, figures, and appendix. Quality assessment of included studies by

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Data Availability

All data relevant to the study are included in the article or uploaded as supplementary information.

Declarations

Ethics Approval and Consent to Participate

Not applicable.

Informed Consent

Not applicable.

Equity, Diversity and Inclusion Statement

Both randomized controlled trials (RCTs) and observational studies were eligible, without restriction by country or region, thereby encompassing diverse racial and ethnic groups. Our author team includes senior and junior scholars from Asia. Currently, our team is all male, which reflects the existing composition rather than deliberate exclusion. The study did not examine the impact of race, ethnicity, or socioeconomic status on outcomes.

Provenance and Peer Review

Not commissioned, externally peer reviewed.

Competing Interests

The authors declare that they have no competing interests.

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The effectiveness of quadrupled semitendinosus graft technique in anterior cruciate ligament reconstruction: A network meta-analysis assessing various graft preparation techniques

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Abstract

Purpose: This study aims to perform a network meta-analysis of hamstring graft preparation techniques to enhance anterior cruciate ligament (ACL) reconstruction guidelines and inform clinical decision-making in patients with primary ACL rupture.

Methods: A review of the literature, from 1 January 1990, to 31 August 2023, was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines, focusing on the clinical outcomes of various hamstring graft preparation techniques. Forty-six studies (over 4800 knees) were analysed. Eight graft compositions: doubled hamstring (ST/G)—1978 grafts, doubled hamstring with augmentation (ST/G+A)—586 grafts, tripled semitendinosus (3ST)—124 grafts, quadrupled semitendinosus (4ST)—1273 grafts, five-strand tripled semitendinosus + doubled gracilis (3ST/2GR)—839 grafts, six-strand tripled semitendinosus + tripled gracilis (3ST/3GR)—335 grafts, seven-strand quadrupled semitendinosus + tripled gracilis (4ST/3GR)—11 grafts and ≥eight strands—24 grafts were compared, considering graft sizes, laxity, muscle strength, range-of-motion, patient-reported outcome measures (PROMs), return to sport (RTS) and adverse events.

Results: The 4ST grafts fared better than the ST/G grafts in the International Knee Documentation Committee (IKDC), Knee Injury and Osteoarthritis Outcome Score (KOOS) Sport, KOOS Q and RTS (mean difference [MD], −1.69; $p = 0.0159$; MD, −1.55; $p = 0.0325$; MD, −1.93; $p = 0.001$; odds ratio: 3.13; $p < 0.0001$). The IKDC differed significantly between the 4ST and ST/G+A groups (MD, 1.88; $p = 0.046$). The ST/G+A resulted in the lowest knee laxity, surpassing the ST/G, 3ST and 4ST. The ST/G had the smallest diameter (ST/G vs. ST/G+A: MD, 1.26; 95% confidence interval [CI]: 0.67–1.86, $p < 0.0001$). Reduced failure rates were noted with the 3ST/2GR (3ST/2GR vs. ST/G: MD, 6.93; $p = 0.009$) and 3ST/3GR (3ST/3GR vs. ST/G: MD, 53.64; $p = 0.006$).

Abbreviations: 3ST/2GR, five-strands graft: tripled semitendinosus + doubled gracilis; 3ST/3GR, six-strands graft: tripled semitendinosus + tripled gracilis; 3ST, three-strands graft: tripled semitendinosus; 4ST/3GR, seven-strands graft: quadrupled semitendinosus + tripled gracilis; 4ST, four-strands graft: quadrupled semitendinosus; 8ST/GR, eight-strands graft and more; ACL, anterior cruciate ligament; ACLR, anterior cruciate ligament reconstruction; PROMs, patient-reported outcome measures; RTS, return to sport; ST/G+A, four-strands graft with augmentation: doubled semitendinosus + doubled gracilis with augmentation; ST/G, four-strands graft: doubled semitendinosus + doubled gracilis.

Conclusion: The ideal hamstring graft for ACLR should be individualized. A 4ST graft is likely to yield good PROMs. For high stability and rapid RTS, adding augmentation to the graft is advisable. The ST/G is the thinnest graft possible.

Level of Evidence: Network meta-analysis of level I–III studies.

KEYWORDS

ACL, ACLR, anterior cruciate ligament, graft with augmentation, hamstring graft, quadrupled semitendinosus graft

INTRODUCTION

The doubled semitendinosus and gracilis tendons (ST/G) and quadrupled semitendinosus tendons (4ST) are the most common graft choices in anterior cruciate ligament (ACL) reconstruction (ACLR) [11]. The need for gracilis tendon harvesting is unclear, as numerous reports have indicated use of this tendon does not significantly improve knee stability and affects muscle strength [12, 17, 30, 54]. Moreover, whether sex has any significant impact on treatment outcomes or on the choice of graft is uncertain [30]. ACLR with the hamstrings augmented with allograft or suture tape to provide additional hamstring support (ST/G+A) has also been proposed [1, 42, 64, 67]. Furthermore, when opting to use both hamstring tendons, the configuration most beneficial to the patient is unclear [12, 16, 17, 21, 30, 37, 55, 56].

This study compared different types of hamstrings used in primary ACLR. Patient-reported outcome measures (PROMs); the International Knee Documentation Committee (IKDC), Lysholm, Tegner, Knee Injury and Osteoarthritis Outcome Score [KOOS]), return to sport (RTS) and instability tests, such as the Lachman test, pivot-shift test, anterior-posterior (AP) laxity, AP laxity >3 mm, muscle strength (flexion force, extension force), graft diameter and failure rates were evaluated with a view to providing a guide for selecting a specific hamstring graft.

The study's hypothesis indicates that there is no single optimal hamstring graft type, but by making both direct and indirect comparisons, the most suitable graft could be chosen based on individual patient's needs and expectations.

METHODS

Study selection

Using a common search strategy, three independent reviewers (anonymized) performed the literature search and independently extracted data, in a manner following the 2009 Preferred Reporting Items for

Systematic Reviews and Meta-Analyses (PRISMA) statement [39]. All discrepancies were reviewed by a fourth author (anonymized) and were discussed to consensus amongst the authors. Titles and abstracts of the identified reports were reviewed. Potentially eligible articles were then analysed by full-text review. All analyses were performed with guidance published by Karlsson et al. [46]. The reference lists of the identified investigations were manually searched to identify further reports. The study protocol was registered in PROSPERO under the number: (357171) before starting the study [53].

Search strategy

Using a comprehensive search strategy, the PubMed, EMBASE, clinicaltrials.gov and Cochrane Library databases were systematically searched for papers published from 1 January 1990 to 31 August 2023. The following search algorithm was used: (semitendinosus OR gracilis OR hamstring) AND (ACL OR ACLR or Anterior Cruciate).

Eligibility criteria

Randomized controlled trials (RCT) and non-RCTs comparing different types of hamstring autografts used in ACLR surgery for patients with primary ACL rupture that were published in peer-reviewed journals were included in this network meta-analysis. The inclusion of both RCTs and non-RCTs aimed to facilitate subgroup analysis (RCT vs. non-RCT) and to broaden the evidence base. This analysis included 5170 grafts involving eight different graft compositions: doubled hamstring (ST/G; 1978 grafts), doubled hamstring with augmentation (ST/G+A; 586 grafts), tripled semitendinosus (3ST; 124 grafts), quadrupled semitendinosus (4ST; 1273 grafts), five-strand tripled semitendinosus + doubled gracilis (3ST/2GR; 839 grafts), six-strand tripled semitendinosus + tripled gracilis (3ST/3GR; 335 grafts), seven-strand quadrupled semitendinosus + tripled gracilis (4ST/3GR; 11 grafts)

TABLE 1 Hamstring graft types analysed.

Abbreviation	Graft composition
ST/G	Four strands: doubled semitendinosus + doubled gracilis
ST/G+A	Four strands with augmentation: doubled semitendinosus + doubled gracilis with augmentation
3ST	Three strands: tripled semitendinosus
4ST	Four strands: quadrupled semitendinosus
3ST/2GR	Five strands: tripled semitendinosus + doubled gracilis
3ST/3GR	Six strands: tripled semitendinosus + tripled gracilis
4ST/3GR	Seven strands: quadrupled semitendinosus + tripled gracilis
8ST/GR	Eight strands and more

and >eight-strand grafts (24 grafts) (Table 1). There were no restrictions on language or patient age. Exclusion criteria included studies of ACL grafts other than hamstring (bone to bone graft, quadriceps femoris graft, etc.), concomitant surgical procedures (osteotomies, multiligament reconstruction) and animal or cadaveric studies.

Data extraction

The following data were extracted from each paper by two independent reviewers using a predetermined data sheet: authors, year of publication, type of intervention, details of interventions used in the study, sample size (randomized and analysed), outcome measurements, follow-up period, main results, failures, percentage and type of adverse events included in the publication and level of evidence.

Level of evidence and risk of bias assessment

The Oxford Centre for Evidence-Based Medicine Levels of Evidence Working Group criteria were used to evaluate the level of evidence. The Cochrane Collaboration Risk-of-bias revised tool (RoB 2) was used to assess the risk-of-bias and methodological quality of evidence of RCTs [38, 58]. The following domains were evaluated and assessed for risk of bias (low, unclear or high): bias domain and signalling question, bias due to deviations from intended interventions, bias due to missing outcome data, bias in the measurement of the outcome, bias in selection of the reported result and

overall bias. Nonrandomized trials were assessed using the ROBINS-I tool for bias. The assessment covered biases due to confounding, in participant selection, in intervention classification, due to deviations from intended interventions, due to missing data, in the measurement of outcomes and in the selection of reported results [20]. Overall bias was categorized as of low, moderate, high or critical risk. Sensitivity analyses were performed to evaluate the impact of biased or large studies on model fit, relative effects and treatment rankings.

Outcome measures

Outcome measures collected included PROMs (IKDC, Lysholm score, Tegner activity scale, KOOS), RTS and instability tests, such as the Lachman test, pivot-shift test, AP laxity and AP laxity >3 mm. Additionally, muscle strength (flexion force, extension force), graft diameter and failures were measured. All collected data were analysed with respect to the following interventions described in Table 1.

Statistical analysis

Qualitative statistical analysis and network meta-analysis were performed using R™ software 4.3.1 (R Foundation for Statistical Computing) [47]. To obtain a single pooled effect from a large number of underpowered studies, both RCTs and non-RCTs were analysed, even though this increased heterogeneity. A frequentist approach to network meta-analysis with a random-effects or common-effect model (for studies with below 25% I^2 and nonsignificant heterogeneity) was used, employing the netmeta package version 2.8-2. Effect sizes were reported as mean differences (MD, for continuous outcomes) or odds ratios (OR, for dichotomous outcomes) with 95% confidence intervals (CIs). The league tables and network plots were generated with the dplyr package version 1.1.2. Publication bias was analysed using a comparison-adjusted Funnel plot, Egger's multivariate test method and Rosenthal's file drawer method, as well as relative risk (RR) for Publication Bias Calculation with the meta and metofo packages, versions 6.5.0 and 4.4.0, respectively.

RESULTS

The initial literature search identified 11,845 potential articles. Of these, 7471 were extracted after screening the titles and abstracts. The full text of 348 studies was retained for further screening according to the inclusion and exclusion criteria. Finally, 46 articles met

the inclusion criteria and were analysed in this network meta-analysis (Figure 1).

Study characteristics

Overall, 4863 patients were included in 46 studies. The number of studies compared alongside every subgroup and the main characteristics of each article are presented in Table 2.

In terms of the outcomes of interest, IKDC scores were reported in 25 studies, Lysholm scores in 25, Tegner activity scale scores in 14, KOOS scores in 11, RTS in eight, Lachman test results in 12, pivot-shift test results in 17, overall AP laxity in 19, AP laxity >3 mm in

14, flexion force in 17, extension force in 15, graft diameter in 24 and failures in 21 studies. Each of these outcomes was analysed using a forest plot diagram and network geometry, comparing repair using a double hamstring approach to the other subgroups. All correlations are presented in league tables (Supporting Information S1: Tables 1–17). No potential conflicts of interest were recorded amongst the authors of the studies included in the analysis.

PROMs

Thirty-two studies were included concerning patient-reported outcomes. Network comparison showed

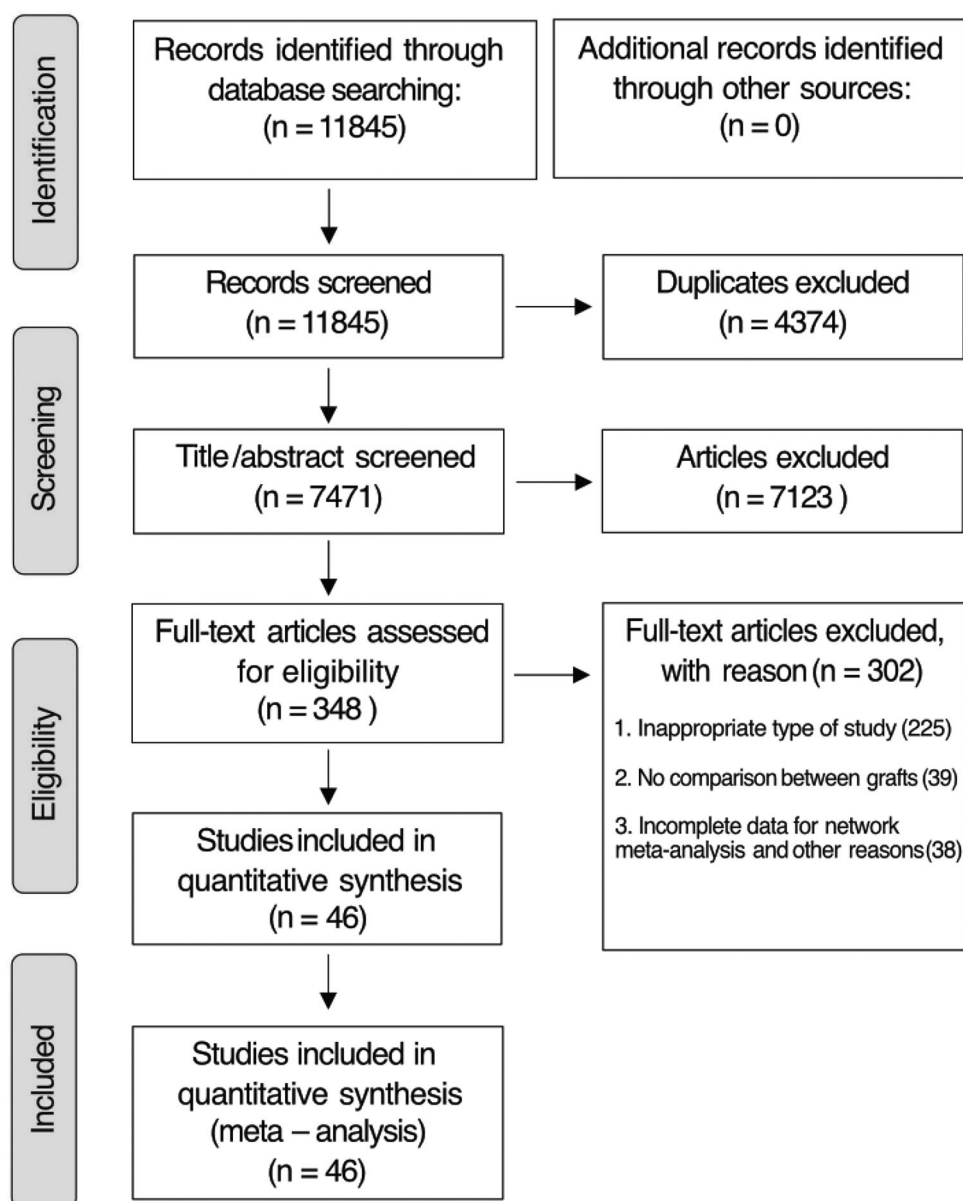


FIGURE 1 Summary of the search process following the PRISMA guidelines. PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

TABLE 2 Main characteristics of the studies included in meta-analysis.

No	Study ID	Study type	Level of evidence	Study period	Country	Type of tendon graft	Number of patients	Patient age (years)	Sex (male%)	Time to surgery	Follow-up period	Outcomes
1	Adams et al. [1]	Retrospective cohort study	III	01.2007–12.2017	USA	ST/G versus ST/G+A	112	29.2 and 28.8	85	Not mentioned	65.3 months	Failure rates, KOOS
2	Alkhalaf et al. [2]	Case series	IV	2012–2018	Kuwait	3–10 stands ST/G versus ST/G+A	711	28.72	Not mentioned	Not mentioned	2 years	Graft diameters, failure rates
3	Ardern et al. [3]	Retrospective cohort study	III	06.2004–06.2007	Australia	4ST versus ST/G	50	26.1 ± 6.9 and 24.4 ± 4.9	70	Not mentioned	31.4 months and 33.5 months	Active and passive knee range-of-motion and anterior knee laxity (KT-1000); IKDC, Isokinetic hamstring strength and isometric hamstring strength
4	Attia et al. [4]	Cohort study	III	01.2011–11.2017	Qatar	4ST versus 3ST2GR versus 3ST/3GR	413	31.54	95	Not mentioned	44.27 months	Failure rates
5	Aujla et al. [5]	Comparative cohort study	III	04.2014–12.2017	Australia	ST/G versus ST/G+A	196	26.80	67	14.8 ± 21.9 weeks 14.2 ± 31.2 weeks	2 years	IKDC, KOOS, Lysholm, Tegner and the NSARS; measures: knee laxity, maximal isokinetic knee flexion and extension strength, four functional hop tests
6	Berenius et al. [7]	Retrospective cohort study	III	Not mentioned	Australia	4ST versus ST/G	20	26 (9) and 26 (7)	50	36 (27–45) months	37 (6) and 36 (4) months	IKDC, RTS, knee range-of-motion, Lachman, pivot-shift, KT-1000; test of hamstring strength, tibial rotation
7	Burrus et al. [9]	Retrospective comparative study	III	01.2007–12.2014	USA	ST/G versus ST/G+A	58	26.6 ± 10.3 and 27.2 ± 9.6	34	3.7 ± 5.6 and 5.2 ± 6.9 months	44.4 ± 16.9 months and 48.0 ± 15.2 months	Failure rates, Lysholm, IKDC
8	Calvo et al. [10]	Retrospective comparative study.	III	01.2012–12.2013	Chile	4ST versus 3ST/2GR	152	13–53	Not mentioned	<3 months	30 months	Reoperation rate, graft diameter, Lysholm, IKDC, anterior laxity measured with a KT-1000

(Continues)

TABLE 2 (Continued)

No	Study ID	Study type	Level of evidence	Study period	Country	Type of tendon graft	Number of patients	Patient age (years)	Sex (male%)	Time to surgery	Follow-up period	Outcomes
9	De Geofroy et al. [13]	Single centre retrospective comparative study	III	2012–2019	France	4ST versus ST/G	133	28.5 ± 6.23 and 29.8 ± 6.46	97	221 ± 437 and 280 ± 336 days	4 years	Return to duty, isokinetic results, sports results (Cooper, walk-run); X-ray analysis, complications and failure rates
10	Demirag et al. [14]	Prospective cohort study	III	11.2005–03.2008	Turkey	ST/G versus ST/G+A	40	30 (19–40)	90	Not mentioned	24.3 months	IKDC, Lysholm, instability tests, patient satisfaction questionnaires and incidences of cyclops lesions and arthrofibrosis; tibial and femoral tunnel widening
11	Drocco et al. [16]	Case-control study	III	2012–2014	Italy	3ST versus ST/G	90	29.5 ± 10.0, 27.7 ± 9.0	44	Not mentioned	12–30 months	IKDC, Tegner, Lysholm, in antero-posterior laxity (Arthrometer KT-1000), the one-leg-hop test, isokinetic test, RTS
12	Gobbi et al. [17]	Randomized controlled trial	I	01.1996–12.2001	Italy	4ST versus ST/G	97	28, 80	57	Not mentioned	36 months (24–52)	Lysholm, Tegner, IKDC, clinical examination, laxity, isokinetic testing and single-leg hop and jump tests
13	Heyworth et al. [19]	Cohort study	III	01.2003–12.2017	USA	ST/G versus ST/G+A	111	15.8 ± 1.9 and 16.1 ± 1.5	20	2.2 [1.2–3.1] and 1.4 [1.1–2.1] months	2.5 years	IKDC, Lysholm, Tegner, pivot-shift test, Lachman test failure rates
14	Hu et al. [21]	Retrospective comparative study	III	01.2009–10.2016	USA	4ST versus ST/G	84	18–40	49	Not mentioned	3.09 ± 0.31 and 3.26 ± 0.23 years	Examination (effusion, joint line tenderness, McMurray, pivot-shift, range-of-motion); laxity test (KT-1000), strength test, functional tests (single-leg-hop test, single-leg-squat test); IKDC, Lysholm, KOOS S
15	Ignazzi et al. [22]	Retrospective cohort study	III	2013–2018	USA	4ST versus 3ST/2GR	74	31.8 ± 13.5/23.6 ± 11.0	54	Not mentioned	3.0 ± 1.5 years and 3.3 ± 1.3 years	graft diameter, revision rates, RTS
16	Jimenez et al. [23]	Research Article	III	2010–2016	USA	ST/G versus ST/G+A	46	31 (17–51) and 33.3 (15–48)	59	Not mentioned	31.5 months (range 12–63 months)	Failure rates, IKDC, KOOS, RTS

TABLE 2 (Continued)

No	Study ID	Study type	Level of evidence	Study period	Country	Type of tendon graft	Number of patients	Patient age (years)	Sex (male%)	Time to surgery	Follow-up period	Outcomes
17	Karimi-Mobarakeh et al. [24]	Randomized controlled trial	I	03.2011–03.2012	Iran	4ST versus ST/G	119	29.9 ± 7.8 and 32.4 ± 6.3	82.4	At least 1.5 months	2.41 ± 0.31 years in the ST/G group compared to 3.26 ± 0.23 years	Surgical complications: IKDC; KOOS; Lysholm; strength of the knee isometric flexors; and flexion and extension loss; laxity
18	Kehribar et al. [25]	Retrospective cohort study										
III	April 2019–March 2021	Turkey	ST/G versus 4ST	39	18–35	not mentioned	6 months	Lysholm, Tegner, IKDC, single-leg-hop test				
19	Kouloumentas et al. [27]	Randomized Clinical Study	I	2015–2016	Greece	4ST versus ST/G	90	28.7 ± 11.3	61	Not mentioned	24 months	IKDC, KOOS, KSS, Anterior tibial translation, isokinetic tests
20	Krishna et al. [28]	Cohort study	II	12.2014–06.2016	Singapore	ST/G versus 3ST/2GR	45	25.6 ± 7.3 and 29.2 ± 8.0	79	Not mentioned	At least 1 year	Lysholm, KOOS symptoms, KOOS pain, KOOS ADLs, KOOS Sports, KOOS quality of life and SF-36 PCS
21	Królikowska et al. [29]	Retrospective comparative study	III		Poland	4ST versus ST/G	40	27.00	100	3–6 months	24 weeks	Isometric torque
22	Kulinski et al. [30]	Randomized controlled trial	I	2015–2017	Poland	4ST versus ST/G	140	31 (18–45)	78.5	8.1 (1.5–209.0) and 7.0 (1.5–67.4) months	4.5 years	VAS, IKDC, KOOS, Tegner, Lysholm, ATT (KT-1000)
23	Kumar et al. [31]	Randomized controlled trial	I	Not mentioned	India	4ST versus ST/G	60	28.50	93	7.6 months and 6.46 months	Not mentioned	Anterior drawer test, Lachman, pivot-shift, mean anterior tibial translation measured in 90° of knee flexion, Lysholm
24	Kyung et al. [32]	Retrospective comparative study	IV	2004–2010	Korea	4ST versus ST/G	144	28.0 ± 10.6 and 31.3 ± 12.3	100	5.8 ± 8.1 and 6.0 ± 7.5 months	26.4 and 27.5 months	Lachman, pivot-shift and one-leg hop tests; an isokinetic test; KT-2000 arthrometer; assessment of return to preinjury activities; Lysholm, Tegner, IKDC

(Continues)

TABLE 2 (Continued)

No	Study ID	Study type	Level of evidence	Study period	Country	Type of tendon graft	Number of patients	Patient age (years)	Sex (male%)	Time to surgery	Follow-up period	Outcomes
25	Lee et al. [33]	Retrospective comparative study	III	06.2009–03.2014	Korea	4ST versus ST/G	120	27.4 ± 6.6 and 26.9 ± 7.3	93	2.7 ± 1.7 months and 2.9 ± 1.5 months	37.5 ± 15.9 and 36.8 ± 6.1 months	Lysholm, IKDC SKF, Postoperative Tegner, side-to-side difference on KT-2000 (mm), graft diameter
26	Mahiroglu et al. [34]	Prospective comparative study										
III	2007–2009	Turkey	4ST versus ST/G	98	31.14 ± 7.80 and 28.69 ± 8.07	Not mentioned	26.04 ± 2.53 and 25.93 ± 2.66	Lysholm, Tegner, IKDC				
27	Manchado et al. [35]	Retrospective observational study	III	2013	Spain	4ST versus ST/G	44	26.8 ± 8.5 and 25.8 ± 8.9	100	Not mentioned	5 years	STSD, Lysholm, IKDC, LSI, HI/Q ratio strength
28	Nazari et al. [37]	Prospective cohort study	III	07.2018–03.2020	United Arab Emirates	3ST, 2ST/G, 3ST/2GR, 3ST/3GR	78	33.7 ± 5.9	100	Not mentioned	Not mentioned	Preoperative anthropometric data, graft diameter
29	Parkes et al. [40]	Retrospective comparative study	III	07.2011–07.2017	USA	ST/G versus ST/G+A	108	25.3 (13–44); 24.9 (13–54)	69	7.6 ± 16.0 (0.2–76.2); 7.1 ± 18.6 (0.3–124.4) months	2 years	RTS, range-of-motion, Tegner, Lysholm, IKDC, Lachman, pivot-shift, presence of effusion
30	Pennock et al. [41]	Cohort study	III	2003–2014	USA	ST/G versus ST/G+A	50	15.7 (12–18)	38	Not mentioned	2 years	Failure rates, need for second surgery, contralateral ACL tear, SANE score, patient satisfaction, VAS, Tegner, Lysholm, RTS
31	Perkins et al. [42]	Cohort study	III	08.2012–04.2016	USA	ST/G, 3ST/2GR, 3ST/3GR, ST/G+A	354	15.3 (10–19)	44	Not mentioned	26 months	Graft diameter, failure rates
32	Plawski et al. [44]	Cohort study	III	03.2012–07.2013	France	4ST versus ST/G	87	25.2 (18–45)	55	6.6 ± 4 and 5.8 ± 4 months	3.2 ± 0.2 years	IKDC, KT-1000 arthrometry, radiologic evaluation of the differential laxity, Tegner, KOOS

TABLE 2 (Continued)

No	Study ID	Study type	Level of evidence	Study period	Country	Type of tendon graft	Number of patients	Patient age (years)	Sex (male%)	Time to surgery	Follow-up period	Outcomes
33	Powell et al. [45]	Retrospective comparative study	III	2017–2021	USA	4ST versus ST/G and ST/G+A	191	18 (17–31.5) and 26 (18–38.25)	48, 35	Not mentioned	6 (3–7) and 4 (2–6) months	Graft diameter
34	Roger et al. [48]	Randomized controlled trial	I	08.2017–08.2018	France	4ST versus ST/G	60	30.5 ± 8.9 and 30.3 ± 8.5	82	1–1/2 months	2 years	IKDC, isokinetic muscle strength recovery (side-to-side differential laxity, balance, loss of flexion/extension), return to work, pain during physical activities, surgical complications
35	Ruffilli et al. [49]	Retrospective comparative study	III	1998–2000	Italy	4ST versus ST/G	51	29.2 ± 3.8	92	Not mentioned	12.1 ± 1.6 years	IKDC, KOOS, Tegner, arthrometric analysis (KT-1000), radiographic evaluation
36	Segawa et al. [54]	Prospective cohort	II	12.1995–12.1998	Japan	3ST versus 4ST versus ST/G	62	20, 80	55	10.2 months	12 months	The peak torque of knee extension and flexion, Lysholm, anteroposterior stability (KT-1000), anteroposterior and lateral radiographs of the knee in maximal extension
37	Sideris et al. [57]	Cohort study	II	12 months	Australia	ST/G versus 3ST/2GR	44	30.0 ± 8.4 and 25.1 ± 4.1	63	16.0 ± 27.8 versus 5.7 ± 3.9 weeks	12 weeks	Graft diameter, flexion/extension in the gait cycle, adduction/abduction in the gait cycle, axial rotation in the gait cycle, relationships between graft size and rotational stability
38	Tashiro et al. [59]	Prospective randomized study	I	03.1996–11.1996	Japan	4ST versus ST/G	85	24.5 ± 7.7 and 24.8 ± 6.4	57	Not mentioned	24 months	IKDC, knee stability (KT-1000), activity level, bilateral thigh muscle strength (Cybex II dynamometer), quadriceps muscle strength, hamstring muscle strength (isokinetic peak torque measurements)
39	Tutkus et al. [61]	Case-control study	III	2014–2015	Lithuania	3ST/3GR versus 3ST/2GR	122	31.57 ± 8.43, 30.29 ± 7.80, 32.09 ± 9.55, 31.14 ± 10.89	74	Not mentioned	Not mentioned	Preoperative anthropometric data, graft diameter

(Continues)

TABLE 2 (Continued)

No	Study ID	Study type	Level of evidence	Study period	Country	Type of tendon graft	Number of patients	Patient age (years)	Sex (male%)	Time to surgery	Follow-up period	Outcomes
40	Walczak et al. [62]	Prospective paired cohort study	III	Not mentioned	USA	ST/G versus 3ST/2GR	46	28, 90	56.5	Not mentioned	Not mentioned	Graft diameter
41	Wan et al. [63]	Prospective cohort study	II	01.2019–06.2021	China	ST/G versus 3ST/2GR	45	18–45	66	Not mentioned	17.5 months	Lysholm, KOOS Symptoms, Pain, ADLs, Sports and Quality of Life; Lachmann, pivot-shift
42	Wang et al. [64]	Retrospective comparative study	III	07.2013–07.2014	China	ST/G versus ST/G+A	57	32.8±9.9; 32.6±10.3	33	1.3±0.8; 1.4±0.8 months	3 years	IKDC, pivot-shift, Lachman, KT-1000 side-to-side differences Tegner, Lysholm, failure rates, physical examinations, magnetic resonance imaging, second-look arthroscopy
43	Yosmaoglu et al. [65]	Prospective cohort	II	Not mentioned	Turkey	4ST versus GST/G	46	29±7 and 28±9	Not mentioned	Not mentioned	12 months	Isokinetic torques of knee extensors and knee flexors, eccentric coordination, concentric coordination, anterior tibial laxity deficits
43	Zaid et al. [66]	Prospective study	II	01.2018–12.2019	China	4ST/ST/G+A	158	28.00	70	25 months	3, 6, 12, 24 months	Lysholm, IKDC, KOOS-ADLS, Tegner, ACL-RSI, NSARS, GRC scale, RTS, arthroscopic outcomes, graft signal intensity, postoperative complications or rerupture rates, objective outcomes included KT-1000 to evaluate graft stiffness and knee stability, Lachman, pivot-shift, knee flexion and extension range-of-motion, the four hop tests, concentric isokinetic quadriceps and hamstring strength
45	Zaid et al. [67]	Randomized clinical study	I	01.2014–04.2023	China	ST/G versus ST/G+A (LARS and	120	26.4±3.5 and 26.1±4.0 and 25.8±4.6	69	25.1±3.2 and 24.8±2.9 and 24.5±3.0	8 years	IKDC, KOOS, Lysholm Tegner, ACL-RSI; Knee Outcome Survey ADL Subscale, NSARS; GRC; KT-1000 to evaluate knee

TABLE 2 (Continued)

No	Study ID	Study type	Level of evidence	Study period	Country	Type of tendon graft	Number of patients	Patient age (years)	Sex (male%)	Time to surgery	Follow-up period	Outcomes
						meshwork of LARS						stability and graft stiffness, pivot-shift, Lachman, knee flexion and extension range-of-motion, four-hop test and concentric isokinetic quadriceps and hamstring strength
46	Zhang et al. [68]	Prospective cohort study	III	02.2016–03.2018	China	ST/G versus ST/G+A	60	Not mentioned	Not mentioned	Not mentioned	18 months	Lysholm, Tegner, IKDC; Lachman, pivot-shift

Abbreviations: ACL, anterior cruciate ligament; ADLs, activities of daily living; GCR, Global Change Rating; IKDC, International Knee Documentation Committee; KOOS, Knee Injury and Osteoarthritis Outcome Score; NSARS, Noyes Sports Activity Rating Scale; RSI, return to sports after injury; RTS, return to sport; SANE, Single Assessment Numeric Evaluation; VAS, visual analogue scale.

significant differences in IKDC, KOOS Q and Sport and RTS. No significant differences in Lysholm scores and KOOS Pain, Symptoms or ADLs were observed. Use of the quadrupled semitendinosus (4ST) approach was superior to that of the ST/G approach in terms of IKDC ($p = 0.0159$), KOOS Sport ($p = 0.0325$), KOOS Q ($p = 0.0012$) and RTS ($p < 0.0001$). The 4ST approach demonstrated significantly better IKDC ($p = 0.046$) and RTS ($p < 0.0001$) results than did the ST/G+A approach. However, the ST/G+A approach yielded significantly better Tegner results than did the ST/G ($p = 0.0014$) and 4ST ($p = 0.0002$) approaches. RTS was significantly faster with the ST/G+A than with the ST/G ($p < 0.0001$) approach (Figure 2).

Laxity

Twenty-six studies reported the results of four different knee instability tests: AP laxity, AP laxity > 3 mm, Lachman test and pivot-shift test. The ST/G+A resulted in the least AP laxity. This approach resulted in significantly better AP > 3 mm and Lachman test results than that of the three other main groups: ST/G, 3ST and 4ST. Additionally, it had a significantly lower prevalence of pivot-shift positive knees than did the 4ST ($p < 0.0001$). The ST/G was more stable than the 3ST and 4ST in terms of AP > 3 mm and the Lachman test. Additionally, in the pivot-shift test, ST/G showed fewer pivot-shift episodes than did the 4ST ($p < 0.0001$) (Figure 3).

Flexion and extension force

Muscle strength, measured by extension force, was reported in 10 studies and showed a significant difference in favour of 3ST compared to ST/G ($p = 0.0066$), ST/G+A ($p = 0.0090$) and 4ST ($p = 0.0085$). However, flexion force was better with 4ST than with ST/G ($p < 0.0001$) (Figure 4).

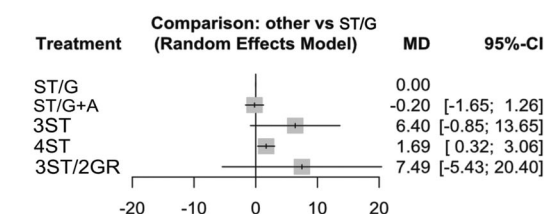
Graft diameter

The smallest graft diameter was assessed in ST/G compared to other groups. Significant differences were found in favour of ST/G+A, 3ST/2GR, 3ST/3GR, 4ST/3GR compared to both groups (ST/G vs. ST/G+A: $p < 0.0001$; ST/G vs. 3ST/2GR: $p = 0.0009$; ST/G vs. 3ST/3GR: $p = 0.0078$; ST/G vs. 4ST/3GR: $p = 0.0442$). Grafts with augment were bigger than 4ST ($p = 0.0139$) (Figure 5).

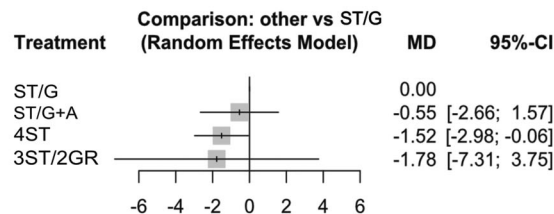
Failures

Interestingly, only the 3ST/2GR and 3ST/3GR groups showed improvements in failures. 3ST/2GR showed

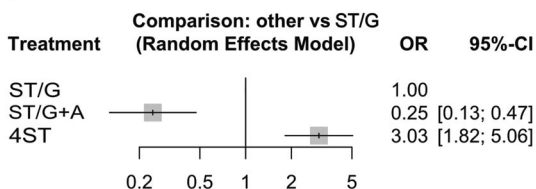
(a) IKDC



(b) KOOS Sport



(c) RTS



(d) Tegner activity score

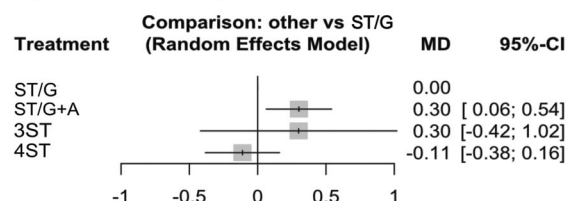
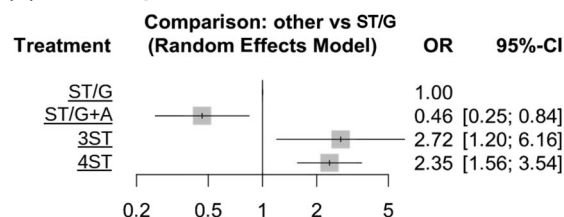


FIGURE 2 Forest plot of International Knee Documentation Committee (IKDC) score (a); Knee Injury and Osteoarthritis Outcome Study (KOOS) Sport (b); return to sport (RTS) (c) and Tegner activity score (d). 95% CI, 95% confidence interval; MD, mean difference; OR, odds ratio.

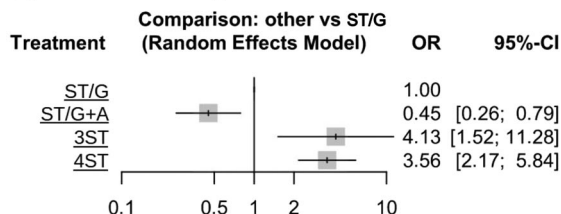
(a) AP laxity (mm)



(b) AP laxity >3mm



(c) Lachmann sign



(d) Pivot-shift

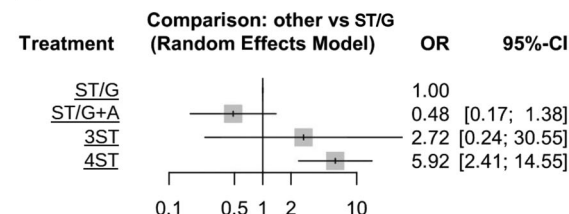
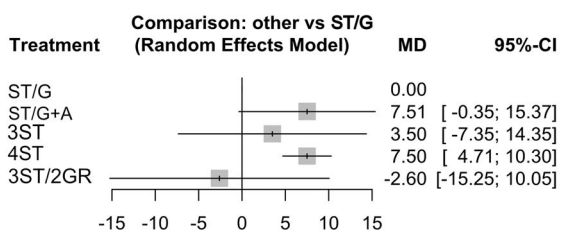


FIGURE 3 Forest plot of AP laxity (mm) (a); AP laxity >3 mm (b); Lachman sign (c) and pivot-shift (d). AP, antero-posterior; 95% CI, 95% confidence interval; OR, odds ratio.

(a) Flexion force



(b) Extension force

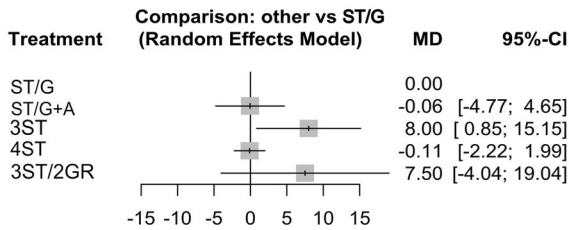


FIGURE 4 Forest plot of flexion force (a) and extension force (b). 95% CI, 95% confidence interval; MD, mean difference.

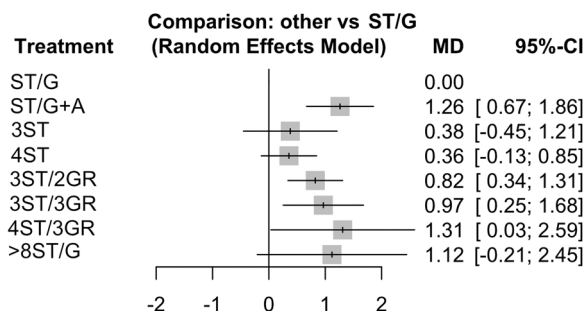


FIGURE 5 Forest plot of graft diameter. 95%-CI, 95% confidence interval; MD, mean difference.

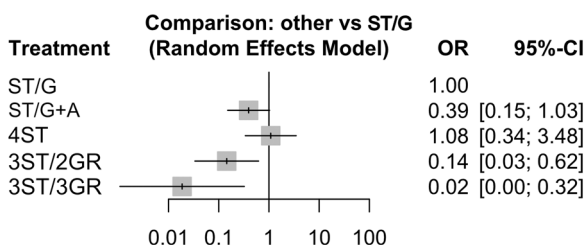


FIGURE 6 Forest plot of failures. 95% CI, 95% confidence interval; OR, odds ratio.

significant improvements as compared to ST/G and 4ST ($p = 0.009$ and $p = 0.005$, respectively). 3ST/3GR also showed significant advantages over ST/G, ST/G +A and 4ST ($p = 0.006$; 3ST/3GR vs. ST/G+A: $p = 0.044$; 3ST/3GR vs. 4ST: $p = 0.007$) in terms of failures (Figure 6).

Risk-of-bias

Of the 46 articles included in this study, eight were randomized trials. They were assessed for risk-of-bias using the ROB2 tool. Two of these articles were categorized as low, four as moderate and two as high risk-of-bias. The remaining 38 articles were non-RCTs. Consequently, they were assessed for bias using the ROBINS-I tool. Ten of these were classified as low, 16 as moderate and 12 as high or very high risk-of-bias. Most of the studies showed potentially high biases in the initial process of randomization, biases due to confounding and unequal treatments in groups, although the final results were improved and presented significantly less high risk-of-bias. The risk-of-bias assessment of all included studies is illustrated in Figure 7 (RCTs) and Figure 8 (non-RCTs).

Sensitivity analysis

Sensitivity analyses were conducted to evaluate the impact of biased or large studies on model fit, relative

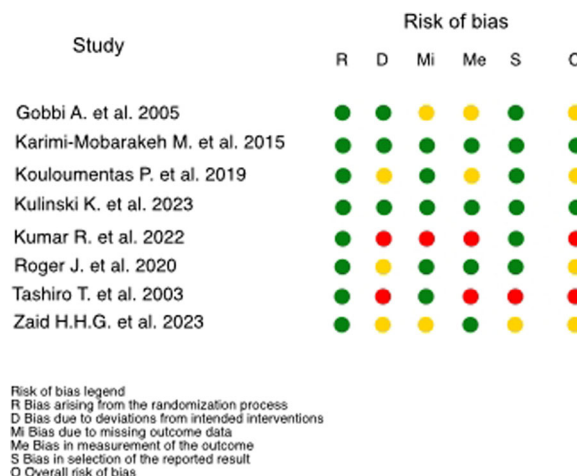


FIGURE 7 Risk-of-bias assessment for the included randomized studies [18, 24, 27, 30, 31, 48, 59, 67].

effects and treatment rankings. To ensure the robustness of the summarized effect sizes, random-effects subgroup analyses were performed for all assessments initially conducted using fixed-effect methods. In the second sensitivity analysis, 14 highly or critically biased studies were excluded from the strength network [2, 4, 5, 14, 16, 31, 32, 41, 59, 61–64, 68]. In the third sensitivity analysis, the largest studies were excluded from the strength network [4, 5, 37, 42, 66]. No significant differences were observed when transforming from fixed-effect to random-effects models. Additionally, upon excluding large or highly biased studies, the results remained grossly unchanged. The estimated failure rate in ST/G+A versus ST/G was significantly lower when large studies were excluded (OR: 0.28; 95% CI: 0.21–0.37, $p = 0.001$). The observed differences in graft diameter upon removal of large or highly biased studies were similar, although nodes with 4ST/3GR and 8ST/GR were removed from the analysis.

The sensitivity analysis supported the robustness of primary findings. The consistency across different models and the negligible impact of excluding large or biased studies indicated that this network meta-analysis provides reliable conclusions.

DISCUSSION

This study indicated that the selection of the appropriate graft for ACLR should be personalized, based on the patient's specific needs, expectations and physical demands. The results of this network meta-analysis confirmed the initial hypothesis that no single, universal, method of folding hamstring grafts is best for all patients undergoing primary ACLR. Nevertheless, the obtained results can be helpful in personalizing and facilitating hamstring graft selection in certain clinical

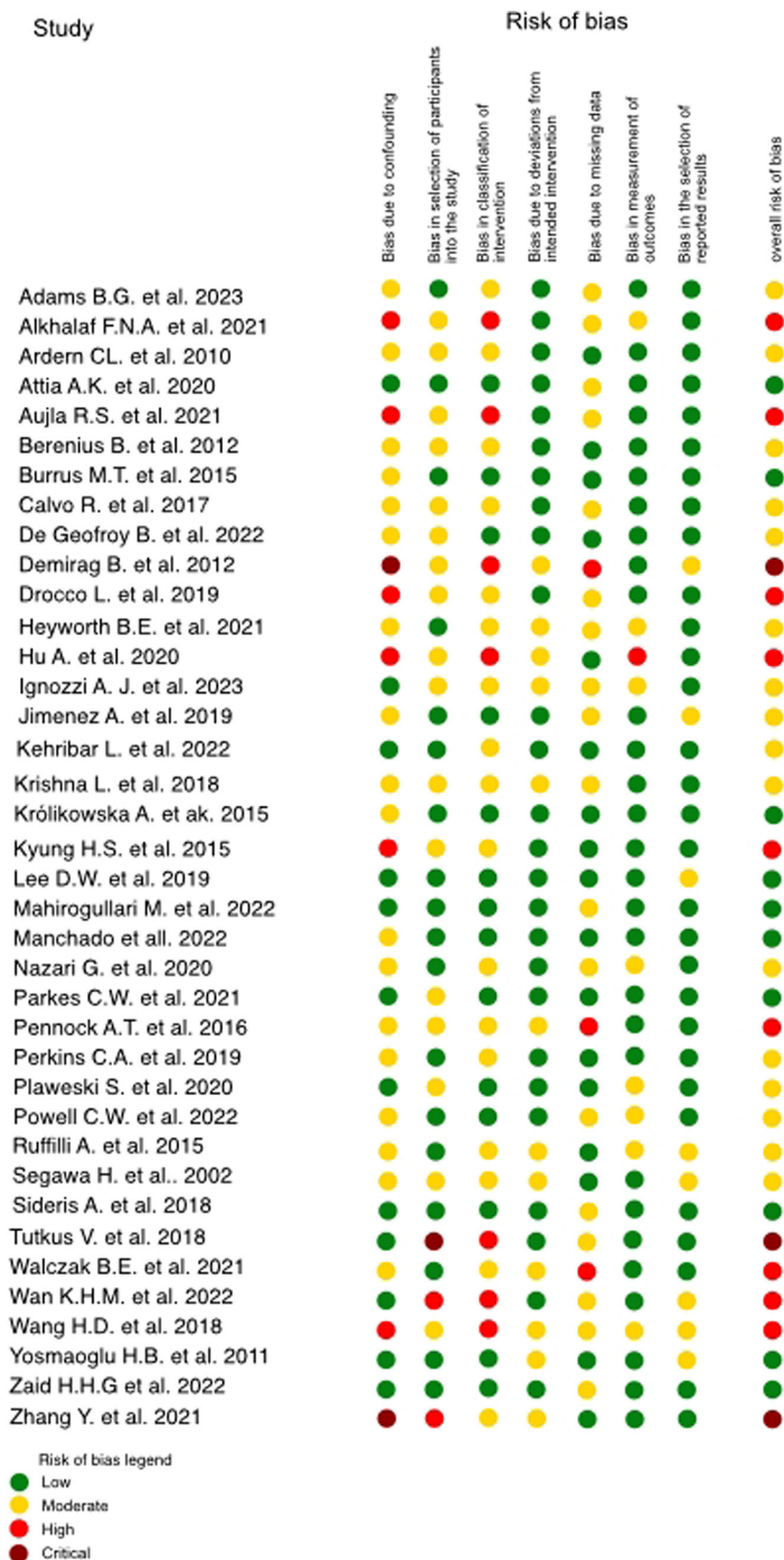


FIGURE 8 Risk-of-bias assessment for the included non-randomized studies [1–5, 7, 9, 10, 13, 14, 16, 19, 21–23, 25, 28, 29, 32–35, 37, 40–42, 44, 45, 49, 54, 57, 61–66, 68].

situations where specific treatment outcomes are desired.

To date, most studies have focused on comparing two specific methods of preparing hamstring grafts. One of the more frequently discussed aspects is the necessity of using the gracilis tendon for ACLR.

One significance found of this study is that doubled hamstring grafts, which are amongst the most commonly chosen hamstring grafts, showed no significant difference in terms of five outcome measures as compared to other hamstring graft preparations. ST/G demonstrated significant advantages as compared to other graft preparations in only four outcomes; more importantly, it showed significant disadvantages in 10 of the outcomes as compared to other groups.

This raises questions that were previously discussed about the need to sacrifice the gracilis tendon. Gobbi et al. found no significant differences between the ST/G and ST groups in terms of knee scores, self-evaluation, clinical findings, knee laxity or functional tests [18]. In a comprehensive analysis, Sharma et al. conducted a systematic review to compare the outcomes of ST grafts and combined ST/G grafts in ACLR. The review revealed no significant differences in PROMs, anterior laxity or isokinetic strength between ST and ST/G groups, particularly at speeds of 180°–240° per second. However, a decrease in hamstring isokinetic strength at 60° per second at flexion angles of 90°, 105° and 110° was noted in the ST/G group [56]. Chin et al. confirmed that, amongst ST and ST/G groups, PROMs and anterior laxity did not differ significantly. However, the ST/G group showed an increased side-to-side deficit in isokinetic peak torque [12].

On the other hand, Kulinski et al. demonstrated that laxity was increased and PROMs were worse amongst females with 4ST than in those with ST/G grafts. Most of the studies included in the present analysis did not provide data on female subgroups [30]. Thus, further well-designed, randomized studies comparing ST/G with ST with a particular focus on differences between the sexes are needed.

The present study expands on these findings by offering a more comprehensive analysis. Unlike previous meta-analyses that compared only ST to ST/G, this study distinguished amongst subtypes of the semitendinosus graft. Given the variability in preparation methods for the semitendinosus graft, such as three- or four-strand configurations, this study provided more detailed outcomes. It revealed statistically significant differences in PROMs and anterior laxity between ST/G and one or multiple groups of ST.

This research supports previous findings that suggested that graft choices should be tailored according to sex, age, activity level and sport [6, 26]. The 4ST presented significantly better PROMs than those of ST/G, but higher laxity than that of ST/G or ST/G+A. In

each of the abovementioned tests, 4ST presented significantly more instances of instability.

Conversely, ST/G+A demonstrated the most stability, but did not show benefits in terms of PROMs, except on the Tegner scale, which rates activity based on work and sport. Here, ST/G+A grafts scored the highest in terms of stability, suggesting better sports performance perceptions in patients. ST/G without augmentation also displayed more stability than did 4ST or 3ST, but fared worse in terms of IKDC, KOOS Sport and Q, and RTS as compared to 4ST, with RTS being significantly faster only than that of ST/G+A. 3ST showed significant improvements only in extension force as compared to ST/G, ST/G+A and 4ST.

These findings suggest that 4ST could well be the preferred option for patients with lower sports expectations who prioritize a pain-free knee for daily activities. ST/G+A might be suitable for active patients for whom returning to high-level sport is crucial. ACLR remains a suitable treatment, particularly for active patients of all age groups, given that an ACL-deficient knee can result in early onset of osteoarthritis, secondary meniscal tears and consequently an increased risk of requiring total knee arthroplasty [15, 43, 52, 60].

The analysis raises many questions. Why do grafts with higher RTS and Tegner scores not perform as well in daily PROMs? Should patients with augmented grafts be allowed to return-to-sport activities faster or would their belief that they can do 'more' increase their risk or is there high heterogeneity within this group?

Other factors, such as graft fixation, graft preparation or graft diameter and length, could have influenced the final outcomes [50, 51]. Graft length was not presented in the included studies, but the ST/G graft is usually longer because, in the ST/G, the ST is only doubled, rather than being folded three, four, five or more times. Length may account for the greater stability of the ST/G and ST/G+A than that of the shorter ST grafts. On the other hand, graft diameter could be related to the better outcome in PROMs. The ST/G is the thinnest graft considered in this meta-analysis and showed inferior outcomes in PROMs. It remains unclear whether differences in PROMs should be attributed to the additional harvest of the gracilis tendon or to the diameter of the graft [8, 36].

This network meta-analysis compared different types of hamstring grafts, which has not been reported previously. The advantage of a network meta-analysis is that it allows for the direct and indirect comparison of treatments using a common comparator and allows for them to be ranked using the *p* value.

However, the study had some limitations. The time frames and length of assessment differed between studies. ACLR surgical technique (graft fixation, tunnel placement, etc.) and postoperative rehabilitation programme may have been sources of heterogeneity. Additionally, the variety of scores, scales and laxity

assessments used in the included studies may have been a reason for high heterogeneity. Comparison of clinical trials with the same follow-up period and measured outcomes may have yielded more consistent findings. The most important outcomes were PROMs and instability assessments. Since the analysed results were derived from studies with mixed cohorts (both female and male), the findings may have been biased due to sex differences in available data. Fourteen studies were at high or critically high risk-of-bias, which could have influenced the outcomes. Too few RCTs included PROMs and outcomes, as illustrated by the network geometries in Figures S1–S4.

Well-designed prospective RCTs incorporating data on anatomical, biomechanical, graft dimensions, length, PROMs, stability, graft preparation and stabilization techniques in both sexes could help to determine the optimal hamstring graft type. This study should prompt clinicians to consider all possible options and consequences when choosing a graft before the ACLR surgery, with a focus on choosing the graft that suits the individual patient best.

CONCLUSION

The ideal choice of a hamstring graft for ACLR remains debated. When selecting a graft, various factors should be considered to ensure the best possible outcome for the patient [50, 51]. According to this network analysis, the 4ST graft offers a higher likelihood of patient satisfaction as measured by some PROMs, making it a strong candidate for those who prioritize overall satisfaction and function. For patients who place a high value on a rapid RTS and knee stability, the ST/G+A graft is advisable. This graft provides the support and strength needed for demanding activities, making it ideal for athletes and highly active individuals. In situations where anatomical or technical conditions necessitate the use of the thinnest possible graft (e.g. children), the ST/G is the recommended option, as this graft can accommodate the need for a slimmer profile without compromising on the effectiveness of the ACLR. Furthermore, it is worth noting that the use of 3ST/2GR or 3ST/3GR grafts statistically resulted in the lowest failure rates. This finding highlighted the reliability and durability of these graft combinations, making them a prudent choice for patients concerned about long-term success and the minimization of reoperation risks. Overall, the selection of the appropriate graft should be personalized, based on the patient's specific needs, expectations and physical demands, to achieve optimal outcomes in ACLR.

AUTHOR CONTRIBUTIONS

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ETHICS STATEMENT

The authors have nothing to report.

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SUPPORTING INFORMATION

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OPEN Bone-patellar tendon-bone versus two- and four-strand hamstring tendon autografts for ACL reconstruction in young adults: a Bayesian network meta-analysis

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Bone-patellar tendon-bone (BPTB), two- and four-strand hamstring tendon (4SHT and 2SHT, respectively) are the most common autografts used for anterior cruciate ligament (ACL) reconstruction. The present study compared BPTB, 2SHT, and 4SHT for ACL reconstruction in terms of joint laxity, patient reported outcome measures (PROMs), rate of failure and anterior knee pain (AKP). The time to return to sport and the peak torque between the autografts were also compared. Finally, prognostic factors leading to worse outcomes were also investigated. It was hypothesized that all grafts yield similar proprieties in terms of joint laxity, patient reported outcome measures (PROMs) and rate of failure, but that the BPTB autograft causes a greater rate of anterior knee pain (AKP). The literature search was conducted. All clinical trials comparing BPTB and/or 2SHT, and/or 4SHT were accessed. Grafts other than BPTB and/or 4SHT and/or 2SHT were not considered. Articles reporting outcomes of allografts or synthetic grafts were not eligible, nor were those concerning revision settings. Articles reporting ACL reconstruction in patients with multi-ligament damage were also not eligible. Data from 95,575 procedures were retrieved. The median length of follow-up was 36 months. The median age of the patients was 27.5 years. With regard to joint laxity, similarity was found in terms of Lachman and Pivot shift tests between all three autografts. The BPTB demonstrated the greatest stability in terms of instrumental laxity. BPTB demonstrated the greatest PROMs. BPTB demonstrated the greatest rate of AKP, while AKP in 2SHT and 4SHT was similar. Concerning failure, statistically significant inconsistency was found ($P = 0.008$). The 4SHT demonstrated the quickest return to sport, followed by BPTB, and 2SHT. There was evidence of a negative association between the time span between injury to surgery, Lysholm score ($P = 0.04$), and Tegner scale ($P = 0.04$). Furthermore, there was evidence of a weak positive association between the time span between injury to surgery and return to sport ($P = 0.01$). BPTB may result in lower joint laxity, greater PROMs, and greater peak flexion torque compared to 2SHT and 4SHT autografts. On the other hand, BPTB reported the lowest peak extension torque and the greatest rate of AKP. Finally, a longer time span between injury and surgery negatively influences outcome.

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The anterior cruciate ligament (ACL) is the primary passive constraint for internal tibial rotation and anterior tibial translation over the femur^{1–4}. ACL injury is one of the most common knee injuries in the young athletic population^{5,6}, most commonly in those performing jumping, twisting and cutting movements⁷. Its estimated incidence worldwide is about 70 per 100,000 people per year^{8–12}. Anterior cruciate ligament rupture affects the knee kinematics^{13–15} resulting in joint instability, articular cartilage injury, and meniscal damage^{14–27}. The optimal management of ACL is still debated^{28,29}. Likewise, despite thousands of clinical articles on ACL surgical treatment, controversies still remain regarding the optimal choice of graft^{30–34}. Bone-patellar tendon-bone (BPTB) and hamstring tendon (HT) autografts are the most common options for primary anterior cruciate ligament reconstruction^{35,36}. The use of the BPTB autograft was introduced in the 1980s³⁷ and it is still one of the most commonly used³⁸. BPTB autografts achieve high patient satisfaction, quick return to sport and bone-to-bone healing^{39,40}. However, concerns have been raised about donor site complications after BPTB autograft, such as anterior knee pain, discomfort, crepitus, loss of sensation, patellar fractures, contracture of the lower patella, and loss of extension strength^{41–43}. To reduce damage to the extensor apparatus, the rates of anterior knee pain and patellar fractures, hamstring tendon (HT) autograft has been advocated^{44–47}. However, ACL reconstruction using HT autograft may lead to a greater tunnel widening, flexor weakness, and knee laxity compared to BPTB^{42,48,49}. In addition, the lack of bone block on the extremities of the HT graft may promote greater laxity leading to higher frequency of rupture⁵⁰. Several clinical studies compare the autografts mentioned above, but the results are inconclusive^{35,51}. In this Bayesian network meta-analysis, BPTB, two- and four-strand HT (4SHT and 2SHT, respectively) autografts for ACL reconstruction in young adults were compared. Joint laxity, patient reported outcome measures (PROMs), rate of failure, and anterior knee pain (AKP) between the autografts were compared, as were the time to return to sport and the peak torque. A multivariate analysis was conducted to investigate possible prognostic factors leading to worse outcomes. It was hypothesized that all grafts yield similar properties in terms of joint laxity, PROMs, and rate of failure, but that the BPTB autograft causes a greater rate of anterior knee pain (AKP).

Material and methods

Search strategy. The present Bayesian network meta-analysis followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) extension statement for reporting of systematic review incorporating network meta-analyses of health care interventions⁵². A PICO guide protocol was preliminary drafted:

- P (population): ACL tears in young adults;
- I (intervention): primary ACL reconstruction;
- C (comparison): BPTB, 4SHT, 2SHT;
- (outcomes): laxity, PROMs, failure, AKP.

Data source and extraction. Two reviewers (**;**) separately performed the literature search in February 2023. PubMed, Google scholar, Embase, and Scopus databases were accessed. The following keywords were used using the Boolean operator AND/OR: *anterior cruciate ligament, ACL, pain, knee, tear, rupture, injury, damage, reconstruction, management, treatment, arthroscopy, surgery, autografts, bone patellar tendon bone, hamstring, strands, patient reported outcome measures, PROMs, laxity, stability, instability, complication, anterior knee pain, failure*. The resulting titles were screened by the same authors independently. If the title and the abstract matched the topic, the article's full-text was accessed. If the full-text was not accessible, the article was excluded from the present study. A cross reference of the bibliographies was also performed. Disagreements were debated and the final decision was made by a third author (**).

Eligibility criteria. All clinical investigations comparing BPTB, and/or 4SHT, and/or 2SHT were accessed. Articles in English, German, Italian, French, and Spanish were eligible. Levels I to III of evidence, according to Oxford Centre of Evidence-Based Medicine (OCEBM)⁵³, were considered. Grafts other than BPTB and/or 4SHT and/or 2SHT were not eligible. Studies which reported data on skeletally immature patients were not considered. Articles reporting outcomes from allograft or synthetic graft reconstructions were not eligible, nor where those concerning revision settings. Articles reporting ACL reconstruction in patients with multi-ligament damage were not eligible. Letters, comments, reviews, opinions, and editorials were not included. Animals and biomechanics studies were also not considered. Only articles reporting quantitative data under the outcomes of interest were considered for inclusion. Missing data under the outcomes of interest warranted the exclusion from this study.

Data extraction. Two authors (**;**) independently examined the resulting articles for inclusion. Generalities and patient demographic were retrieved: author, year, journal, study design, length of the follow-up, type of graft, number of included patients, mean age, BMI, sex, time span from injury to surgery, and size of the graft. To investigate knee stability, data from the manual (Pivot shift and Lachman tests) and instrumental laxity were extracted. The instrumental laxity was evaluated using the arthrometers KT-1000 and KT-2000 (MEDmetric Corp, San Diego, California). Both of these devices applied a force of 134N on the tibial plateau over the femoral condyles, directed anteriorly. Concerning PROMs, data from the Tegner activity scale and Lysholm score at the last follow-up were extracted. The Lysholm score and Tegner activity scale have been validated for knee ligament surgery^{54–56}. Data concerning the peak torque and the return to sport were also retrieved. The rates of failure and AKP were also investigated.

Methodology quality assessment. The methodological quality assessment was made using the risk of bias graph of the Review Manager Software (The Nordic Cochrane Collaboration, Copenhagen). The following risks of bias were evaluated: selection, detection, reporting, attrition, and other sources of bias.

Statistical analysis. The statistical analyses were performed by the main author (FM) using STATA Software/MP, Version 14.1 (StataCorporation, College Station, Texas, USA). For descriptive statistics, mean and standard deviation were calculated. The analysis of variance (ANOVA) was performed to evaluate the baseline comparability, with P values > 0.1 considered satisfactory.

To assess the return to sport, the ANOVA test with Tukey post-hoc test and honestly significant difference (HSD) were performed, with values of P < 0.05 were considered statistically significant. The confidence interval (CI) was set at 95%.

The NMA was performed through the STATA routine for Bayesian hierarchical random-effects model analysis. The inverse variance method was used for analysis of continuous variable, with standardized mean difference (STD) effect measure. The Log odd ratio (LOR) effect measure was used for binary data. The overall inconsistency was evaluated through the equation for global linearity via the Wald test. If the P value was > 0.5, the null hypothesis could not be rejected, and the consistency assumption could be accepted at the overall level of each treatment. Both confidence (CI) and percentile (PrI) intervals were set at 95%. Edge plot, interval plots, and funnel plots were obtained and evaluated.

For the multivariate analysis, a multiple linear model regression with Pearson Product-Moment Correlation Coefficient (r) was used to establish whether patient characteristics (age, BMI, women, time from injury to surgery, and graft size) are associated with the outcome (Pivot shift and Lachman tests, instrumental laxity, Lysholm score, Tegner scale, return to sport, failure, and anterior knee pain). The Cauchy–Schwarz formula was used for inequality: + 1 was considered as positive linear correlation, while – 1 was a negative one. Values of $0.1 < |r| < 0.3$, $0.3 < |r| < 0.5$, and $|r| > 0.5$ were considered to have respectively small, medium, and strong association. The overall significance was assessed through the χ^2 test, with values of P < 0.05 considered statistically significant.

Ethical approval. This study complies with ethical standards.

Results

Search result. The literature search resulted in 1035 articles. Of them, 306 were excluded as they were duplicates. Furthermore, 636 articles were not eligible: not matching the topic (N = 403), reporting data on allografts or synthetic grafts (N = 41), study type (N = 154), revision or multi-ligament settings (N = 37), language limitation (N = 1). Additionally, 32 articles were excluded as they did not report quantitative data under the outcomes of interest. This left 61 clinical trials for the present study. The literature search results are shown in Fig. 1.

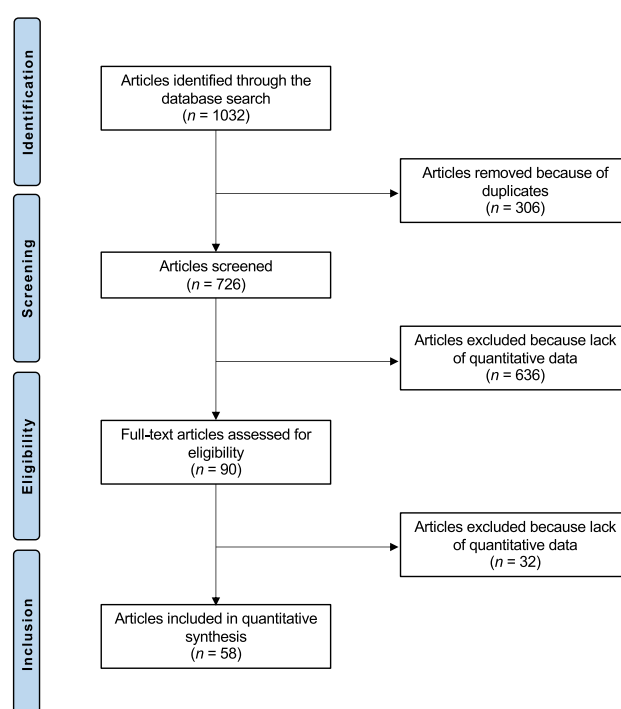


Figure 1. Flow chart of the literature search.

Methodological quality assessment. The prospective design of 85% (52 of 61) of the included investigations was an important strength of the present study. Of them, 62% (32 of 52) performed randomisation. Since most of the studies performed assessor blinding, the risk of detection bias was moderate to low. The proper analyses of most of the included studies, along with the intention to treat, clear definition of the timing of assessing outcomes, as well the use of validated tools for assessing outcomes, lead to a low risk of reporting and attrition bias. The risk of other biases was moderate to low. In conclusion, the methodological quality assessment demonstrated a moderate to low risk of bias (Fig. 2).

Patient demographics. Data from 102,573 procedures were retrieved. The median length of follow-up was 51.5 ± 49.4 months. The median age of the patients was 27.9 ± 4.2 years. The median time span from injury to surgery was 14.4 ± 11.2 months. The mean BMI was 24.6 ± 1.6 . The median size of the graft was 9.7 ± 0.7 mm. The ANOVA test found moderate baseline comparability among age, length of the follow-up, time span from injury to surgery, BMI, and graft size ($P > 0.05$). Patient demographics is shown in Table 1.

Network comparisons. With regard to joint laxity, similarities were found in terms of Lachman and Pivot shift tests between all three autografts. The BPTB demonstrated the greatest stability in terms of instrumental laxity. The equation for global linearity found no statically significant inconsistency ($P = 0.06$, $P = 0.08$, and $P = 0.1$, respectively). These results are shown in greater detail in Fig. 3.

Concerning PROMs, BPTB demonstrated the greatest Lysholm score and Tegner activity scale, followed by 2SHT and 4SHT, which scored similarly (Fig. 4). The equation for global linearity found no statically significant inconsistency ($P = 0.3$ and $P = 0.5$, respectively).

Patients who underwent reconstruction of the ACL using a BPTB graft demonstrated the greatest rate of anterior knee pain, while both 2SHT and 4SHT ranked similarly. No statistically significant inconsistency was found ($P = 0.2$). The equation for global linearity found statistically significant inconsistency for the comparison of graft failure ($P = 0.008$), thus no further conclusion could be inferred. The network comparisons of complications are shown in greater detail in Fig. 5.

Peak torque. Given the lack of quantitative data concerning the 2SHT group, only BPTB and 4SHT were considered for analysis of peak torque. BPTB demonstrated greater peak flexion torque at 60° ($P < 0.0001$) and 180° ($P < 0.0001$). No difference was found at 120° ($P = 0.06$). BPTB demonstrated lower peak extension torque at 60° ($P = 0.01$), 120° ($P = 0.008$), and 180° ($P = 0.006$). These results are shown in greater detail in Table 2.

Return to sport. The 4SHT demonstrated the quickest return to sport, followed by BPTB, and 2SHT (Table 3).

Multivariate analysis. There was evidence of a negative association between the time span between injury to surgery and Lysholm score ($r = -0.50$; $P = 0.04$) and Tegner scale ($r = -0.26$; $P = 0.04$). Furthermore, there was evidence of a weakly positive association between the time span between injury to surgery and return to sport ($r = -0.06$; $P = 0.01$). The results of the multivariate analysis are shown in Table 4.

Discussion

According to the main findings of the present study, BPTB may promote lower joint laxity, greater PROMs, and greater peak flexion torque compared to 2SHT and 4SHT autografts in young adults. The ACL is one of the most important constraints against anteroposterior translation of the knee^{113,114}. In the present study, BPTB was associated with the lowest peak extension torque and the greatest rate of AKP. Peak flexion torque is used to assess knee flexor muscle strength after reconstruction, as a quantitative outcome measure, particularly when comparing hamstring autografts to alternative graft options. Knee flexor weakness in knee flexion is relevant in certain sports such as gymnastics, judo, or wrestling, and it is useful to assess the return to sport¹¹⁵. Knee torque is significantly affected after ACL injury. Both extension and flexion isokinetic strength are important outcomes to evaluate after surgical reconstruction¹¹⁶. AKP remains a major complication after ACL reconstruction, and potentially recognizes several aetiologies, including bone-harvesting pain, neuroma of the infrapatellar branch of the medial saphenous following its lesion, and rarely, patellar tendinopathy³³. Finally, a longer time span between ACL rupture and reconstruction may represent a negative factor influencing the outcome.

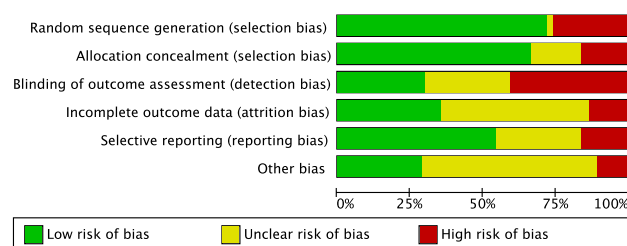


Figure 2. Methodological quality assessment.

Author, year	Journal	Study design	Follow-up (months)	Type of graft	Patients	Mean Age	Female (%)
Aglietti et al. 1994 ⁵⁷	Am. J. Sports Med.	Prospective	28	BPTB	30		
				4SHT	30		
Aglietti et al. 2004 ⁵⁸	J. Bone Jt. Surg. Am.	RCT	24	BPTB	60	25	23.00
				4SHT	60	25	23.00
Aune et al. 2001 ⁴⁵	Am. J. Sports Med.	RCT	24	BPTB	35	25	45.71
				4SHT	37	27	43.23
Barenius et al. 2010 ⁵⁹	Am. J. Sports Med.	RCT	100.8	BPTB	84	33	57.30
				4SHT	80	35	42.70
Beynnon et al. 2002 ⁶⁰	J. Bone Jt. Surg. Am.	RCT	36	BPTB	28	28.5	35.71
				2SHT	28	29.9	53.57
Biz et al. 2019 ⁶¹	Acta Biomed	Prospective	44.8	BPTB	22	31.9	
				4 SH	21	31	
Bizzini et al. 2006 ⁶²	Clin J Sport Med	Prospective	11	BPTB	87	34	37.90
				4SHT	66	31.3	31.82
Carter et al. 1999 ⁶³	Arthroscopy	RCT	6	BPTB	38		
				2SHT	35		
Corry et al. 1999 ⁶⁴	Am. J. Sports Med.	Prospective	24	BPTB	90	25	47.00
				4SHT	90	25	48.00
Cristiani et al. 2018 ⁶⁵	Knee Surg. Sports Traumatol. Arthrosc.	Retrospective	12	BPTB	692	28.8	28
				4 SH	4770	28.1	46.5
Denti et al. 2006 ⁶⁶	Knee Surg. Sports Traumatol. Arthrosc.	Prospective	24	BPTB	39	23.5	15.39
				2SHT	22	40.1	68.20
Drogset et al. 2009 ⁶⁷	Knee Surg. Sports Traumatol. Arthrosc.	RCT	24	BPTB	58	26	
				2SHT	57	27	
Feller et al. 2001 ⁶⁸	Knee Surg. Sports Traumatol. Arthrosc.	RCT	4	BPTB	31	26.2	25.81
				2SHT	34	27.1	29.41
Gifstad et al. 2013 ³⁵	Knee Surg. Sports Traumatol. Arthrosc.	RCT	84	BPTB	58	27	
				4SHT	56	27	
Gifstad et al. 2014 ⁶⁹	Am. J. Sports Med.	Prospective	48	BPTB	6736	29	37.10
				4SHT	38,666	26	42.80
Gobbi et al. 2003 ³⁹	Arthroscopy	Prospective	36	BPTB	40	28	35.00
				4SHT	40	29	45.00
Gobbi et al. 2006 ⁷⁰	Knee Surg. Sports Traumatol. Arthrosc.	RCT	24	BPTB	50	28	
				4SHT	50	28	
Gudas et al. 2018 ⁷¹	Med. Sci. Monit.	Retrospective	24	BPTB	88	26	30.68
				4SHT	95	25.1	30.53
Guglielmetti et al. 2021 ⁷²	Orthop J Sports Med	Prospective Rand-omized	24	PT	31	25.2	26
				4 SH	31	24.64	39
Harilainen et al. 2006 ⁷³	Knee Surg. Sports Traumatol. Arthrosc.	RCT	60	BPTB	40		
				2SHT	39		
Heijne et al. 2009 ⁷⁴	Knee Surg. Sports Traumatol. Arthrosc.	RCT	24.7	BPTB	34	29	35.29
				4SHT	34	30	58.82
Heijne et al. 2013 ⁷⁵	Knee Surg. Sports Traumatol. Arthrosc.	RCT	61.5	BPTB	34	29	35.29
				4SHT	34	30	58.82
Holm et al. 2010 ⁷⁶	Am. J. Sports Med.	RCT	120	BPTB	28	25	35.17
				4SHT	29	27	48.27
Ibrahim et al. 2005 ⁷⁷	Arthroscopy	RCT	81	BPTB	40	22.3	0
				4SHT	45	22.3	0
Jansson et al. 2003 ⁷⁸	Am. J. Sports Med.	RCT	21	BPTB	43		
				4SHT	46		
Kautzner et al. 2014 ⁷⁹	Int Orthop	RCT	24	BPTB	75	26	
				4SHT	75	26	
Continued							

Author, year	Journal	Study design	Follow-up (months)	Type of graft	Patients	Mean Age	Female (%)
Keays et al. 2007 ⁸⁰	Am. J. Sports Med.	Retrospective	72	BPTB	31	27	29.00
				4SHT	31	27	29.00
Laxdal et al. 2006 ⁸¹	Knee Surg. Sports Traumatol. Arthrosc.	Prospective	25	BPTB	45	26	0.00
				4SHT	78	28	0.00
Leitgeb et al. 2014 ⁸²	Wien Klin Wochenschr	RCT	60	BPTB	56	28.4	19.64
				4SHT	40	29.2	42.50
Leys et al. 2011 ⁸³	Am. J. Sports Med.	Prospective	180	BPTB	90	25	46.67
				4SHT	90	24	47.78
Lidén et al. 2007 ⁸⁴	Am. J. Sports Med.	RCT	86	BPTB	34	28	32.35
				4SHT	37	29	29.73
Machado et al. 2018 ⁸⁵	Phys Sportsmed	RCT	6	BPTB	17	31.9	29.41
				4SHT	17	37.7	17.65
Maletis et al. 2007 ⁸⁶	Am. J. Sports Med.	RCT	24	BPTB	46	27.2	32.61
				4SHT	53	27.7	15.09
Maletis et al. 2013 ⁸⁷	Bone Joint J	Retrospective	18	BPTB	2791	25.4	30.60
				4SHT	3012	27.2	39.70
Marder et al. 1991 ⁸⁸	Am. J. Sports Med.	Prospective	29	BPTB	37	21.6	35.14
				4SHT	35	23.8	25.71
				4SHT	23	18	57.00
Matsumoto et al. 2006 ⁵¹	Am. J. Sports Med.	RCT	80	BPTB	37	23.7	43.24
				4SHT	35	24.4	57.14
Mohtadi et al. 2015 ⁸⁹	Clin J Sport Med	RCT	24	BPTB	110	28.7	42.72
				4SHT	110	28.5	46.36
Pasquini et al. 2017 ⁹⁰	Acta Biomater	Prospective		BPTB	15	26	0.00
				4SHT	15	30.3	0.00
Persson et al. 2013 ⁹¹	Am. J. Sports Med.	Retrospective	48	BPTB	3428	29	41.10
				4SHT	9215	28.3	43.20
Persson et al. 2015 ⁹²	Am. J. Sports Med.	Retrospective	54	BPTB	3806	28.8	41.70
				4SHT	10,228	28.36	44.08
Pinczewski et al. 2007 ⁹³	Am. J. Sports Med.	Prospective	120	BPTB	90		
				4SHT	90		
Pinczewski et al. 2016 ⁹⁴	Am. J. Sports Med.	Prospective	60	BPTB	90	25	47.00
				4SHT	90	24	48.00
Predescu et al. 2010 ⁹⁵	IEEE CS	Prospective	12	BPTB	76		
				4SHT	59		
Rahr-Wagner et al. 2013 ⁹⁶	Am. J. Sports Med.	Prospective	36	BPTB	1971		34.00
				4SHT	11,676		42.00
Razi et al. 2014 ⁹⁷	Med J Islam Repub Iran	RCT	36	BPTB	37	30.8	21.62
				4SHT	34	28.2	14.71
Sadoghi et al. 2011 ⁹⁸	Int Orthop	Prospective	24	BPTB	41	30	
				4SHT	51	29	
Sajovic et al. 2006 ⁹⁹	Am. J. Sports Med.	RCT	60	BPTB	32	27	46.15
				4SHT	32	24	53.57
Sajovic et al. 2011 ¹⁰⁰	Am. J. Sports Med.	RCT	132	BPTB	32	38	36.00
				4SHT	32	36	48.15
Sajovic et al. 2018 ¹⁰¹	Am. J. Sports Med.	RCT	204	BPTB	32	45.5	37.50
				4SHT	32	42.5	45.83
Shaieb et al. 2002 ⁴⁷	Am. J. Sports Med.	RCT	24	BPTB	33	32	21.21
				4SHT	37	30	43.24
Stanczak et al. 2017 ¹⁰²	J Int Med Res	RCT	12	BPTB	48	31.6	16.66
				4SHT	48	31.6	25.00
Svensson et al. 2005 ¹⁰³	Knee Surg. Sports Traumatol. Arthrosc.	Prospective	24	BPTB	28	28	100.00
				4SHT	31	25	100.00
Tajima et al. 2020 ¹⁰⁴	J Knee Surg	Prospective	26	BPTB-G	32	22.7	93.75
				2 SH	43	24.8	93
Continued							

Author, year	Journal	Study design	Follow-up (months)	Type of graft	Patients	Mean Age	Female (%)
Taylor et al. 2009 ¹⁰⁵	<i>Am. J. Sports Med.</i>	RCT	36	BPTB	32	21.7	21.90
				4SHT	32	22.1	12.50
Thompson et al. 2016 ¹⁰⁶	<i>Am. J. Sports Med.</i>	Prospective	240	BPTB	90	25	46.67
				4SHT	90	24	47.78
Wagner et al. 2005 ¹⁰⁷	<i>Am. J. Sports Med.</i>	Prospective	24	BPTB	55	33.6	27.27
				4SHT	55	31.1	27.27
Webster et al. 2015 ¹⁰⁸	<i>Am. J. Sports Med.</i>	RCT	180	BPTB	22	26.6	27.27
				4SHT	25	26.1	20.00
Wipfler et al. 2011 ¹⁰⁹	<i>Arthroscopy</i>	RCT	105	BPTB	31	29.87	38.71
				4SHT	31	34.23	41.94
Witvrouw et al. 2001 ¹¹⁰	<i>Int Orthop</i>	Prospective	12	BPTB	17	24.3	41.18
				4SHT	32	24.6	46.88
Zaffagnini et al. 2006 ¹¹¹	<i>Knee Surg. Sports Traumatol. Arthrosc.</i>	RCT	60	BPTB	25	30.5	36.00
				4SHT	25	31.3	40.00
Zoran et al. 2015 ¹¹²	<i>Inj Epidemiology</i>	Prospective	24	BPTB	54	28	27.70
				4SHT	58	26	27.70

Table 1. Generalities and patient baselines from the included studies.

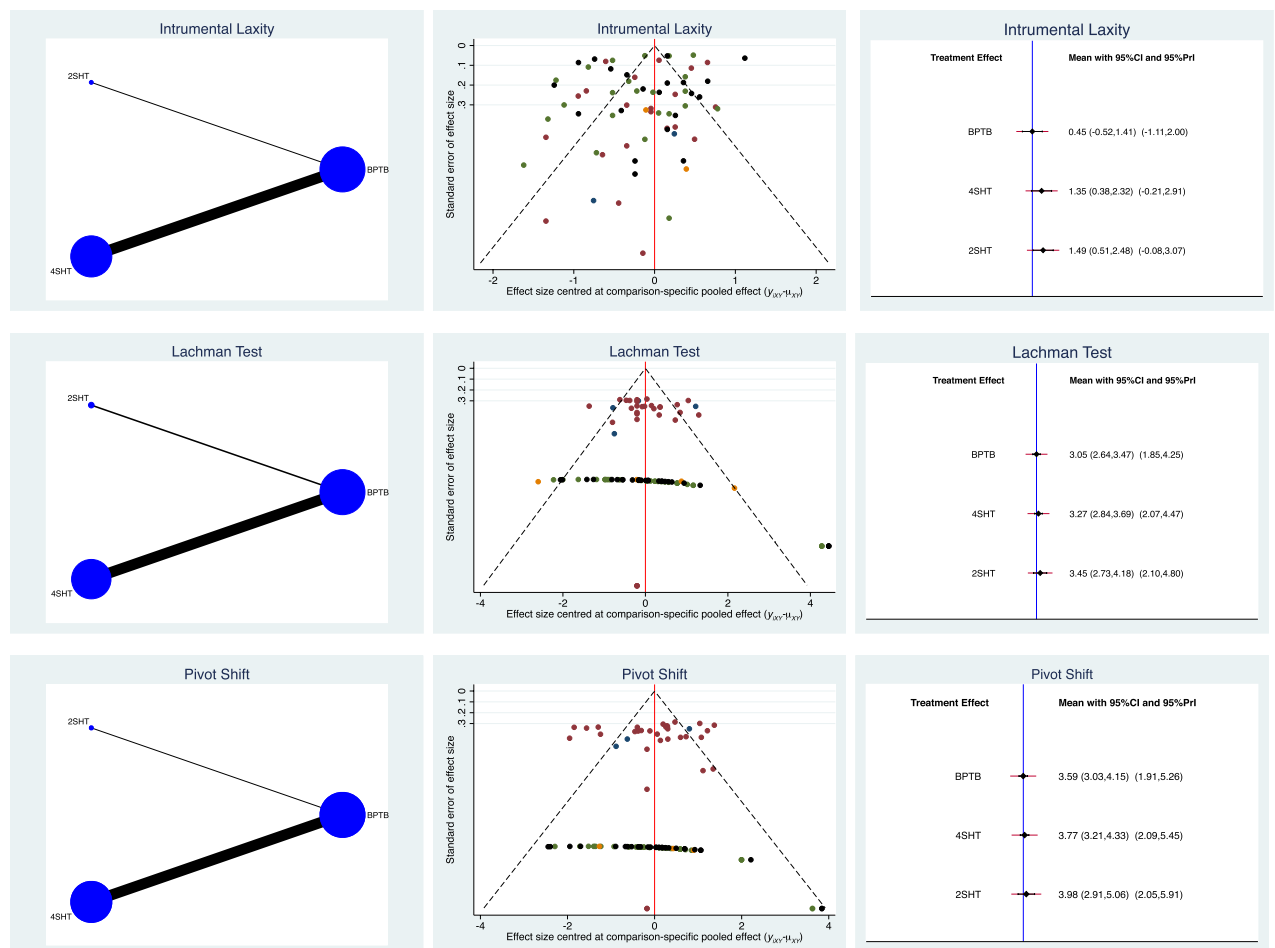


Figure 3. Edge, funnel, and interval plots of the network comparisons: joint laxity.

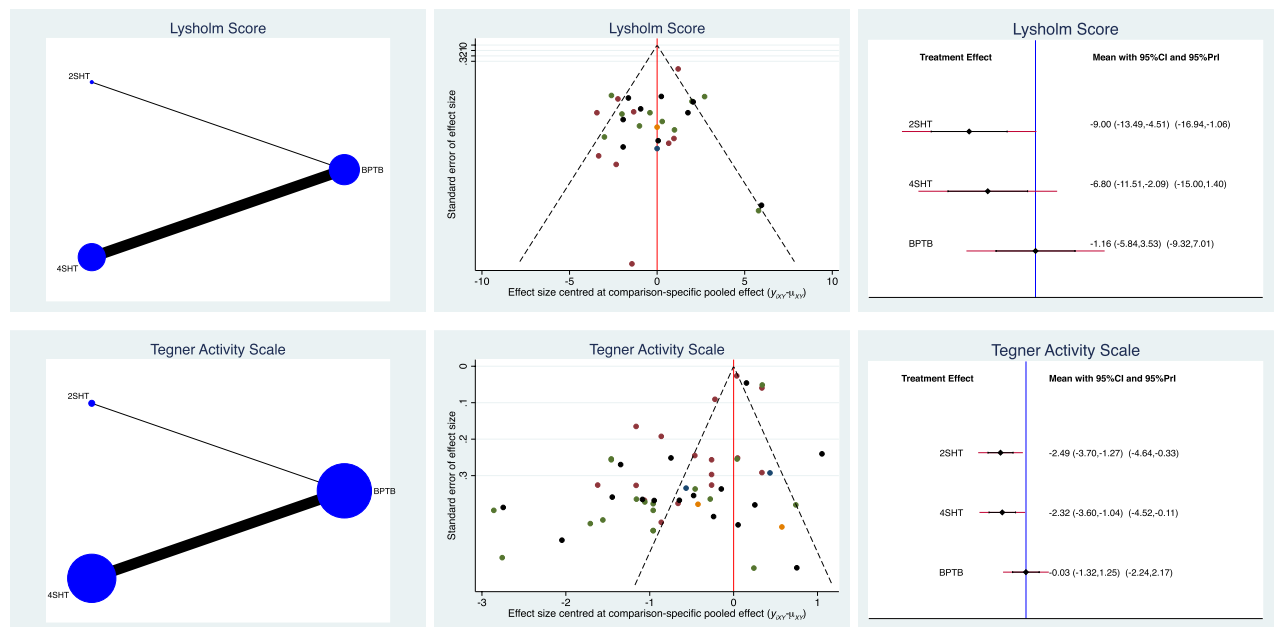


Figure 4. Edge, funnel, and interval plots of the network comparisons: PROMs.

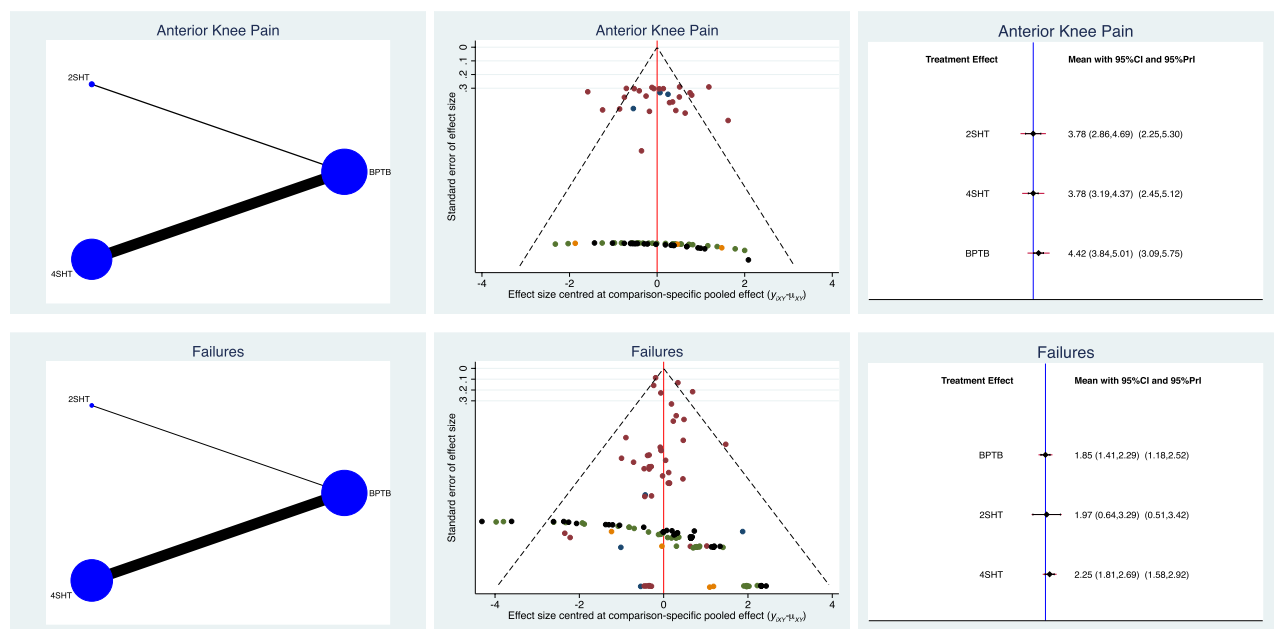


Figure 5. Edge, funnel, and interval plots of the network comparisons: complication.

Variables	4SHT	BPTB	MD	95% CI	P
Peak flexion torque 60° (deg/sec)	93.2 ± 6.2	99.1 ± 1.5	5.9	- 7.1 to - 4.6	<0.0001
Peak flexion torque 120° (deg/s)	97.9 ± 2.8	98.8 ± 3.9	0.9	- 1.8 to 0.1	0.06
Peak flexion torque 180° (deg/s)	94.3 ± 4.9	100.3 ± 3.0	6.0	- 7.1 to - 4.8	<0.0001
Peak extension torque 60° (deg/s)	93.0 ± 6.7	77.0 ± 31.3	- 16.0	9.6 to 22.3	0.01
Peak extension torque 120° (deg/s)	98.2 ± 3.2	96.4 ± 2.9	- 1.8	0.9 to 2.6	0.008
Peak extension torque 180° (deg/s)	96.2 ± 4.2	94.3 ± 5.4	- 1.9	0.5 to 3.2	0.006

Table 2. Peak torque.

	2SHT	4SHT	BPTB
2SHT	1		
4SHT	MD - 1.1; 95% CI - 1.45 to - 0.74; P<0.0001	1	
BPTB	MD - 0.9; 95% CI - 1.25 to - 0.54; P<0.0001	MD 0.2; 95% CI 0.03-0.36; P=0.01	1

Table 3. Return to sport.

Endpoints	Age		BMI		Women		Time: injury to surgery		Graft size	
	r	P	r	P	r	P	r	P	r	P
Pivot shift test	- 0.14	0.08	- 0.21	0.8	- 0.22	0.1	0.28	0.04	- 0.24	0.6
Instrumental laxity	0.13	0.7	- 0.02	0.8	- 0.09	0.3	0.00	0.00002	- 0.14	0.5
Lachman test	0.11	0.3	- 0.80	1.0	- 0.40	0.5	- 0.48	1.0	0.00	0.6
Lysholm score	0.18	0.5	- 0.89	0.5	0.31	0.2	- 0.50	0.04	- 0.30	1.0
Tegner scale	0.13	0.4	- 1.00	0.05	- 0.40	0.26	- 0.26	0.04	- 0.73	0.4
Return to sport	- 0.04	0.6	0.09	0.3	0.06	0.5	0.06	0.01	- 0.35	0.2
Failure	0.33	0.7	- 0.06	0.7	0.04	0.6	0.17	0.6	0.12	0.07
AKP	0.00	0.02	0.58	0.9	0.67	0.8	0.74	0.3	0.86	0.6

Table 4. Results of the multivariate analysis (AKP = Anterior knee pain).

Concerning joint laxity, similarity was found in terms of Lachman and Pivot shift tests between all three autografts. The Lachman test evaluates the anterior translation of the tibia in relation to the femur with the knee in static flexion¹¹⁷. The Pivot shift test instead assesses the rotatory instability of the joint during its dynamic flexion¹¹⁸. Similarly, a previous meta-analysis found no difference in IKDC score, Lachman and Pivot shift tests between BPTB and hamstring autografts¹¹⁹. However, BPTB autograft resulted in a higher incidence of AKP, kneeling pain, and rate of osteoarthritis¹¹⁹. The literature on osteoarthritis of patients undergoing reconstruction with BPTB or HT autografts is controversial^{120,121}. In the present study, patients receiving a BPTB graft demonstrated the lowest instrumental laxity and the greatest Lysholm score and Tegner activity scale, followed by 2SHT and 4SHT, which scored similarly. The Lysholm score and Tegner activity scale are outcomes measurements of a subjective nature that evaluate performance and activity restrictions both before and after surgery¹²². These PROMs have been validated for knee ligament surgery^{54–56}.

On the other hand, BPTB demonstrated the greatest rate of AKP compared to both the 2SHT and 4SHT autografts, which showed a similar rate. Concerning failure, no statistically significant inconsistency was found. The equation for global linearity found statistically significant inconsistency for the comparison failure; thus, no further conclusion could be inferred. In a study on 5462 patients with primary ACL reconstruction, HT autografts resulted in greater anterior knee laxity and failures compared with BPTB autografts⁶⁵. In a previous meta-analysis including 25 studies (47,613 ACL reconstructions), HT autografts failed at a higher rate than BPTB autografts¹²³. Similar results have been evidenced in another meta-analysis involving 15 RCTs (1298 patients)¹²¹. A further meta-analysis including 20 RCTs compared BPTB versus 4SHT. The BPTP cohort evidenced lower laxity and failure rupture, but a greater risk of kneeling pain and AKP¹²⁴.

Given the lack of quantitative data concerning the 2SHT group, only BPTB and 4SHT were considered in our study for analysis of peak torque. BPTB demonstrated greater peak flexion torque at 60° and 180°. No difference was found at 120°. BPTB demonstrated lower peak extension torque at 60°, 120°, and 180°. While BPTB exhibits some better outcome measures, it should be noted that BPTB also demonstrated the greatest rate of AKP. These findings agreed with previous studies comparing HT and BPTB, which stated that the latter restores greater knee stability, but also results in greater postoperative complications^{121,125,126}. AKP is common following ACL reconstruction and can persist for a long time in athletes. The removal of the central third of the patellar tendon and its subsequent repair might cause a lowering of the patella and lead to increased sensitivity and pain during kneeling or squatting¹²⁷. In this regard, in our study, the 4SHT graft demonstrated the quickest return to sport, followed by BPTB, and 2SHT. This should be considered when making a decision with athletes whose goal is to return to play as soon as possible. Lastly, results from the multivariate analysis demonstrated that a longer time span between initial injury and surgery was associated with lower Lysholm scores, Tegner scale, and longer return to sport. This worse outcome associated with a longer time from injury to surgery should be considered when planning the reconstruction. It should also be noted that some insurance companies currently require a dedicated physiotherapy trial for ACL injuries before surgery is authorized¹²⁸. This delay in treatment can lead to suboptimal results^{129,130}.

This study has certainly limitations. The retrospective nature of most studies is an important limitation which increases the risk of selection bias. Demographic data of the patients were collected, but further information regarding their general health were seldom reported in the included studies. Most of the authors did not specify whether the surgeon who performed the procedure was the investigator himself, and whether the assessor was blinded to the procedure performed. Many studies did not clearly specify the surgical technique (arthroscopic,

open, or both) or postoperative management. Rehabilitation protocols following ACL reconstruction are associated with significant differences in outcome¹³¹. Several new modalities of rehabilitation after ACL reconstruction such as strengthening, and functional exercises, resistance training, neuromuscular exercise, high-level dynamic functional tasks and sport-specific training have been proposed^{132–136}. However, given the lack of quantitative data, the various rehabilitation protocols could not be analysed separately. Most authors did not specify whether patients had undergone MRI preoperatively, thus providing poor information on preoperative diagnostic methods. Most authors did not report information on the sporting activity and level of the patients; therefore, further subgroup analyses were not possible. Given the lack of quantitative data, it was not possible to investigate additional autografts^{137,138}. Allografts have been advocated as they avoid donor site morbidity^{139–141}. However, the greater risk of graft-versus-host reaction, disease transmission, and delayed graft incorporation limits the use of allografts^{142–144}. There is also a growing trend of using quadriceps tendon grafts, which may provide another viable and safe alternative for autografts options^{145–148}. This autograft may result in a lower rate of failure compared to both BPTB and HT grafts, as well as a reduced rate of AKP compared to the BPTB autograft¹⁴⁹. Further high-quality investigations should validate the present results also in skeletally immature patients. Furthermore, the aetiology of the AKP following ACL surgery still remains debated, and international recommendations on the management and classification of this condition are required.

Conclusion

BPTB may promote lower joint laxity, greater PROMs, and greater peak flexion torque compared to 2SHT and 4SHT autografts. On the other hand, BPTB resulted in the lowest peak knee extension torque and the greatest rate of AKP. Concerning PROMs and AKP, similar scores were obtained in the comparison between SHT2 and 4SHT. However, the 4SHT demonstrated the quickest return to sport, followed by BPTB, and 2SHT. Finally, longer time span between injury and ACL reconstruction negatively influences the outcomes.

Data availability

The datasets generated during and/or analysed during the current study are available throughout the manuscript.

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Author contributions

F.M.: literature search, data extraction, risk of bias assessment, conception and design, drafting, final approval; N.M.: supervision, revision, final approval; A.T.: writing, final approval; E.T.: literature search, data extraction, risk of bias assessment, final approval; F.H.: supervision, final approval; J.E.: drafting, final approval. All authors have agreed to the final version to be published and agree to be accountable for all aspects of the work.

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REVIEW ARTICLES
SYSTEMATIC REVIEWS

Comparison of Autograft Types in Anterior Cruciate Ligament Reconstruction

A Systematic Review and Bayesian Network Meta-Analysis of Randomized Clinical Trials

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Background: The literature regarding optimal autograft choice for anterior cruciate ligament (ACL) reconstruction (ACLR) remains inconclusive. This network meta-analysis (NMA) compares common autografts for primary ACLR.

Methods: PubMed, Scopus, Web of Science, and Embase were searched up to May 3, 2025, for randomized clinical trials (RCTs) on primary ACLR in adults that compared ≥ 2 of the following tendon autografts: 4-strand semitendinosus (4SST), 4-strand semitendinosus-gracilis (4SSTG), its 5-strand variant (5SSTG), bone-patellar tendon-bone (BPTB), quadriceps tendon with bone (QTB), and free quadriceps tendon (FQT). Outcomes analyzed in the NMA were the International Knee Documentation Committee (IKDC) subjective score, Lysholm score, Tegner Activity Scale, anteroposterior (instrumented) and rotational (pivot-shift) stability, and rerupture or revision ACLR rate. Autografts were ranked using surface under the cumulative ranking (SUCRA) values.

Results: A total of 44 RCTs with 3,491 patients were included in the NMA. With respect to the IKDC, QTB was statistically superior to BPTB (mean difference = 3.46, 95% credible interval [CrI]: 0.29 to 6.77), although the difference was likely not clinically meaningful. QTB ranked highest for the IKDC (SUCRA = 90.1%) and Tegner (SUCRA = 85.3%), while BPTB ranked lowest for the IKDC and Lysholm. With respect to knee laxity, QTB ranked second in anteroposterior and first in rotational stability, and it carried a significantly lower risk of a 2+ or higher pivot-shift than 4SST (risk ratio = 0.26, 95% CrI: 0.07 to 0.85). QTB was associated with a decreased risk of rerupture/revision compared with other autografts (SUCRA = 83.3%).

Conclusions: Based on the autograft rankings, QTB was found to lead to improved functional, activity-related, and stability outcomes overall, while also reducing the risk of graft failure.

Level of Evidence: Therapeutic Level I. See Instructions for Authors for a complete description of levels of evidence.

Anterior cruciate ligament (ACL) injuries are common orthopaedic sports injuries and a significant global health-care burden^{1,2}, often leading to secondary injuries, long-term instability, and osteoarthritis³⁻⁵. While ACL reconstruction (ACLR) using autografts is a standard treatment, the optimal choice of autograft, typically among hamstring tendon (HT), bone-patellar tendon-bone (BPTB), and quadriceps tendon (QT), remains controversial^{6,7}. Conventional studies on this topic have often relied on pairwise meta-analyses, which are limited in their ability to account for the complexity of multiple

autograft comparisons⁸⁻¹¹. Subsequent network meta-analyses (NMAs) attempted to address this issue.

To our knowledge, 7 previous NMAs have addressed graft selection for ACLR¹²⁻¹⁸. However, their findings were limited by significant methodological inconsistencies, including the inclusion of heterogeneous graft types, such as allografts, synthetic grafts, and autografts, in 1 network¹⁵⁻¹⁸, an approach that might provide a broad overview but does not yield a clear consensus, as these graft types differ substantially in their clinical indications. Given that autografts remain the preferred

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choice for ACLR¹⁹⁻²², conducting an NMA dedicated to their comparison using standardized graft classifications is essential. Previous NMAs have often misclassified autograft types, by grouping different HT variants together^{12,13,16}, not specifying bone block use in patellar tendon autografts^{13,15,16}, and either omitting QT autografts^{14,16,18} or not distinguishing between QT variants with versus without bone^{12,13,15}.

This systematic review and Bayesian NMA aimed to address these gaps by comparing 6 well-defined autograft subtypes for primary ACLR in adults with respect to patient-reported outcome measures (PROMs), knee stability, graft failure, and return to sport. We hypothesized that there would be no significant differences among the compared autografts with respect to these outcomes.

Materials and Methods

This systematic review and NMA was performed according to the PRISMA-NMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension Statement for Reporting of Systematic Reviews Incorporating Network Meta-Analyses of Health Care Interventions)²³, and the protocol was registered in PROSPERO (CRD420251117682).

Search Strategy

An electronic search was conducted in PubMed, Scopus, Web of Science, and Embase, from inception to May 3, 2025, without language restrictions. The search strategy was developed using keywords and MeSH terms relating to ACLR, different autograft types, and randomized clinical trials (RCTs). The detailed search strategy is available in Appendix 1.

Eligibility Criteria and Outcome Definition

We included RCTs comparing ≥ 2 autografts for primary ACLR in adults, with eligibility criteria detailed in Table 1. Included studies had to report data on at least 1 predefined outcome. These included PROMs, namely the International Knee Documentation Committee (IKDC) Subjective Knee Form²⁴, Lysholm score²⁵, and Tegner Activity Scale²⁵. Additional outcomes involved knee stability measures, namely instrumented anteroposterior (AP) laxity and the pivot-shift test result. The other outcomes were ACL rerupture or the need for revision surgery and return to sport. For homogeneity, studies reporting AP laxity data were limited to those employing a standardized 134 N or maximum manual force. For RCTs with multiple published reports, the one with the most recent follow-up was used; however, if an outcome was missing from that report, the next latest report was included.

Study Selection and Data Extraction Process

Three authors (S.Y., S.M.M., and F.S.) independently performed title and abstract screening, followed by full-text screening of relevant studies. A senior author (F.V.) was consulted to resolve any disagreements. Data extracted included study details (title, first author, year, country, follow-up, design), patient characteristics (sample size, age, sex, body mass index [BMI], time from injury to surgery), inclusion and exclusion criteria, athlete level (general, professional), graft details (type, subtype, diameter, fixation), and outcome data from the latest follow-up.

Risk-of-Bias Assessment

Risk of bias was assessed using the Cochrane risk-of-bias tool (RoB 2) for randomized trials²⁶. RoB plots were generated using the robvis web app (Luke McGuinness)²⁷.

Statistical Analysis

We performed an NMA within a Bayesian framework using Markov-chain Monte Carlo (MCMC) simulation. Model convergence was assessed using trace plots and the Gelman-Rubin diagnostic. Treatment effects are reported as the mean difference (MD) for continuous outcomes and as the risk ratio (RR) for binary outcomes, with the 95% credible interval (CrI). When the effect estimate for a continuous outcome reached significance, we further assessed whether it exceeded the minimal clinically important difference (MCID). Publication bias and small-study effects were assessed using funnel plots and the Egger test. Treatment rankings were estimated using surface under the cumulative ranking (SUCRA) values. Full, detailed descriptions of our statistical analysis methods are available in Appendix 2.

Results

Study Selection and Characteristics of the Included Studies

The study selection process is illustrated in Figure 1. A total of 48 reports representing 45 RCTs were included in the systematic review²⁸⁻⁷⁵, and 44 of the 45 were ultimately included in the NMA. These reports included 3,491 patients and were from 25 countries. For 3 RCTs, data were extracted from 2 separate reports, as different follow-up reports provided data on different outcomes^{29,52,55,58,61,70}. Characteristics of the included studies are summarized in Table 2 and in Appendix 3.

Network Geometry

Network characteristics and plots for all outcomes are provided in Appendix 4, including Supplementary Figure 1. The

“QT autografts performed the best overall with respect to the PROMs... QTB [QT with bone] was the most favorable autograft with respect to the risk of ACL rerupture or revision ACLR.”

Table 1. Inclusion and Exclusion Criteria*

Inclusion Criteria	Exclusion Criteria
Population Patients aged 18 years or older Patients diagnosed with ACL rupture Intervention Primary ACL reconstruction Comparison Studies comparing at least 2 of the specified graft types: - Quadrupled (4-strand) semitendinosus, abbreviated as <i>4SST</i> - Double-looped (4-strand) semitendinosus and gracilis, abbreviated as <i>4SSTG</i> 5-strand hamstring graft consisting of both semitendinosus and gracilis, abbreviated as <i>5SSTG</i> - Bone-patellar tendon-bone graft with bone blocks on both ends, abbreviated as <i>BPTB</i> - All-soft-tissue (free) quadriceps tendon, abbreviated as <i>FQT</i> - Quadriceps tendon with patellar bone block, abbreviated as <i>QTB</i> Outcome Studies reporting at least 1 of the following outcomes: - IKDC - Lysholm - Tegner - Instrumented laxity - Pivot-shift test - Rerupture or revision ACL reconstruction - Return to sport Study design Randomized clinical trials	Population Patients with a history of ACL rupture Patients with a history of relevant chronic diseases Intervention Revision ACL reconstruction Any ACL repair technique ACL remnant preservation technique Graft augmentation technique Double-bundle reconstruction Comparison Studies involving none of the specified graft types Studies that do not compare at least 2 graft types of interest Studies that do not specify which hamstring tendons were used for the graft Studies that do not report the number of strands used in the hamstring graft Studies that do not specify the number of bone block ends in the patellar graft Studies involving allografts Studies involving artificial grafts Outcome Studies with insufficient outcome data Studies reporting none of the specified outcomes Study design Reports with shorter follow-up reporting data on the same outcomes as a publication with longer follow-up Nonrandomized clinical trials Observational studies, including case reports, case series, cohort studies, and case-control studies Conference abstracts, pilot studies, commentaries

*IKDC = International Knee Documentation Committee Subjective Knee Form.

pivot-shift analysis lacked data for 1 of the autografts, QT without bone (free QT [FQT]). All 6 autografts were included in the analyses of the remaining outcomes.

Risk of Bias

RoB assessment is provided in Appendix 5, including Supplementary Figures 2-A and 2-B. Of the 48 included reports, 35 were deemed to have a low risk of bias, 7 exhibited some concerns regarding risk of bias^{30,48,50,54,55,58,67}, and 6 were judged to have a high risk of bias^{46,61,62,64-66}.

Network Meta-Analysis

Network estimates for pairwise comparisons are provided in Table 3 and shown in forest plots in Appendix 6, including Supplementary Figures 3-A through 3-F. Additionally, SUCRA

plots illustrating the ranking probabilities of the autografts for each outcome are shown in Figure 2.

IKDC Subjective Knee Evaluation

The IKDC NMA included 22 studies with a total of 1,419 patients^{28,29,31,33,35-41,43-46,48,50,51,54,57,70,72}. The largest MD was observed between quadriceps tendon with bone block (QTB) and BPTB (MD = 3.46 in favor of QTB, 95% CrI: 0.29 to 6.77), which was statistically significant but likely not clinically meaningful. Other pairwise comparisons did not reveal a significant difference. As shown in Figure 2-A, QTB (SUCRA = 90.1%) and 4-strand semitendinosus (4SST) (SUCRA = 70.7%) demonstrated the highest probability of being the most favorable autograft, followed by 5-strand semitendinosus-gracilis (5SSTG) (SUCRA = 45.0%), 4-strand semitendinosus-

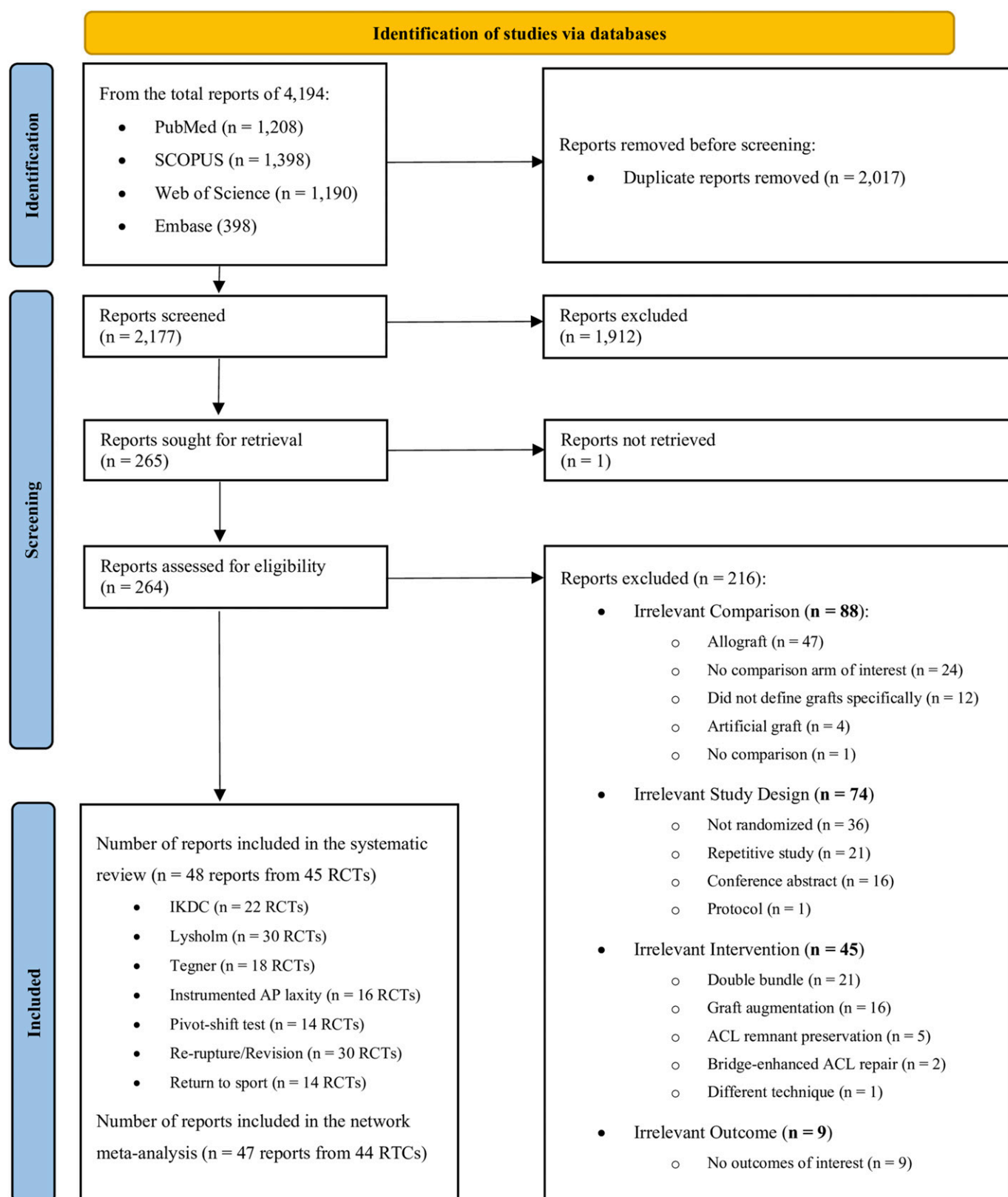


Fig. 1
PRISMA flowchart of the included randomized clinical trials (RCTs) in the systematic review and network meta-analysis.

Table 2. Characteristics of the Included Randomized Clinical Trials*

Study	Latest Follow-up	Included Outcomes	Comparison Arms	No. of Participants at Final Follow-up	Age (yr)	Sex, F/M
García-Linage, 2025, Mexico ³⁰	1 yr	Lysholm Tegner Return to sport	FQT 4SST	13 13	26.2 ± 6.2 28.5 ± 7.2	7/6† 6/7†
Lucidi, 2025, Italy ²⁹ ‡	20 yr	IKDC Lysholm Tegner Rerupture/revision	BPTB 4SSTG	19 20	29.3 ± 7.2 31.3 ± 7.6	6/13 7/13
Solanki, 2025, India ²⁸ §	2 yr	IKDC Lysholm Return to sport	FQT	30	Overall numbers: 20 to 25 yr: 22; 26 to 30 yr: 16; 31 to 35 yr: 14; 36 to 40 yr: 8#	Overall: 20/40
Ebert, 2024, Australia ³⁵	2 yr	IKDC Lysholm Tegner Instrumented laxity Rerupture/revision Return to sport	5SSTG FQT 4SST**	30 48 49	28.1 ± 8.2 29.4 ± 7.7	29/28 27/28
Mo, 2024, Norway ³³	2 yr	IKDC Instrumented laxity Pivot-shift Rerupture/revision	4SSTG 4SST	44 45	34.1 ± 10.1 34.2 ± 9.5	20/29 26/22
Fallah, 2024, Iran ³⁴	1 yr	Lysholm Rerupture/revision	QTB BPTB 4SSTG	25 25 25	25.2 ± 3.9 26.3 ± 4.1 25.8 ± 3.9	0/25 0/25 0/25
Popovic, 2024, Norway ³²	18 yr	Lysholm Tegner Instrumented laxity Pivot-shift Rerupture/revision	BPTB 4SSTG	47 49	Overall median (range): 27 (18 to 49)	19/28† 14/35†
Khatri, 2023, India ³⁷ §	2 yr	IKDC Lysholm Return to sport	FQT 5SSTG	17 17	NS	Overall: 7/27
Kuliński, 2023, Poland ³⁶	4.5 yr	IKDC Lysholm Tegner Pivot-shift Rerupture/revision	4SSTG 4SST	68 72	Median (range): 31 (18 to 45) Median (range): 31 (18 to 44)	15/64 18/57
Tang, 2024, Turkey ³¹	2 yr	IKDC Lysholm	FQT 4SSTG	17 16	28.1 ± 6.2 28.3 ± 8.6	0/17† 3/13†
Horstmann, 2022, Germany ³⁹	2 yr	IKDC Lysholm Instrumented laxity Rerupture/revision Return to sport	QTB 4SSTG	23 26	24.1 ± 3.6 32.7 ± 11.4	3/21 15/12
Komzák, 2022, Czech Republic ³⁸	28 mo	IKDC Lysholm	QTB BPTB	40 40	29.8 ± NS 31.0 ± NS	19/21 17/23
Arida, 2021, Greece ⁴³	1 yr	IKDC Lysholm Tegner Rerupture/revision	BPTB 4SSTG	29 24	29.8 ± 11.1 30.0 ± 11.7	7/23 12/18

continued

Table 2. (continued)

Study	Latest Follow-up	Included Outcomes	Comparison Arms	No. of Participants at Final Follow-up	Age (yr)	Sex, F/M
Guglielmetti, 2021, Brazil ⁴¹	2 yr	IKDC Lysholm Pivot-shift Rerupture/revision Return to sport	BPTB 4SSTG	31 31	25.2 ± 5.5 24.6 ± 5.3	8/23 12/19
Krishna, 2021, Singapore ⁴⁰	2 yr	IKDC Lysholm Tegner Instrumented laxity Pivot-shift Rerupture/revision	4SSTG 5SSTG	28 28	27.6 ± 7.3 26.3 ± 7.3	8/20 5/23
Vilchez-Cavazos, 2020, Mexico ⁴⁴	1 yr	IKDC Lysholm	QTB 4SSTG††	14 14	Mean (IQR): 23 (19.5 to 30.5) Mean (IQR): 23 (20 to 30)	2/11†† 3/12††
Cristiani, 2021, Sweden ⁴²	2 yr	Rerupture/revision	BPTB 4SSTG	56 50	28.9 ± 5.9 28.4 ± 6.3	21/59 24/56
Smith, 2020, U.S.A. ⁴⁵	2 yr	IKDC Instrumented laxity Rerupture/revision Return to sport	BPTB 4SST	32 27	17.8 ± 2.3 17.6 ± 2.5	17/15 19/13
Roger, 2020, France ⁴⁶	2 yr	IKDC Instrumented laxity Pivot-shift Rerupture/revision Return to sport	4SSTG 4SST	27 33	30.3 ± 8.5 30.5 ± 8.9	4/23† 7/26†
Rajput, 2020, Pakistan ⁴⁷	2 yr	Lysholm	BPTB 4SSTG	22 19	37.5 ± 11.2 37.0 ± 12.4	0/22 0/19
Barié, 2020, Germany ⁴⁸	10 yr	IKDC Lysholm Tegner Instrumented laxity Rerupture/revision	QTB BPTB	21 22	30.5 ± 7.8 30.6 ± 7.5	13/17 13/17
Lind, 2020, Denmark ⁷²	2 yr	IKDC Tegner Instrumented laxity Pivot-shift Rerupture/revision	QTB 4SSTG	44 47	27.2 ± 6.4 27.1 ± 6.1	21/29 24/25
Mohtadi, 2019, Canada ⁷⁰ §§	5 yr	IKDC Pivot-shift Rerupture/revision	BPTB 4SSTG	103 105	33.8 ± 9.8† 33.7 ± 10.0†	43/60† 47/58†
Gupta, 2019, India ⁷⁴	1 yr	Lysholm Rerupture/revision	BPTB 4SSTG	20 20	25.9 ± 6.2# 25.8 ± 7.5#	NS NS
Gupta, 2020, India ⁷³	2 yr	Lysholm Rerupture/revision Return to sport	BPTB 4SSTG	80 80	25.0 ± 5.8# 24.8 ± 5.0#	1/79# 2/78#
Martin-Alguacil, 2018, Spain ⁷¹	2 yr	Rerupture/revision	QTB 5SSTG	19 18	18.7 ± 3.6 19.2 ± 3.6	3/23 9/16
Sajovic, 2018, Slovenia ⁶⁹	17 yr	Lysholm Instrumented laxity Rerupture/revision	BPTB 4SSTG	24 24	45.5 ± 8.7† 42.5 ± 7.5†	9/15† 11/13†
Lu, 2017, People's Republic of China ⁴⁹	18 mo	Lysholm Tegner Pivot-shift	BPTB 4SSTG	40 59	35.2 ± 14.9# 37.5 ± 17.1#	11/29# 19/40#

continued

Table 2. (continued)

Study	Latest Follow-up	Included Outcomes	Comparison Arms	No. of Participants at Final Follow-up	Age (yr)	Sex, F/M
Webster, 2016, Australia ⁵⁰	15 yr	IKDC Instrumented laxity Rerupture/revision Return to sport	BPTB	22	26.6 ± 6.7	6/16†
			4SSTG	25	26.1 ± 5.9	5/20†
Mohtadi, 2015, Canada ^{52§§}	2 yr	Instrumented laxity	BPTB	106	28.7 ± 9.7	47/63
			4SSTG	108	28.5 ± 9.9	51/59
Karimi-Mobarakeh, 2015, Iran ^{75##}	1 yr	Return to sport	4SSTG	61	29.7 ± 7.9	11/50†
			4SST	58	28.8 ± 8.2	10/48†
Barenius, 2014, Sweden ^{55***}	14 yr	Lysholm Tegner Rerupture/revision	BPTB	69	39.2 ± 5.6†	34/35†
			4SST	65	41.6 ± 7.0†	21/44†
Lund, 2014, Denmark ⁵⁴	2 yr	IKDC Pivot-shift Rerupture/revision	QTB	21	30 ± 9	5/20
			BPTB	18	31 ± 8	4/22
Papalia, 2015, Italy ⁵¹	1 yr	IKDC Lysholm	BPTB	20	Median (range): 24.7 (17 to 30)	20/0
			4SSTG	20	Median (range): 25.4 (18 to 36)	20/0
Razi, 2014, Iran ⁵³	3 yr	Tegner Pivot-shift	BPTB	37	30.8 ± 4.5	8/38
			4SSTG	34	28.2 ± 3.7	5/36
Wipfler, 2011, Germany ⁵⁶	8.8 yr	Lysholm Tegner Instrumented laxity Rerupture/revision	BPTB	29	29.9 (25 to 55)	12/19
			4SSTG	25	34.2 (26 to 64)	13/18
Barenius, 2010, Sweden ^{58***}	8 yr	Instrumented laxity Pivot-shift	BPTB	78	33.0 ± 6.3†	40/38†
			4SST	75	35.0 ± 7.5†	24/51†
Moraiti, 2010, Greece ⁵⁷	21 mo for BPTB, 25 mo for 4SSTG	IKDC Lysholm	BPTB	6	23 ± 2	0/6
			4SSTG	6	24 ± 3	0/6
Taylor, 2009, U.S.A. ⁵⁹	2.7 yr for BPTB, 3.2 yr for 4SSTG	Lysholm Tegner Rerupture/revision	BPTB	21	Median (range): 21.7 (18 to 37)	7/25
			4SSTG	24	Median (range): 22.1 (17 to 44)	4/28
Maletis, 2007, U.S.A. ⁶⁰	2 yr	Rerupture/revision	BPTB	46	27.2 (15 to 42)	15/31
			4SSTG	50	27.7 (14 to 48)	8/45
Zaffagnini, 2006, Italy ^{61†}	5 yr	Pivot-shift Return to sport	BPTB	25	30.5 (22 to 47)	9/16
			4SSTG	25	31.3 (26 to 49)	10/15
Harilainen, 2006, Finland ⁶²	5 yr	Lysholm Tegner Pivot-shift Rerupture/revision	BPTB	37	Overall mean: females, 33; males, 29	NS
			4SSTG	37		NS
Laxdal, 2005, Sweden ⁶³	2 yr	Lysholm Tegner Rerupture/revision	BPTB	38	Median (range): 28 (16 to 52)	11/29
			4SSTG	38	Median (range): 26 (15 to 35)	11/28
Ibrahim, 2005, Kuwait ⁶⁴	81 mo	Lysholm Tegner Return to sport	BPTB	40	Overall: 22.3 (17 to 34)	0/40
			4SSTG	45		0/45
Aglietti, 2004, Italy ⁶⁵	2 yr	Instrumented laxity	BPTB	60	25 (16 to 39)	14/46
			4SSTG	60	25 (15 to 39)	14/46
Shaieb, 2002, U.S.A. ⁶⁶	2.5 yr	Rerupture/revision Return to sport	BPTB	31	30 (14 to 53)	7/26
			4SSTG	35	32 (14 to 48)	16/21

continued

Table 2. (continued)

Study	Latest Follow-up	Included Outcomes	Comparison Arms	No. of Participants at Final Follow-up	Age (yr)	Sex, F/M
Beard, 2001, U.K. ⁶⁷	1 yr	Lysholm Tegner	BPTB	22	NS	NS
			4SSTG	23	NS	NS
Aune, 2001, Norway ⁶⁸	2 yr	Instrumented laxity	BPTB	29	25 ± 7	16/19
		Rerupture/revision	4SSTG	32	27 ± 9	16/21

*Ages are given as the mean ± standard deviation or mean (range) unless otherwise indicated. Age and sex data reflect baseline unless otherwise specified. 4SST = 4-strand semitendinosus, 4SSTG = 4-strand semitendinosus and gracilis, 5SSTG = 5-strand semitendinosus and gracilis, BPTB = bone-patellar tendon-bone, FQT = free quadriceps tendon, QTB = quadriceps tendon with bone block, IKDC = International Knee Documentation Committee Subjective Knee Form, NS = not specified, IQR = interquartile range. †Data reflect final follow-up. ‡2 reports of 1 RCT. §We had been concerned that these 2 studies might share overlapping data; however, as that could not be verified and they were conducted in different settings, we treated them as separate studies. #Whether data reflect baseline or final follow-up was not specified. **Although 4SSTG was used in <10% of cases, all cases were ultimately analyzed as 4SST based on the intention-to-treat principle. ††The sum of the 2 sexes contradicts the number of participants. ‡‡Although 5SSTG was used in some cases, all cases were ultimately analyzed as 4SSTG based on the intention-to-treat principle. §§2 reports of 1 RCT. ##This RCT was not included in the network meta-analysis. ***2 reports of 1 RCT.

gracilis (4SSTG) (SUCRA = 40.6%), FQT (SUCRA = 31.1%), and BPTB (SUCRA = 22.6%).

Lysholm Score

The Lysholm NMA included 30 studies with 1,950 patients^{28-32,34-41,43,44,47-49,51,55-57,59,62-64,67,69,73,74}. The greatest difference was observed between FQT and BPTB (MD = 3.44 in favor of FQT, 95% CrI: -0.60 to 7.46). However, none of the pairwise comparisons for the Lysholm score were significant. According to the SUCRA values (Fig. 2-B), FQT was the most likely to be the best autograft (SUCRA = 87.4%) on the basis of the Lysholm score, followed by 5SSTG (SUCRA = 78.8%), 4SST (SUCRA = 49.2%), 4SSTG (SUCRA = 40.9%), and QTB (SUCRA = 32.7%), while BPTB was the least likely (SUCRA = 11.1%).

Tegner Scale

The Tegner NMA included 18 studies with 1,326 patients^{29,30,32,35,36,40,43,48,49,53,55,56,59,62-64,67,72}. The comparison between QTB and 5SSTG showed the greatest difference (MD = 1.10 in favor of QTB, 95% CrI: -1.13 to 3.29). No significant differences in the Tegner scale were observed between autografts. In the autograft rankings (Fig. 2-C), QTB scored highest (SUCRA = 85.3%). The moderately performing autografts 4SST, FQT, and BPTB achieved SUCRA values of 51.9%, 49.5%, and 46.2%, respectively. In contrast, 5SSTG was the least likely to be the best autograft (SUCRA = 30.8%), followed by 4SSTG (SUCRA = 36.4%).

Instrumented AP Laxity Measurement

A total of 16 studies with 1,298 patients were incorporated into the NMA of AP laxity^{32,33,35,39,40,45,46,48,50,52,56,58,65,68,69,72}. The largest difference was observed between BPTB and 5SSTG (MD = 0.57 mm in favor of BPTB, 95% CrI: -0.73 to 1.88 mm). All autografts were comparable with respect to AP laxity, with no significant differences observed. The rankings indicated that BPTB was the top-performing autograft (SUCRA = 85.7%), as shown in Figure 2-D. The remaining autografts, in order of

descending rank, were QTB (SUCRA = 52.0%), 4SST (SUCRA = 47.5%), 4SSTG (SUCRA = 43.5%), and FQT (SUCRA = 39.5%), with 5SSTG showing the least relative performance (SUCRA = 31.7%).

Pivot-Shift Test

The pivot-shift NMA included 14 studies with 1,250 patients^{32,33,36,40,41,46,49,53,54,58,61,62,70,72}. QTB showed a significantly lower risk of a pivot-shift grade of 2+ or higher compared with 4SST (RR = 0.26, 95% CrI: 0.07 to 0.85). Based on the rankings (Fig. 2-E), QTB demonstrated the highest probability of outperforming other autografts (SUCRA = 88.3%), followed by 5SSTG (SUCRA = 58.1%), BPTB (SUCRA = 57.4%), and 4SSTG (SUCRA = 38.5%); in contrast, 4SST ranked lowest (SUCRA = 7.7%).

Rerupture and Revision ACLR

The NMA for rerupture or revision ACLR included 30 studies with 2,462 patients^{29,32-36,39-43,45,46,48,50,54-56,59,60,62,63,66,68-74}. No pairwise comparison indicated a significant difference in the risk of rerupture or revision ACLR. According to the rankings, QTB was the top-performing autograft (SUCRA = 83.3%). Autografts with intermediate performance were 4SSTG (SUCRA = 52.5%), BPTB (49.9%), FQT (45.5%), and 4SST (40.7%). With a SUCRA value of 28.1%, 5SSTG was identified as the least favorable autograft (Fig. 2-F).

Return to Sport

The qualitative synthesis of the return to sport comparisons included 14 studies^{28,30,35,37,39,41,45,46,50,61,64,66,73,75} and is provided in Appendix 7. No evidence of significant differences was observed between the autografts.

Convergence, Inconsistency, Publication Bias, and Sensitivity Analysis

All outcomes showed good convergence and no inconsistency between direct and indirect evidence (see Appendices 8 and 9,

Table 3. League Table Showing the Mean Difference or Risk Ratio for Each Pairwise Comparison*

IKDC: Mean Difference (95% Credible Interval)						
QTB	−1.50 (−5.72 to 2.36)	−2.59 (−7.71 to 2.25)	−2.70 (−6.11 to 0.39)	−3.18 (−8.27 to 1.31)	−3.46 (−6.77 to −0.29)*	
1.50 (−2.36 to 5.72)	4SST	−1.09 (−5.30 to 3.12)	−1.20 (−3.96 to 1.42)	−1.68 (−5.62 to 1.92)	−1.95 (−5.13 to 1.28)	
2.59 (−2.25 to 7.71)	1.09 (−3.12 to 5.30)	5SSTG	−0.11 (−4.09 to 3.66)	−0.59 (−3.91 to 2.18)	−0.86 (−5.27 to 3.59)	
2.70 (−0.39 to 6.11)	1.20 (−1.42 to 3.96)	0.11 (−3.66 to 4.09)	4SSTG	−0.48 (−4.14 to 2.97)	−0.76 (−3.01 to 1.65)	
3.18 (−1.31 to 8.27)	1.68 (−1.92 to 5.62)	0.59 (−2.18 to 3.91)	0.48 (−2.97 to 4.14)	FQT	−0.27 (−4.27 to 4.07)	
3.46 (0.29 to 6.77)*	1.95 (−1.28 to 5.13)	0.86 (−3.59 to 5.27)	0.76 (−1.65 to 3.01)	0.27 (−4.07 to 4.27)	BPTB	
Lysholm: Mean Difference (95% Credible Interval)						
FQT	−0.28 (−1.84 to 1.39)	−2.55 (−6.60 to 1.61)	−2.87 (−6.82 to 1.09)	−3.03 (−7.17 to 1.04)	−3.44 (−7.46 to 0.60)	
0.28 (−1.39 to 1.84)	5SSTG	−2.27 (−6.23 to 1.74)	−2.59 (−6.43 to 1.25)	−2.75 (−6.80 to 1.20)	−3.16 (−7.05 to 0.74)	
2.55 (−1.61 to 6.60)	2.27 (−1.74 to 6.23)	4SST	−0.32 (−2.05 to 1.08)	−0.48 (−2.59 to 1.19)	−0.89 (−2.68 to 0.65)	
2.87 (−1.09 to 6.82)	2.59 (−1.25 to 6.43)	0.32 (−1.08 to 2.05)	4SSTG	−0.17 (−1.57 to 1.09)	−0.57 (−1.35 to 0.25)	
3.03 (−1.04 to 7.17)	2.75 (−1.20 to 6.80)	0.48 (−1.19 to 2.59)	0.17 (−1.09 to 1.57)	QTB	−0.41 (−1.67 to 1.06)	
3.44 (−0.60 to 7.46)	3.16 (−0.74 to 7.05)	0.89 (−0.65 to 2.68)	0.57 (−0.25 to 1.35)	0.41 (−1.06 to 1.67)	BPTB	
Tegner: Mean Difference (95% Credible Interval)						
QTB	−0.60 (−1.88 to 0.70)	−0.66 (−2.72 to 1.43)	−0.71 (−1.87 to 0.44)	−0.81 (−1.95 to 0.35)	−1.10 (−3.29 to 1.13)	
0.60 (−0.70 to 1.88)	4SST	−0.05 (−2.11 to 2.02)	−0.11 (−1.25 to 1.01)	−0.21 (−1.33 to 0.91)	−0.49 (−2.69 to 1.70)	
0.66 (−1.43 to 2.72)	0.05 (−2.02 to 2.11)	FQT	−0.05 (−1.88 to 1.75)	−0.15 (−1.89 to 1.57)	−0.44 (−3.01 to 2.11)	
0.71 (−0.44 to 1.87)	0.11 (−1.01 to 1.25)	0.05 (−1.75 to 1.88)	BPTB	−0.10 (−0.65 to 0.46)	−0.38 (−2.35 to 1.58)	
0.81 (−0.35 to 1.95)	0.21 (−0.91 to 1.33)	0.15 (−1.57 to 1.89)	0.10 (−0.46 to 0.65)	4SSTG	−0.28 (−2.15 to 1.60)	
1.10 (−1.13 to 3.29)	0.49 (−1.70 to 2.69)	0.44 (−2.11 to 3.01)	0.38 (−1.58 to 2.35)	0.28 (−1.60 to 2.15)	5SSTG	
Instrumented Anteroposterior Laxity in mm: Mean Difference (95% Credible Interval)						
BPTB	0.26 (−0.38 to 0.92)	0.30 (−0.23 to 0.85)	0.31 (−0.07 to 0.68)	0.40 (−0.59 to 1.40)	0.57 (−0.73 to 1.88)	
−0.26 (−0.92 to 0.38)	QTB	0.04 (−0.79 to 0.85)	0.05 (−0.57 to 0.64)	0.14 (−1.04 to 1.30)	0.31 (−1.11 to 1.68)	
−0.30 (−0.85 to 0.23)	−0.04 (−0.85 to 0.79)	4SST	0.01 (−0.60 to 0.60)	0.10 (−0.74 to 0.96)	0.27 (−1.12 to 1.64)	
−0.31 (−0.68 to 0.07)	−0.05 (−0.64 to 0.57)	−0.01 (−0.60 to 0.60)	4SSTG	0.08 (−0.93 to 1.14)	0.26 (−0.99 to 1.51)	
−0.40 (−1.40 to 0.59)	−0.14 (−1.30 to 1.04)	−0.10 (−0.96 to 0.74)	−0.08 (−1.14 to 0.93)	FQT	0.17 (−1.43 to 1.77)	
−0.57 (−1.88 to 0.73)	−0.31 (−1.68 to 1.11)	−0.27 (−1.64 to 1.12)	−0.26 (−1.51 to 0.99)	−0.17 (−1.77 to 1.43)	5SSTG	
Pivot-Shift Grade ≥2+: Risk Ratio (95% Credible Interval)						
QTB	1.58 (0.22 to 11.23)	1.73 (0.71 to 4.62)	2.10 (0.90 to 5.64)	3.81 (1.17 to 14.78)*	No data	
0.63 (0.09 to 4.59)	5SSTG	1.10 (0.18 to 6.96)	1.33 (0.24 to 8.00)	2.40 (0.35 to 18.20)	No data	
0.58 (0.22 to 1.41)	0.91 (0.14 to 5.43)	BPTB	1.20 (0.74 to 2.07)	2.17 (0.86 to 6.22)	No data	
0.48 (0.18 to 1.12)	0.75 (0.12 to 4.19)	0.83 (0.48 to 1.35)	4SSTG	1.80 (0.77 to 4.62)	No data	
0.26 (0.07 to 0.85)*	0.42 (0.05 to 2.82)	0.46 (0.16 to 1.16)	0.56 (0.22 to 1.30)	4SST	No data	
No data	No data	No data	No data	No data	FQT	
Rerupture/Revision ACLR: Risk Ratio (95% Credible Interval)						
QTB	1.73 (0.64 to 5.09)	1.77 (0.65 to 5.41)	2.18 (0.04 to 107.88)	2.04 (0.57 to 8.01)	2.86 (0.63 to 17.10)	
0.58 (0.20 to 1.57)	4SSTG	1.02 (0.69 to 1.54)	1.26 (0.03 to 51.17)	1.17 (0.51 to 2.77)	1.66 (0.35 to 10.02)	
0.57 (0.18 to 1.54)	0.98 (0.65 to 1.45)	BPTB	1.24 (0.02 to 51.13)	1.15 (0.47 to 2.85)	1.62 (0.33 to 10.13)	
0.46 (0.01 to 24.94)	0.80 (0.02 to 38.65)	0.81 (0.02 to 41.16)	FQT	0.94 (0.02 to 42.89)	1.37 (0.02 to 91.48)	
0.49 (0.12 to 1.74)	0.85 (0.36 to 1.96)	0.87 (0.35 to 2.12)	1.06 (0.02 to 41.92)	4SST	1.43 (0.24 to 9.57)	
0.35 (0.06 to 1.59)	0.60 (0.10 to 2.88)	0.62 (0.10 to 3.03)	0.73 (0.01 to 44.76)	0.70 (0.10 to 4.18)	5SSTG	

*A positive value indicates that the graft named in the column is superior to the graft named in the row by the stated amount. 4SST = 4-strand semitendinosus, 4SSTG = 4-strand semitendinosus and gracilis, 5SSTG = 5-strand semitendinosus and gracilis, BPTB = bone-patellar tendon-bone, FQT = free quadriceps tendon, QTB = quadriceps tendon with bone block, IKDC = International Knee Documentation Committee Subjective Knee Form.

including Supplementary Figs. 4-A through 5-E). No publication bias was detected except in the pivot-shift outcome, where the Egger test showed significant bias (see Appendix 10, including Supplementary Figs. 6-A through

6-E). After excluding 3 outlier studies^{32,61,62} in a sensitivity analysis, SUCRA rankings for the pivot-shift remained unchanged, supporting the robustness of the primary findings.

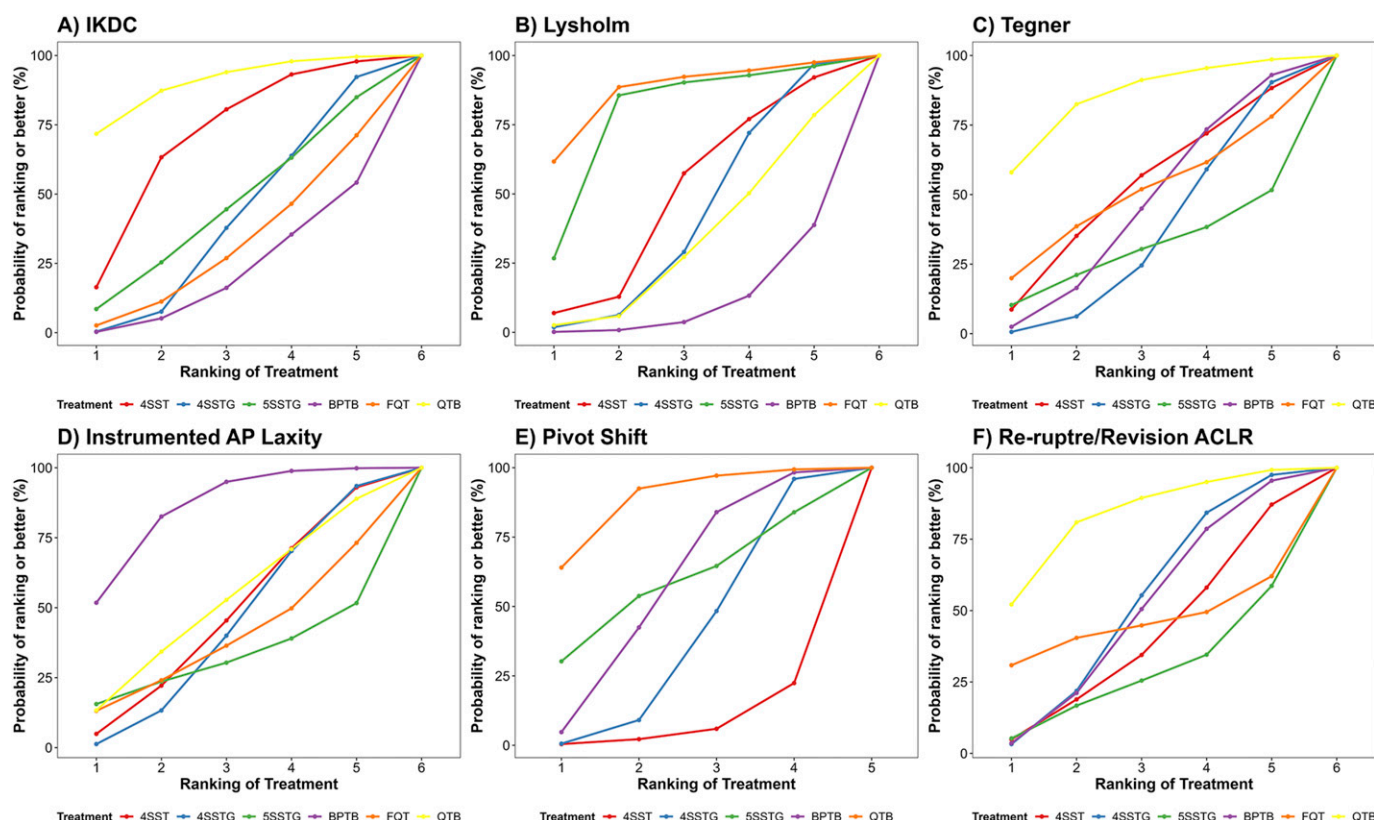


Fig. 2

SUCRA plots illustrating the ranking probabilities of the 6 autografts for each analyzed outcome: IKDC score, Lysholm score, Tegner scale, instrumented anteroposterior laxity, pivot-shift, and rerupture or revision risk.

Network Meta-Regression

The interaction between autograft type and follow-up time for rerupture or revision ACLR risk revealed no significant evidence that the RR of any comparison changed over time (see Appendix 11, including Supplementary Fig. 7).

Discussion

This study represents the most comprehensive evidence synthesis to date on autograft selection for primary ACLR. Based on our findings, QTB was identified as the most favorable autograft option overall, consistently ranking first or second in all outcomes except the Lysholm score. In contrast, BPTB was the best autograft option with respect to AP laxity but was among the least favorable options with respect to PROMs, ranking the lowest for both the IKDC and Lysholm scores. Also, considerable performance variability was observed among the HT subtypes, with no single variant emerging as consistently superior.

QT autografts performed the best overall with respect to the PROMs. Specifically, QTB was the most favorable autograft according to the IKDC score and Tegner scale, whereas FQT achieved the highest Lysholm scores. QT autografts minimize bone-block harvest burden, avoid saphenous nerve injury inherent to patellar tendon harvesting⁷⁶, and preserve superior histological integrity of the graft-tendon interface⁷⁷,

resulting in less anterior knee pain and preserved kneeling tolerance, which translate to superior PROMs⁷⁸. BPTB was the least favorable autograft with respect to both the IKDC and Lysholm scores. BPTB harvesting may directly damage the extensor mechanism, resulting in increased anterior knee symptoms, kneeling pain, and early extensor weakness or extensor lag⁷⁹, which weigh heavily on PROMs. Therefore, although BPTB demonstrates mechanical superiority, it results in comparatively worse PROMs. In the present study, a significant difference was observed between QTB and BPTB, with an MD of 3.5 in favor of QTB for the IKDC score. However, this MD falls below the established MCID, indicating that it is unlikely to be clinically meaningful^{80,81}. Previous NMAs have reported conflicting PROM findings for QT autografts, likely because some did not distinguish between QTB and FQT variants^{13,17}. In the current study, we could not identify any RCT directly comparing QTB and FQT autografts; however, our NMA indicated that, unlike QTB, FQT was not a top-performing autograft with respect to the IKDC score and Tegner scale. Therefore, future RCTs directly comparing FQT and QTB autografts for primary ACLR are encouraged.

With respect to knee stability, 4SST use was associated with a fourfold higher risk of a grade 2+ or higher pivot-shift, which was significant, compared with QTB, which was the best autograft option for this outcome. A previous meta-analysis reported that a grade 2+ or higher pivot-shift was less

frequently observed for QT compared with HT⁸². In addition, an NMA comparing a heterogeneous set of autografts, allografts, and synthetic grafts also revealed that QT achieved the highest SUCRA value compared with all other graft types based on the pivot-shift¹⁵. Our findings revealed that BPTB ranked best for instrumented AP laxity (SUCRA = 86%, highest MD = 0.6 mm). BPTB autograft may have demonstrated non-significantly superior stability primarily through bone-to-bone healing, which occurs more reliably than tendon-to-bone healing seen with soft-tissue autografts⁸³. The bone plugs undergo direct osseous integration into the bone tunnels, providing immediate mechanical fixation that may prevent graft slippage⁷. Likewise, a previous NMA comparing a heterogeneous set of grafts reported BPTB to be the most favorable option with respect to instrumented KT-1000/2000 (MEDmetric) results, with lower performance regarding all other knee stability outcomes. Importantly, our analysis demonstrated that QTB ranked as the first and second most favorable autograft for pivot-shift and AP laxity, respectively, whereas that previous NMA did not include any subtype of QT¹⁸.

In our NMA, QTB was the most favorable autograft with respect to the risk of ACL rerupture or revision ACLR. This interpretation, however, should be made with caution, as none of the comparisons reached significance. Likewise, a previous NMA noted that QTB use was associated with the lowest risk of rerupture¹⁷. Also, our systematic review revealed that return-to-sport comparisons remain limited by heterogeneity in study methods and insufficient evidence regarding newer autografts such as FQT.

This study has some limitations. Donor-site morbidity and complication rates were not included as evaluated outcomes. In addition, the included studies varied with respect to autograft preparation, fixation methods, and rehabilitation protocols, with many lacking clear reporting of these details. Furthermore, the follow-up durations varied considerably among studies, ranging between 1 and 20 years. To account for this, however, we performed a meta-regression on follow-up duration for the ACL rerupture outcome.

In summary, this study indicated that QTB consistently outperformed other autografts, showing more favorable results for most outcomes compared with BPTB and the HT subtypes. Our findings suggest that using QTB autograft may lead to improved functional, activity-related, and stability outcomes, while also reducing the risk of graft failure. BPTB performed well with respect to AP knee laxity measures but poorly with respect to PROMs, whereas comparisons among HT subtypes were inconsistent and showed no clear superiority relative to each other in any specific domain. Given the subtle and mainly nonsignificant differences among the autografts, autograft selection should remain a multifaceted decision that balances the favorable profile of QTB against the risks and benefits associated with each alternative autograft option, tailored to patient-specific goals. Future RCTs focusing more on return to sport outcomes and directly comparing FQT outcomes with QTB are encouraged.

Appendix

 Supporting material provided by the authors is posted with the online version of this article as a data supplement at <http://links.lww.com/JBJS/J136>. ■

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Network meta-analysis of knee outcomes following anterior cruciate ligament reconstruction with various types of tendon grafts

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Abstract

Purpose Various grafts were selected for restoring knee stability and regaining pre-injury activity levels after anterior cruciate ligament (ACL) rupture. This review aimed to compare the short-term knee outcomes of different tendon grafts for primary ACL reconstruction (ACLR).

Methods Databases of PubMed, Embase, and CENTRAL were retrieved for identifying the published RCT comparing the short-term (≤ 2 years) knee outcomes of different tendon grafts for ACLR. The main outcomes include the International Knee Documentation Committee (IKDC) subjective and objective scores, side-to-side difference on knee laxity, Lachman test, pivot-shift test, Lysholm score, Tegner score, all recorded complications and graft re-ruptures.

Results A total of 32 trials involving 2962 patients and 17 grafts were included. Superior IKDC subjective score was found for quadriceps tendon autograft (QTB) when compared with autografts including bone-patellar tendon-bone (BPTB), four-strand hamstring tendon (4S-HT), and double-bundle HT. The 4S-HT autograft was associated with poorer anteroposterior and rotational knee stability than the BPTB autograft. Artificial ligament also was found to provide lower knee stability compared with the autografts including the BPTB, patellar tendon, 4S-HT, 4S-HT with preserved tibial insertion, 6S-HT, and four-strand semitendinosus tendon. Additionally, the BPTB allograft showed a statistically higher knee laxity than 6S-HT allograft, while no difference was found between the genuine autografts and non-irradiated allografts.

Conclusions QTB was recommended to be used even over the BPTB and 4S-HT autografts. BPTB autograft was more effective on restoring the knee stability than 4S-HT autograft. Non-irradiated allografts could be safely used as alternatives to autografts. However, artificial ligaments were not recommended for primary ACLR for significantly increased risk of knee laxity.

Keywords Anterior cruciate ligament · Tendon autograft · Tendon allograft · Artificial ligament · Knee function · Knee stability

Xiong-gang Yang and Feng Wang contributed equally to this work.

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Introduction

The anterior cruciate ligament (ACL) tear is one of the most frequently encountered knee injuries in orthopaedic surgery. As such, the incidence of ACL reconstruction (ACLR) has increased over the past few decades [1, 2]. Following the disruption of the ACL, patients typically report symptoms including pain, difficulty in sports, and the knee giving way [3]. Non-surgical treatments have been demonstrated to be associated with suboptimal knee outcomes and often lead to joint deterioration in ACL-deficient patients [4]. Therefore, to adequately restore knee stability and regain pre-injury activity levels, many operative reconstruction managements, including autogenous or allogeneic tendon grafts and artificial

ligaments, have been proposed. However, each graft has its proponents, and orthopedic surgeons continue to debate about which graft is optimal for ACLR.

The central third bone-patellar tendon-bone (BPTB) autograft was firstly introduced in the 1980s for ACLR [5]. It has potential for accelerated bone-to-bone healing and a predictable success rate for restoration of knee stability, for which it was regarded as the “golden standard” in ACLR. However, concerns about the donor-site complications after BPTB autograft harvesting, such as anterior knee pain, crepitus, loss of sensation, patellar fracture, inferior patellar contracture, and loss of extension force, have been proposed in numerous studies [6, 7]. Therefore, the hamstring tendon (HT) autograft has been developed as an alternative to BPTB autograft, resulting in less damage to the extensor apparatus as well as less anterior knee pain and patellar fractures [8]. However, this autograft is also not without shortcomings, as numerous studies have reported that tunnel widening, flexor weakness, and increased knee laxity were more likely to result from the HT than the BPTB autograft [6, 9]. The quadriceps tendon (QT) autograft is another alternative graft of BPTB autograft, which was firstly suggested for ACLR by Marshall et al. [10] in 1979. This autograft has aroused wide concern about the post-operative quadriceps muscle weakness and QT rupture [11]. However, this autograft has recently gained more popularity, as it has demonstrated similar post-operative knee function and stability as well as less donor-site pain when compared with the BPTB and HT autografts [12].

Despite the high success rate in the reconstruction procedures with autograft, surgeons have gradually accepted the use of allogeneic tendons and artificial ligaments, following the advent of some advanced fixation methods and processing technics. All donor site complications would be avoided, and the surgery and recovery time could be shortened. However, the risks of disease transmission, immune rejection, and delayed tendon-bone healing for allogeneic tendon transplantation have become the major challenges for clinicians [13]. Additionally, pre-treatment procedures on the tendon allografts, including both cryopreservation and irradiation sterilization, inevitably cause damage to the mechanical and histological properties, leading to an increased likelihood of re-rupture of the reconstructed ACL [14, 15]. With the use of synthetic ligaments, there are no risks of donor site complications and disease transmission, and fast rehabilitation is permitted [16]. However, the high rate of re-rupture, the depressed ligamentization process, and the increased knee laxity have limited the routine application of such ligaments in clinical practice.

Though many researchers have now compared the treatment outcomes of ACLR with various tendon grafts, disagreements were ubiquitous among the available evidences and there are no directive and informative guidelines, meaning that difficulties remain for surgeons in their choice of an optimal graft. In this study, we will perform a systematic review and network meta-analysis basing on available evidences from randomized controlled trials

(RCTs), with the aim of providing ranking plots for the tendon grafts on the clinical outcomes in a short-term follow-up.

Materials and methods

Study eligibility and selection

The current study was conducted following the guidelines outlined in the Preferred Reporting Items for Systematic Reviews and Meta-analysis (PRISMA) statement [17]. Our systematic review was prospectively registered, and the protocol is available to be read on PROSPERO (registration number CRD42019120661).

RCTs which included patients who had a diagnosis of ACL tears and who had been treated with operative reconstruction using various tendon grafts, comparing short-term (≤ 2 years) knee outcomes, were eligible. Studies were excluded according to the following criteria: (1) duplicated studies; (2) studies which were designed as observational or non-randomized clinical studies, literature reviews, systematic review and/or meta-analyses, case series, case reports or letters to editors; (3) studies without a full-text available. We only included studies which had been published in English.

Two researchers independently conducted database searches on PubMed, Embase, and the Cochrane Central Register of Controlled Trials (CENTRAL) from the inception to November 2018. Detailed searching strategies used in the three electronic platforms are available in Appendix 1.

After initial database searches using these strategies, retrieved studies were checked for duplicate. Following the exclusion of duplicated studies, the titles/abstracts and full-texts of the remaining records were screened for final eligibility. Finally, some other potentially related articles were manually searched after screening the reference sections of head-to-head meta-analyses and other studies.

Data collection and assessment of risk of bias

Two reviewers independently extracted the following data from the included studies: (1) study details: lead author's name, year of publication, study design, location, length of follow-up, and study period; (2) participant details: number of patients, number of participants lost to follow-up, age, gender, injured side, interval between injury and surgery, and meniscal and cartilage status; (3) intervention details: graft type used, fixation methods for both the femoral and the tibial sides, and post-operative rehabilitation program; (4) outcome details: International Knee Documentation Committee (IKDC) subjective score, IKDC objective score, Lachman and pivot-shift tests, side-to-side difference of knee laxity between the operated and contralateral sides by arthrometer (such as KT-1000 or KT-2000), Lysholm score, Tegner score,

overall incidence of complications, and frequency of graft re-rupture. Intent to treatment data was extracted where it was available.

The risk of bias assessment was carried out by two researchers utilizing the Cochrane Collaboration tool [18]. This detects the following seven potential biases for RCTs: randomization sequence generation, allocation concealment, blinding of both participants and personnel, blinding of outcome assessment, incomplete outcome data, selective reporting, and other bias. Each parameter is set as either low risk of bias, high risk of bias, or unclear.

Data synthesis and statistical analysis

We generated network plots for each outcome to illustrate which interventions were compared in available RCTs directly, utilizing the “network” suite of commands for Stata. When outcomes were reported as ordered categorical data, such as the IKDC objective score, Lachman test, pivot-shift test, and side-to-side difference, they were transferred to dichotomous variables as normal (including both normal and near normal grades) and abnormal (including abnormal and obviously abnormal grades) groups, and the odd ratio (OR) was selected as the effect size for further combining, while the mean difference (MD) was used for continuous outcomes. When the outcomes were presented for multiple time points within two years, only the longest follow-up period was selected. In terms of the transferred dichotomous outcomes, a classic half integer continuity correction (adding a 0.5 to each cell) was performed when studies reported zero events in any of the treatment arms.

We conducted all network meta-analyses using the “network” suite of commands in Stata. Prediction interval plots were used to present treatment effect estimates, therefore displaying estimated summary effects with both the 95% credibility interval (95% CI) and the 95% prediction interval (95% PrI). Following this, treatments were ranked according to their estimated effect sizes using the surface under the cumulative ranking curves (SUCRA), showing which treatment ranked the highest, second highest, and so on. Additionally, SUCRA values of the interventions were plotted into radar maps, which coherently displayed treatment effects on various outcome evaluations.

In order to evaluate the inconsistency, the “node-splitting approach” was used to detect the presence of heterogeneity between direct and indirect estimates [19]. Inconsistency plots and heterogeneity variance parameters (τ^2) were used to display the magnitude of inconsistency between the direct and indirect estimates for each loop. Ratio of the two odds ratios (RoR) in the loop were estimated, and the lower 95%CI limit of RoR value close to one was interpreted to mean there was no local inconsistency. We also used a comparison-adjusted funnel plot to assess the presence of small-sample effect in the

network meta-analyses. This study only included a subgroup of patients who had been followed for no more than two years.

Statistical procedures were conducted using the Stata version 14.0 (StataCorp LLC, College Station, TX, USA). Statistical significance was defined as a two-sided *P* value of less than 0.05.

Results

Studies included

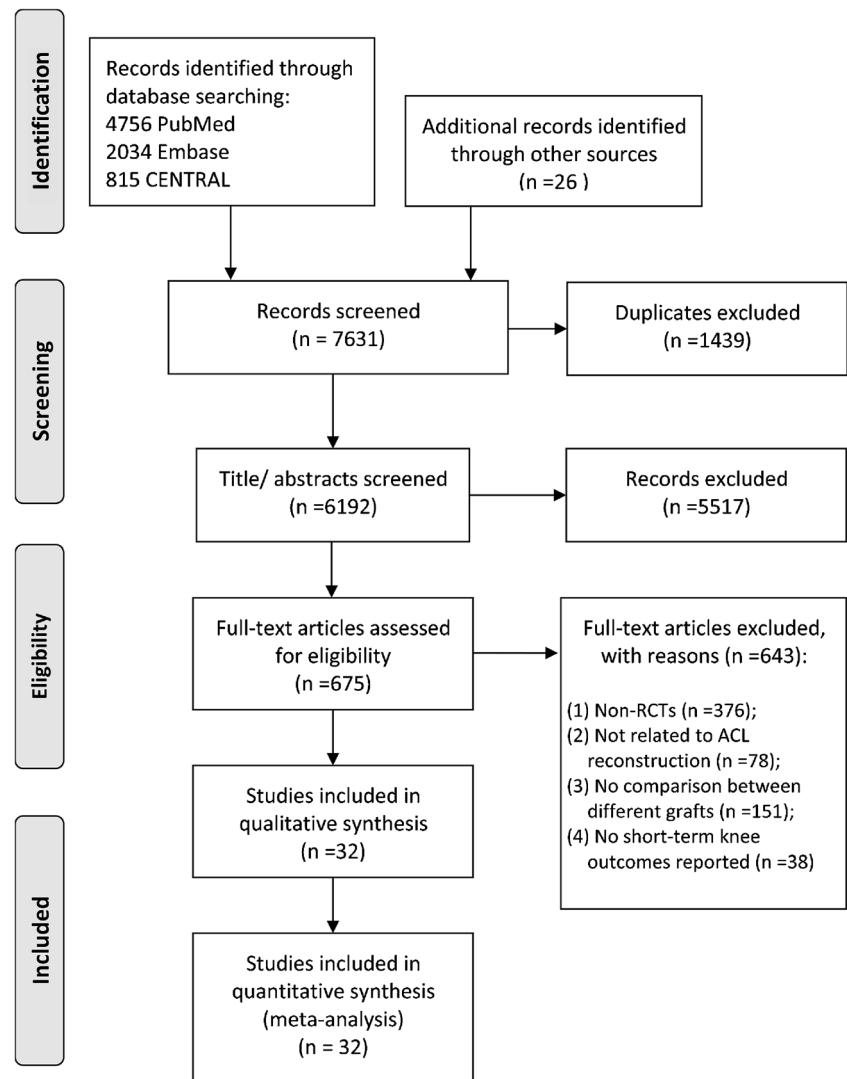
Figure 1 shows the process of searching and selecting studies. The initial retrieval from databases yielded a total of 7605 studies, while 26 potentially eligible studies were manually found, giving a total of 7631 studies. After excluding 1439 duplicates, abstracts and titles of the remaining 6192 studies were screened, while 5517 were excluded as they were not relevant. Following this, 675 articles were considered to be potentially relevant and thus, their full texts were screened. Finally, a total of 32 RCTs [20–51] were included, both for qualitative and quantitative syntheses.

A total of 2962 patients, including 1680 (62.5%) males and 1009 (37.5%) females (the gender of 273 patients was not available), were enrolled. Seventeen different types of grafts were used for ACLR in these patients (Table 1). Table 2 shows the summary of the included studies. Two-arm studies predominated ($n = 30$, 93.8%), with only two studies [41, 43] (6.3%) having three arms. The mean/median age of patients tended to be between 25 and 30 years. The most common treatment arms were the BPTB and four-strand HT autografts, both of which were involved in 22 arms (33.3%). Pre-operative evaluations of the knee function and stability were available in 20 studies [21, 24–27, 30, 31, 33, 34, 36, 38, 39, 41–45, 47, 50, 51] and can be seen in Table S1. Almost all of these comparisons were found to be similar between groups.

Different fixation methods were chosen on the femoral and tibial sides as shown in Figure S1. A summary of the intra-operative meniscal and cartilage status and the rehabilitation protocol are listed in Table S2 and S3, respectively. The application of the protective knee brace was available in 22 studies [20–31, 33, 35, 37, 38, 40, 41, 44–46, 51], eight of which [20, 21, 24, 25, 30, 33, 38, 45] used a brace for protection during the early rehabilitation period.

Summaries of the risk of bias and the risk of bias graph for each study are presented in Fig. 2. In general, the studies included were of a low risk of bias in terms of the random sequence generation, incomplete outcome data, selective reporting, and other bias. However, some studies were found to be have a high risk of bias for allocation concealment, blinding of participants, and personnel as well as blinding of outcome assessment.

Fig. 1 PRISMA flowchart for the searching and selecting of studies



Network meta-analysis of knee function and stability outcomes

The studies, number of patients, and graft types involved in the network for each outcome can be seen in Table S4. Figure 3a–h shows the network plots for knee function and stability outcomes, and the corresponding prediction interval forest plots are presented in Figure S2A–H. There was no evidence that one graft was better than another in terms of the short-term IKDC score, Lachman test, side-to-side difference (categorical data), or Tegner score. Significant comparisons for each outcome are presented in Fig. 4a–d. This shows that the QT autograft with bone block (QTB) has a significantly superior IKDC subjective score than the BPTB autograft, the four-strand HT autograft, and the double-bundle HT autograft, while the four-strand HT autograft has a higher anteroposterior and rotational knee laxity than the BPTB autograft, according to the side-to-side difference (continuous data) and pivot-shift test, respectively. The artificial ligament

was found to be inferior to many other autografts, including the BPTB, PT, four-strand HT, four-strand HT with preserved insertions (HTB), six-strand HT, and four-strand semitendinosus tendon (ST) in terms of the anteroposterior stability according to side-to-side difference, while it was found to be similar to the two-strand HT and three-strand ST autografts. The artificial ligament was found to be superior to the BPTB, BPT, PT, and three-strand ST on the Lysholm score. The BPTB allograft also showed a higher anteroposterior and rotational knee laxity than the six-strand HT allograft according to side-to-side difference and pivot test.

SUCRA ranking plots are shown in Figure S3A–H. We also plotted radar maps depicting the magnitude of SUCRA values for the knee outcomes after they were reconstructed with various tendon grafts (see Fig. 5). When compared with other grafts, patellar autografts (BPTB, BPT, and PT) were shown to have modest short-term knee function outcomes, while the stability is favourable for BPTB autograft but is lowered for

Table 1 Types of the grafts used for ACLR in the included studies

No.	Abbreviated names of grafts	Full names of grafts
1	BPTB-auto	Bone-patella tendon-bone autograft
2	BPT-auto	Patella tendon autograft with bone block on one side
3	PT-auto	Patella tendon autograft without bone block
4	NonIrra-BPTB-allo	Non-irradiated bone-patellar tendon-bone allograft
5	2S-HT-auto	Two-strand hamstring tendon autograft
6	4S-HT-auto	Four-strand hamstring tendon autograft
7	6S-HT-auto	Six-strand hamstring tendon autograft
8	4S-HTB-auto	Four-strand hamstring tendon autograft with Preserved tibial insertion
9	DB-HT-auto	Double-bundle hamstring tendon autograft
10	4S-Contral-HT-auto	Four-strand hamstring tendon autograft harvested from contralateral knee
11	6S-NonIrra-HT-allo	Six-strand non-irradiated hamstring tendon allograft
12	3S-ST-auto	Three-strand semitendinosus tendon autograft
13	4S-ST-auto	Four-strand semitendinosus tendon autograft
14	2S-NonIrra-Achilles-allo	Two-strand non-irradiated Achilles' tendon allograft
15	NonIrra-TA-allo	Non-irradiated tibialis anterior tendon allograft
16	QTB-auto	Quadriceps tendon autograft with bone block
17	Artificial-ligament	Artificial ligament

A total of 17 types of tendon grafts were involved, and this table describes the code, abbreviation, and full name of each graft correspondingly

BPT and PT autografts. When compared with patellar autografts, no obvious differences on knee functional outcomes were found in hamstring grafts, while HT autografts which used no more than four strands were likely to be inferior to patellar tendons on knee stability. However, four-strand HT autografts with preserved bone insertions were associated with better stability than patellar tendons. With the increase on the cross-sectional area of HT autograft, there was a tendency for corresponding elevated anteroposterior and rotational stability. Artificial and allogeneic grafts were generally associated with inferior stability but relatively better short-term knee function.

Network meta-analysis of all recorded complications and re-ruptures

A total of 28 (2645 patients, 16 grafts) of the 32 studies reported data about the number of recorded complications while 22 (1859 patients, 13 grafts) reported data about re-ruptures (Table S4). Network plots and SUCRA ranking plots are shown in Figure S4 and Figure S5. Prediction interval plots for all recorded complications and re-ruptures are presented in Figure S2I–J, while significant comparisons are displayed in Fig. 4E. The double-bundle HT autograft was found to be associated with significantly more complications than the two- or four-strand HT and the BPTB autografts. There was no significant difference in the incidence of re-rupture among the grafts. According to the SUCRA ranking plot, the four-strand contralateral HT autograft had the highest probability of re-rupture. In addition, allogeneic grafts (especially for BPTB,

TA, and Achilles) were more likely to suffer re-rupture than autogenous grafts (patellar tendons, QTB, and HT). Patellar tendons had a tendency of lower re-rupture incidence than three- or four-strand ST and four-strand HT, while the six-strand HT and QTB grafts had the lowest re-rupture rate.

Inconsistency assumption and small sample effect

The local inconsistency results for the closed loops are shown in Fig S6, S4C, and S5C. All the lower limits of the 95% CI of RoRs are shown as “1,” and Z tests on the heterogeneity were all non-significant ($P > 0.05$). All local inconsistency detection results based on the “node-splitting model” were also non-significant (Table S5).

The comparison-adjusted funnel plots for knee functional/stability outcomes, complications, and re-rupture events are shown in Figure S7, S4B, and S5B. A subjective symmetry was found, indicating that no obvious small-sample effect exists to increase the risk of publication bias.

Discussion

To our knowledge, this is the first network meta-analysis of various tendon grafts for ACLR that has provided a rank order for those grafts' individual clinical outcomes. The currently published studies have not given a legible tutorial on graft selection for ACLR, meaning evidence is needed for the development of clinical treatment guidelines. Based on the available direct and indirect evidence, a general description for the

Table 2 Summary of included studies

Study ID	Study period	Country	Type of tendon graft	Patients (N)	Dropped patients (N)	Male (%)	Age (years)	Time from injury to surgery	Fixation approach (F = femoral, T = tibial)	Follow-up
Maletis GB, 2007 [20]	2000–2003	USA	BPTB-auto	46	0	67.4	Mean, 27.2 (range 15–42)	< 3 m, 11; 3–6 m, 14; > 6 m, 21	F&T: poly L-lactide bioabsorbable interference screws	Mean, 26.2 (range 23–45)m
Beynon BD, 2002 [21]	1990–1991	USA	4S-HT-auto	53	3	85.0	Mean, 27.7 (range 14–48)	< 3 m, 9; 3–6 m, 12; > 6 m, 32	F&T: interference screw	36 m
			BPTB-auto	28	0	64.3	Mean, 28.5 (range 18–46)	Median, 3.6 (range 0.6–10.2)m	F: a belt-buckle-type double-staple technique; T: staple	
			2S-HT-auto	28	0	46.4	Mean, 29.9 (range 19–42)	Median, 2.6 (range 0.6–9)m	F&T: bioabsorbable interference screw (poly-L-lactic acid screw)	Mean, 19.3 (range 12–25)m
Laoruengthana A, 2009 [22]	2002–2004	Thailand	6S-HT-auto	17	4	88.2	Mean, 29.9	Mean, 35.46 (range 2–108)m	F&T: bioabsorbable interference screw (poly-L-lactic acid screw)	Mean, 18.9 (range 12–31)m
Aune AK, 2001 [23]	N/A	Norway	BPTB-auto	17	2	94.1	Mean, 28.7	Mean, 20.0 (range 3–30)m	F&T: titanium interference screw	24 m
			BPTB-auto	35	6	54.3	Mean, 25.0 ± 7.0	Mean, 41.3 ± 41.0 m	F: titanium button; T: interference screw	
			4S-HT-auto	37	5	56.8	Mean, 27.0 ± 9.0	Mean, 40.5 ± 41.6 m		
Barenius B, 2010 [24]	1995–1997	Sweden	BPTB-auto	84	6	42.7	Mean, 33.0 ± 6.3	Mean, 16.0 ± 24.0 m	F&T: interference screw	Mean, 100.8 ± 11.8 m
			4S-ST-auto	80	5	57.3	Mean, 35.0 ± 7.5		F: EndoButton; T: suture-over-post fixation	
Dai C, 2015 [25]	2007–2009	China	6S-NonIrra-HT-allo	69	8	59.0	Mean, 30.0 ± 6.0	Mean, 10.0 ± 7.0 m	F: EndoButton; T: bioabsorbable interference screw	Mean, 52 (range 30–68)m
			NonIrra-BPTB-allo	60	8	67.3	Mean, 29.0 ± 5.0	Mean, 8.0 ± 6.0 m	F&T: bioabsorbable interference screw	
Drogset JO, 2009 [26]	2001–2004	Norway	4S-HT-auto	57	10	63.5	Median, 27 (range 18–50)	Mean, 12.0 (range 0–240)m	F: Bone Mulch Screw; T: WasherLoc fixation	24 m
			BPTB-auto	58	8		Median, 26 (range 18–49)	Mean, 13.0 (range 2–180)m	F: cannulated interference screw; T: interference screw	
Ejerhed L, 2003 [27]	N/A	Sweden	BPTB-auto	34	2	65.6	Median, 26 (range 14–49)	Median, 11 (range 2–252)m	F&T: Acuflex “silk” interference screw	24 m
			4S-ST-auto	37	3	73.5	Median, 29 (range 15–59)	Median, 17.5 (range 3–360)m	F&T: RCI interference screw	
Feller JA, 2001 [28]	N/A	Australia	4S-HT-auto	31	0	74.2	Mean, 26.2 ± 6.2	Na	F: EndoButton; T: cannulated metallic interference screw	4 m
			BPTB-auto	34	0	70.6	Mean, 27.1 ± 6.2			
Feller JA, 2003 [29]	1996–1998	Australia	4S-HT-auto	31	5	74.2	Mean, 25.8 ± 6.0	Mean, 3.3 ± 3.5 m	F: EndoButton; T: cannulated metallic interference screw	36 m
			BPTB-auto	34	2	70.6	Mean, 26.3 ± 6.0	Mean, 4.6 ± 5.4 m		
	1995–1999	UK		24	2	87.5	Mean, 31.7 ± 1.4		N/A	60 m

Table 2 (continued)

Ghalayini SR, 2010 [30]	Artificial- ligament	PT-auto	26	73.1	Mean, 30.9 ± 1.4	Mean, 55.0 (range 35–75)m	
						Mean, 33.0 (range 19–47)m	
Gifstad T, 2013 [31]	Norway	BPTB-auto	58	63.2	Median, 27 (range 18–49)	NA	F&T: metallic interference screws
		4S-HT-auto	56				F&T: Bone Mulch Screw and the WasherLoc
							N/A
Gobbi A, 2006 [32]	Italy	PT-auto	50	N/A	N/A	NA	24 m
		4S-HT-auto	50				
		4S-HTB-auto	55	96.4	Mean, 27.0 ± 7.5	More than 3 weeks	F: EndoButton; T: based on preserved tibial insertions only
Gupta R, 2017 [33]	India	4S-HT-auto	55	94.5	Mean, 27.2 ± 5.7		F: EndoButton; T: bioabsorbable interference screw
							F&T: interference screw
							N/A
Beard DJ, 2001 [34]	UK	4S-HT-auto	30	N/A	N/A	NA	12 m
		PT-auto	30				
		BPTB-auto	34	64.7	Mean, 29.0 ± 7.0	Median, 7.8 (range 5–12)m	24 m
Heijne A, 2010 [35]	Sweden	4S-HT-auto	34	41.2	Mean, 30.0 ± 9.0	Median, 8.5 (range 5–18)m	
Iliopoulos E, 2016 [36]	Greece	BPTB-auto	10	100.0	Mean, 24.8 ± 5.0	NA	F&T: interference screws
		4S-HT-auto	10	100.0	Mean, 26.7 ± 7.2		F: Endobutton; T: interference screw
Jansson KA, 2003 [37]	Finland	BPTB-auto	51	N/A	N/A	Median, 19.0 (range 0.2–240)m	F&T: interference metal screws
		4S-HT-auto	48			Median, 16.0 (range 0.5–120)m	F: small metal plate; T: cortical screw secured with a spiked washer
Kang HJ, 2014 [38]	China	NonIrra-BPTB- allo	43*	46.5	Mean, 30.0 ± 5.0	Mean, 2.1 ± 1.4 m	F&T: interference screws
		NonIrra-TA-allo	41*	51.2	Mean, 28.0 ± 5.0	Mean, 2.3 ± 1.2 m	F: EndoButton; T: bioabsorbable interference screws
							F: cross-pin fixation; T: interference screw
Kautzner J, 2015 [39]	Czech Republic	BPTB-auto	75	0.0	Mean, 26.0 (range 17–47)	Mean, 2.5 (range 0.1–6) m	F: suspension fixation; T: interference screw
		4S-HT-auto	75	0.0			F: cross-pin device; T: washer with multiple spikes and a compression screw
Lawhorn KW, 2012 [40]	USA	4S-HT-auto	74	59.3	Mean, 32.0	Within 12 m	More than 24 m
		NonIrra-TA-allo	73	79.2	Mean, 33.3		
Laxdal G, 2005 [41]	Sweden	BPTB-auto	42	72.5	Median, 28 (range 16–52)	Median, 14.5 (range 2–240) m	Mean, 26.0 (range 20–43)m

Table 2 (continued)

																			F&T: Acufex “silk” interference screw
																			F&T: RCI interference screw
																			F&T: RCI interference screw
																			F&T: interference screws
																			More than 24 m

Table 2 (continued)

Eriksson K, 2001 [51]	1995– 1997	Sweden	BPTB-auto	50	8	64.5	Mean, 27.0 ± 6.0	NA	F&T: titanium interference screw F: Endobutton; T: titanium interference screws	6 m
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*Only the number of patients at the final follow-up was reported in the primary studies; The abbreviations as well as the full names for the grafts are listed in Table 1

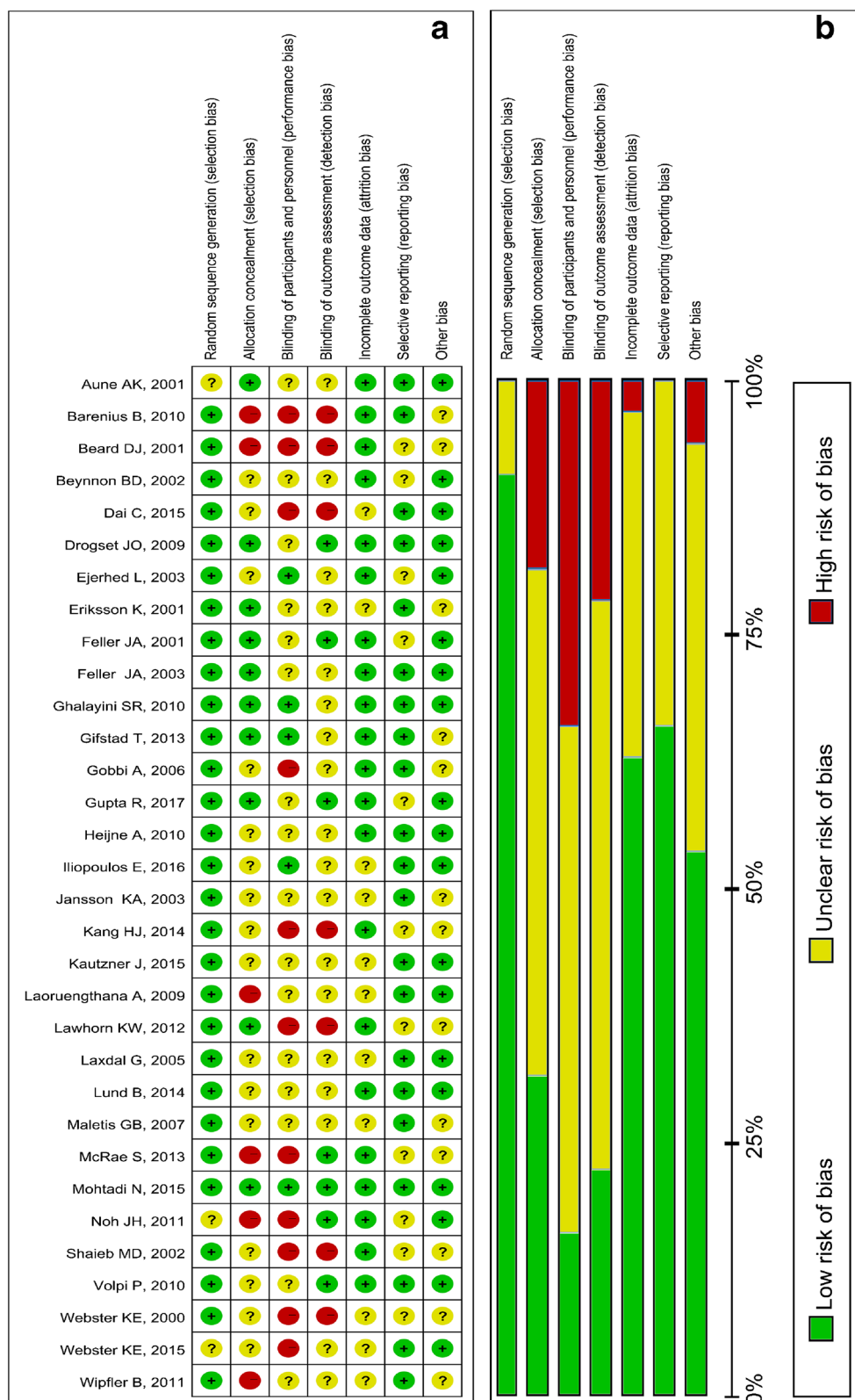
knee function, stability, complication, and re-rupture considerations of each graft was provided, offering a guidance that indicates which graft should be used.

Generally, reconstruction procedures using the BPTB or four-strand HT autografts are the most commonly discussed two treatments. Similar to observations from previous comparative studies, we confirmed that the BPTB autograft provides better knee stability than the four-strand HT autograft [52, 53]. Theoretically, as shown in former studies, the initial failure load of an evenly tensioned four-strand HT is approximately 4090 N, which is stronger than 2977 N for a 10-mm patellar tendon or 2160 N for an intact ACL [54–56]. This meets the principle consideration of strength for choice of grafts. From a biologic standpoint, however, it is well accepted that bone-to-bone healing (typically 4–6 weeks) is more accelerated than tendon-to-bone healing (typically 8–12 weeks) [57]. Recent research has also reported more fibrocartilage healing in the PT-bone interface while more fibrous healing in the HT-bone interface [58]. Therefore, the BPTB autograft is associated with more rigid healing both at the bone block-to-tunnel and the tendon section-to-tunnel interfaces. Additionally, when using suspensory fixation as the main approach to fix the HT graft, the distance between the articular surface and the point of fixation is increased, causing greater longitudinal (bungy effect) and transverse (windshield-wiper effect) motions. Therefore, a larger number of patients in the four-strand HT group were reported to have femoral tunnel widening [59].

As the least studied and used autografts, recent studies have shown excellent clinical results and low morbidity using the QTB autograft, especially when it is harvested as a free tendon without a bone block [60, 61]. In our results, the QTB autograft showed similar knee function and stability, while more of these patients reported satisfaction according to subjective IKDC score when compared with the BPTB, four-strand HT, and double-bundle HT autografts. The advantages of the QTB autograft include preservation of hamstring anatomy and function as well as less damage to the infrapatellar branch of saphenous nerve, thereby decreasing incidences of anterior knee pain and loss of sensitivity [62]. In addition, tendon defect after BPTB harvest may cause scar tissue, which leads to fat pad hypersensitivity and increased anterior knee pain when receiving a pressure load, such as during kneeling [63]. Though no more overall complications were related to BPTB, four-strand HT, and double-bundle HT autografts than the QTB autograft, complications of these grafts are generally more painful, resulting in an adverse effect on patients' subjective satisfaction. We therefore recommend that QTB autograft may be one of the best choices and should be used even over the BPTB and four-strand HT autografts.

In 1997, Kim et al. [64] proposed a new treatment option by preserving tibial insertions of HT when harvesting the graft. Since the blood supply of the tendon remains, the

Fig. 2 Risk of bias graph: the percentages of “high risk of bias,” “low risk of bias,” and “unclear risk of bias” for each item are presented in a bar diagram



healing of the tendon may be promoted, which allows rapid incorporation into the bone tunnel, minimizing risk of graft elongation. In addition, this technique helps preserve

mechanoreceptors in grafts [33, 57]. Our results fit well with former studies which show better anteroposterior stability for the four-strand HTB than for the four-strand HT autograft,

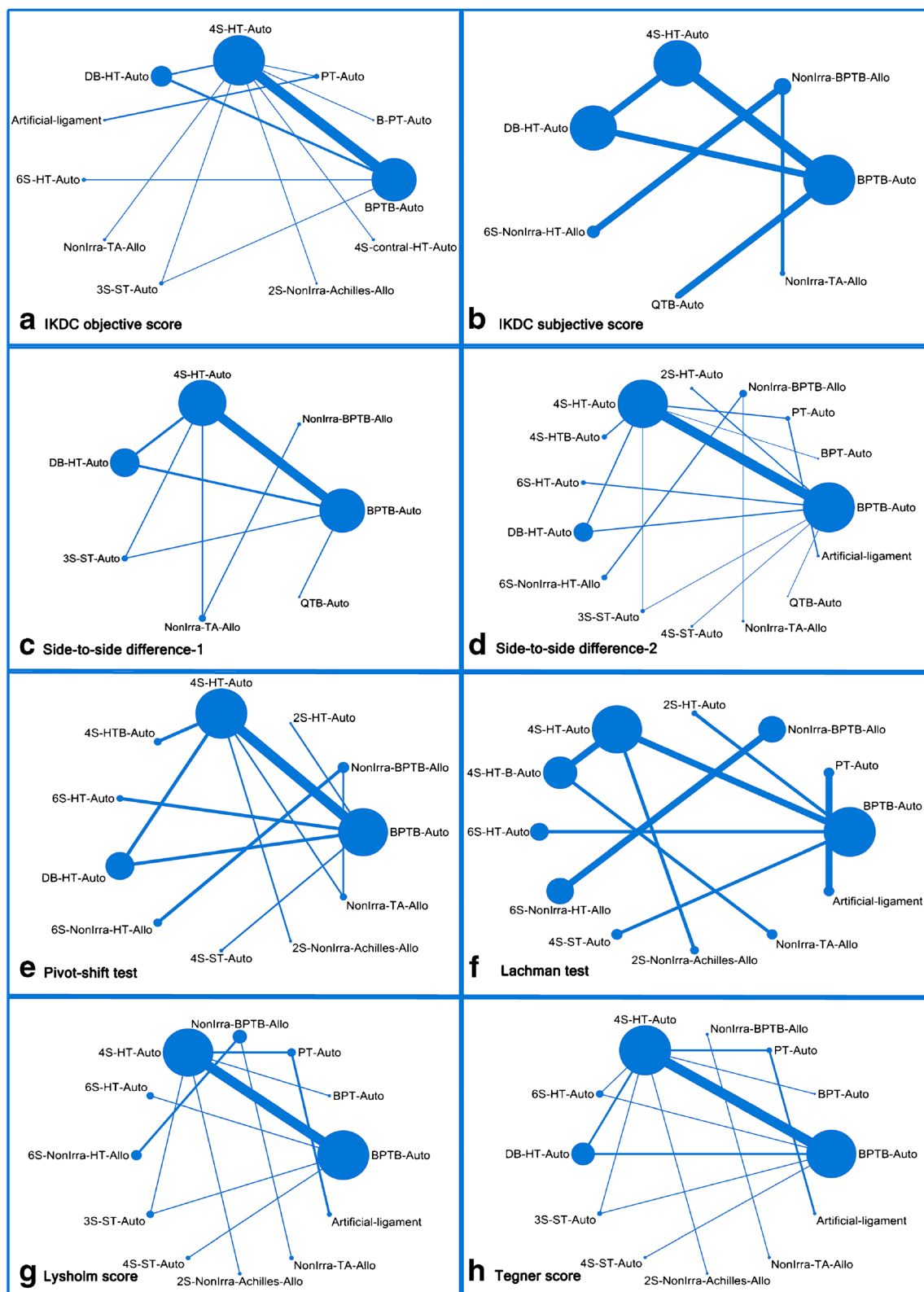
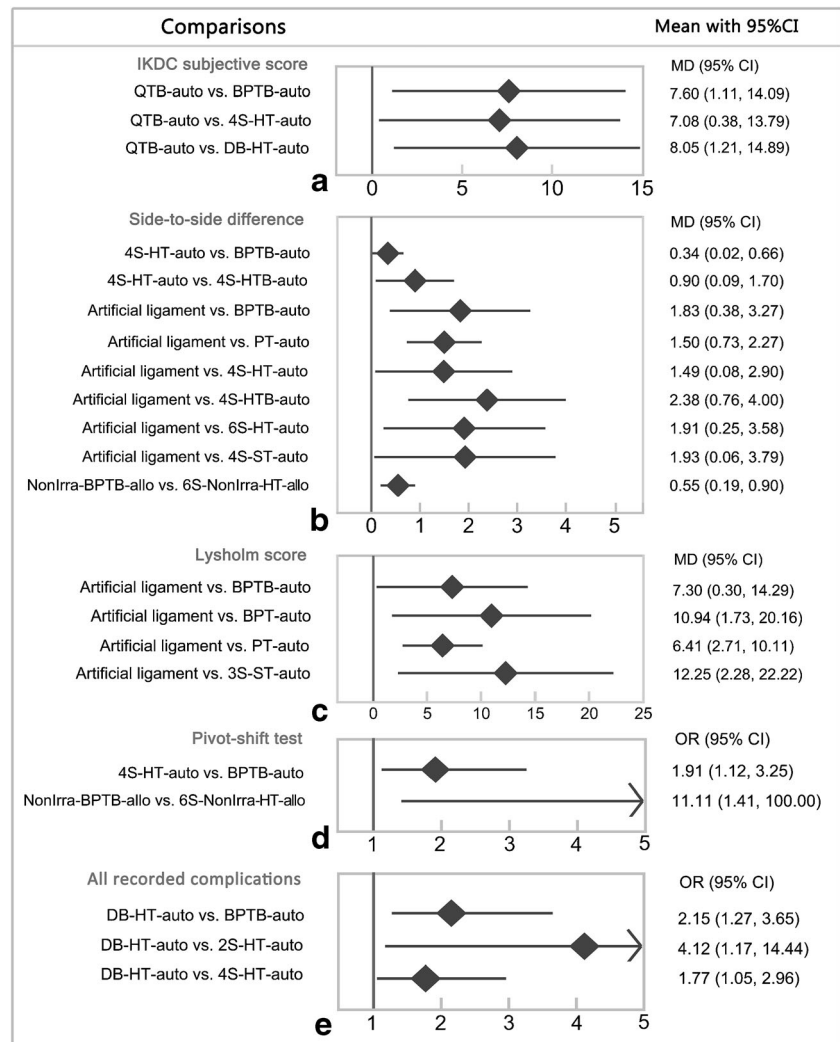


Fig. 3 Network plots for knee function and stability outcomes, including IKDC objective score (**a**), IKDC subjective score (**b**), side-to-side difference (reported as categorical data) (**c**), side-to-side difference (reported as continuous data) (**d**), pivot-shift test (**e**), Lachman test (**f**), Lysholm score (**g**), and Tegner score (**h**). Each node represents a treatment modality used

for ACLR, while each line represents a direct comparison between two treatments. The nodes and lines are weighted by the numbers of related patients and studies. Abbreviations as well as the corresponding full names of grafts are listed in Table 1

Fig. 4 Forest plots depicting the significant comparisons for each outcome in the network meta-analysis. No comparison was presented to be significant for IKDC objective score, side-to-side difference (continuous data), Lachman test, Tegner score, and re-ruptures. Abbreviations and full names of the grafts are listed in Table 1



indicating that the HTB autograft may be a better option for the reconstruction technique.

Many studies have focused on using allografts for ACLR following the advent of improved fixation device and tunnel techniques. The benefits of allografts include an absence of donor site morbidity, shortened surgical time, availability of large grafts, and no damage to the extensor and flexor mechanism [65]. However, to eliminate risks of rejection and disease transmission, treatments using cryopreservation and irradiation are often introduced to the grafts. These have been widely proved to be damaging to the biomechanical and histological properties [14, 15]. Recent meta-analyses have concluded that allografts are associated with less-than-acceptable knee outcomes when compared with autografts [66, 67]. However, in the subgroup of non-irradiated allograft, most studies reported a similar clinical outcome [67–69]. In our study, only a few non-irradiated allografts were available to be compared with autografts, and no difference in clinical outcome was found. We therefore suggest that non-irradiated allografts might be used as an alternative to autogenous

tendon grafts, while a conservative rehabilitation program should be advocated for allografts as the healing is relatively slow.

Artificial ligament is a synthetic material designed to provide a scaffold for the in-growth of neo-ligament and is related to a lack of donor site morbidity and shortened surgical time [70]. However, several histology and arthroscopy studies have suggested that artificial ligaments were not particularly useful as a scaffold-type graft, as they lack collagenization and neo-ligament in-growth [70, 71]. Chronic inflammatory responses have been reported in ruptured Leeds-Keio (LK) artificial ligaments to inhibit in-growth of the neo-ligament [72]. This may be why artificial ligaments were shown to provide inferior knee stability than almost all autografts. Thus, we do not suggest that artificial ligaments be selected for primary ACLR over other grafts, while they could be used for revision procedures if other options are not available, or used in combination with other grafts when carrying out complex ACLR. Additionally, to promote tendon graft integration and ligamentization following ACLR with artificial ligament,

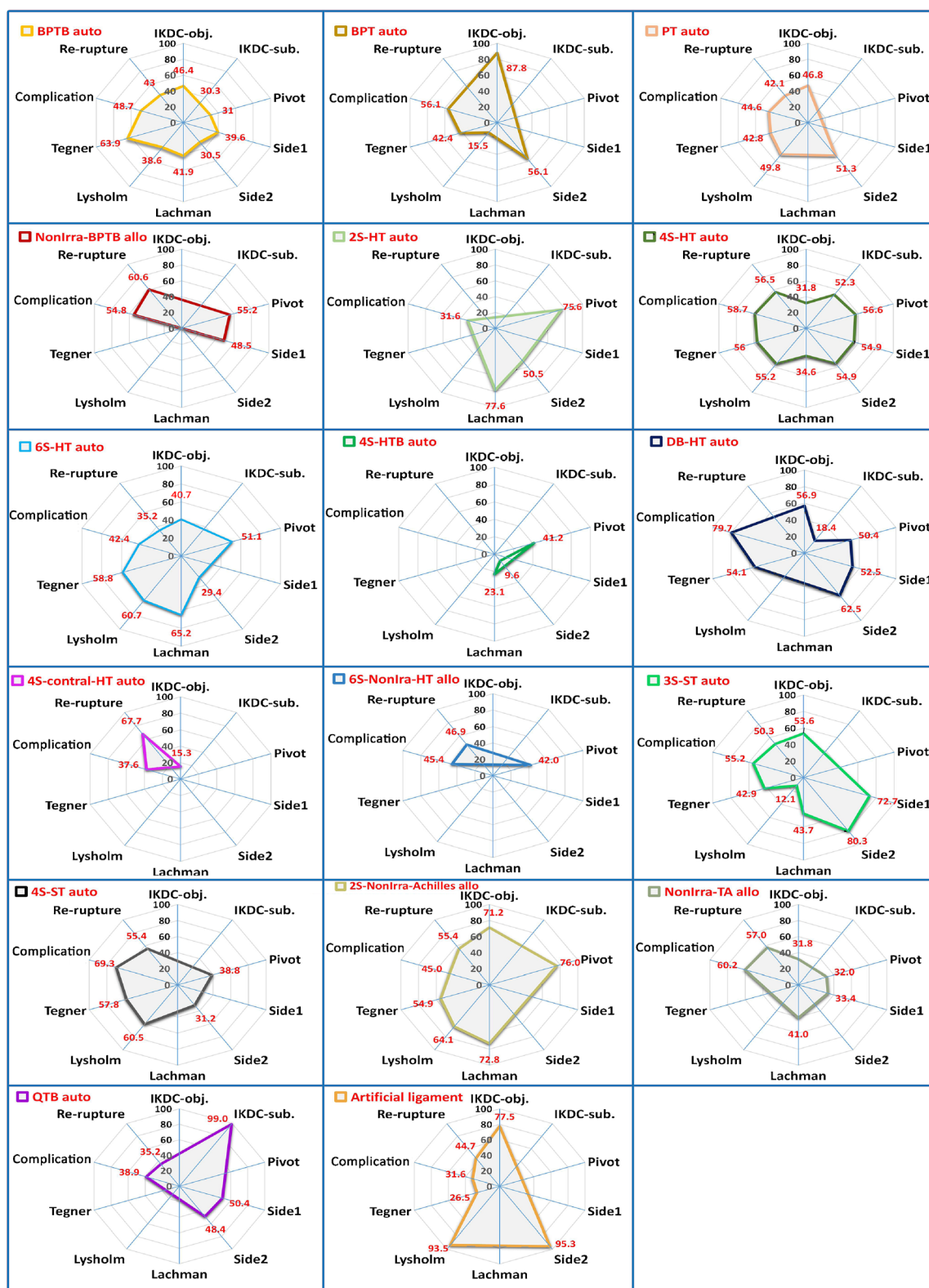


Fig. 5 Radar plots for the grafts with the axes presenting SUCRA values of the ten outcomes. IKDC-obj., IKDC objective score; IKDC-sub., IKDC subjective score; Pivot, pivot-shift test; Side1, side-to-side difference (categorical data); Side2, side-to-side difference (continuous data).

The SUCRA values are positively related to the magnitude of absolute measurement on the outcomes. Abbreviations and full names of the grafts are listed in Table 1

some biological agents or cells could be selected for augmentation of the reconstruction process [7, 73].

Limitations

Several weaknesses of this review must be stated. Firstly, available primary trials are predominately focused on comparing the BPTB and HT autografts, remaining a relatively small sample investigated other intervention arms. This may not allow for the detection of small differences between treatments, meaning there is risk of type II error. Secondly, different internal fixation devices and operation procedures were introduced for the various tendon transplants in the primary trials which have been recognized to confound treatment effects of different grafts themselves [74–76]. Finally, inconsistency on post-operative rehabilitation protocols may produce another potential risk of bias.

Conclusions

A recommendation for graft selection was proposed on the basis of the available direct and indirect evidence. We found that the four-strand HT autograft has an increased knee joint laxity than the BPTB autograft. The QTB autograft is recommended as one of the best choices and should be used even over the BPTB and four-strand HT autografts. We also recommend that non-irradiated allografts may be used as alternatives to autogenous tendon grafts. However, artificial ligaments are not recommended for primary ACLR, but could be used for revision procedures if other options are not available, or used in combination with other graft types.

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Compliance with ethical standards

Conflict of Interest The authors declare that they have no conflict of interest.

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